



Hospital Readmission Prediction

TA: Rana

Team ID: 68

Team Members:

أحمد مدحت مصطفى سيد محمد	2023170063
احمد محمد سيد احمد محمد شرف الدين	2023170057
احمد محمد شوقي امين	2023170058
السجاد شمس الكمال نور الهادي أحمد	2022170070
محمود حماده محمود علي	2023170553
زياد اياد فهد	2023170821

Project Overview

This project aims to analyze and predict hospital readmissions for diabetic patients using the dataset provided. The task involves comprehensive data preprocessing, exploratory data analysis (EDA), and implementing several machine learning models to predict readmissions effectively.

Data Source

The dataset, **diabetic_data.csv**, contains information on diabetic patients hospitalized, including demographic information, admission details, diagnoses, and medications.

Objectives

- To preprocess and clean the dataset for accurate analysis.
- To perform exploratory data analysis to understand the data structure and distributions.
- To apply machine learning models to predict patient readmissions.
- To evaluate model performance using appropriate metrics.

Steps and Methodologies

Data Preprocessing

1. **Reading the Data:** Import the data using **pandas**.
2. **Handling Missing Values:** Replace placeholder missing values ('?') with **NaN** and drop columns with excessive missing data.
3. **Removing Irrelevant Columns:** Drop columns that are deemed not useful for prediction based on domain knowledge.
4. **Encoding Categorical Features:** Use **LabelEncoder** to convert categorical variables into numerical form.
5. **Feature Engineering:** Create additional features like **med_change** and **num_med** to capture medication variations.

Exploratory Data Analysis (EDA)

1. **Descriptive Statistics:** Use **describe()** to get a statistical summary of the numerical columns.
2. **Distribution Visualization:** Use histograms and bar plots in **seaborn** and **matplotlib** to understand the distribution of key features.
3. **Correlation Analysis:** Create a heatmap of correlations between numerical attributes to detect potential relationships.

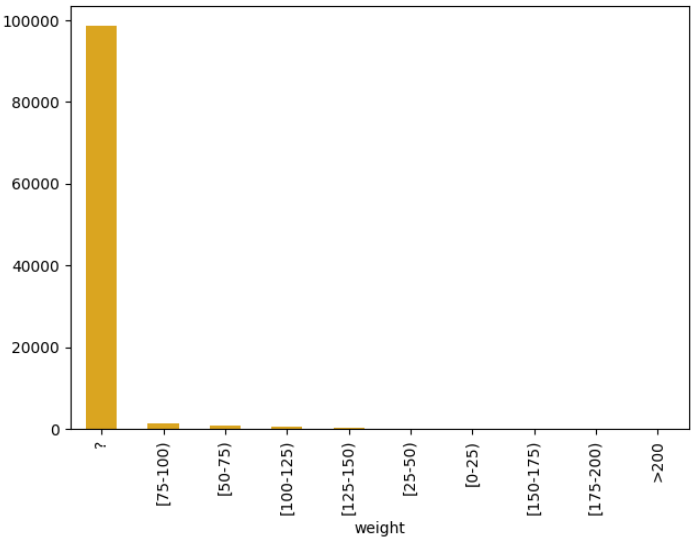
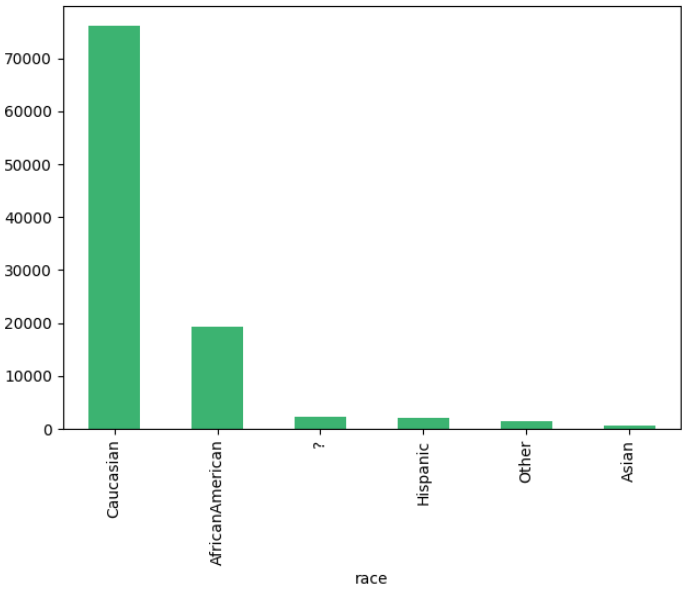
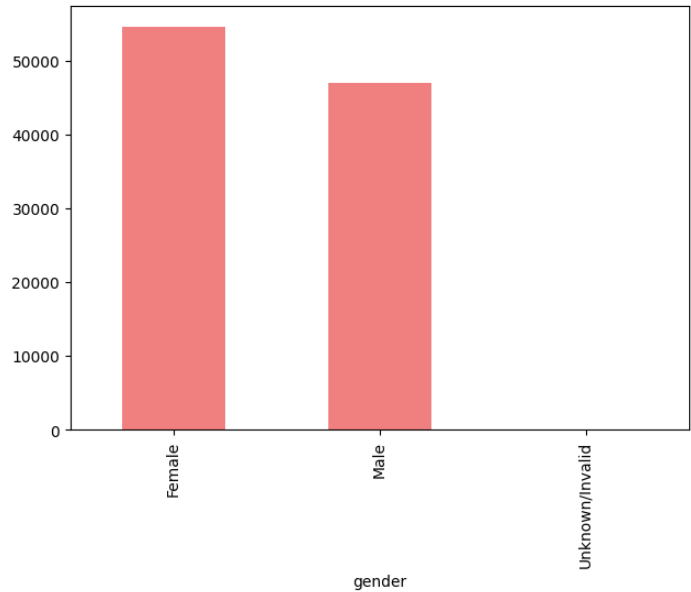
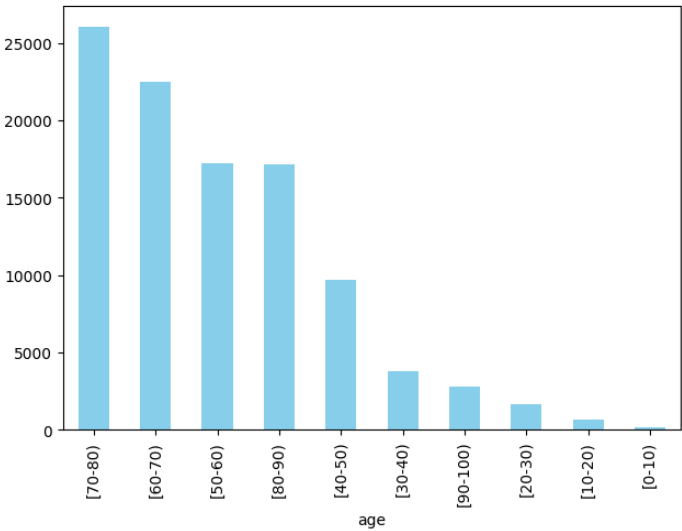
Model Implementation

1. **Data Splitting:** Split the dataset into training and testing sets using `train_test_split`.
2. **SMOTE:** Apply SMOTE to balance the class distribution in the training set.
3. **Model Training:**
 - **Random Forest:** A robust ensemble method known for handling complex data patterns and reducing overfitting.
 - **Logistic Regression:** Used for baseline performance, beneficial for interpreting linear relationships.
4. **Model Evaluation:** Use metrics such as accuracy, precision, recall, F1 Score, and confusion matrix to evaluate model performance.

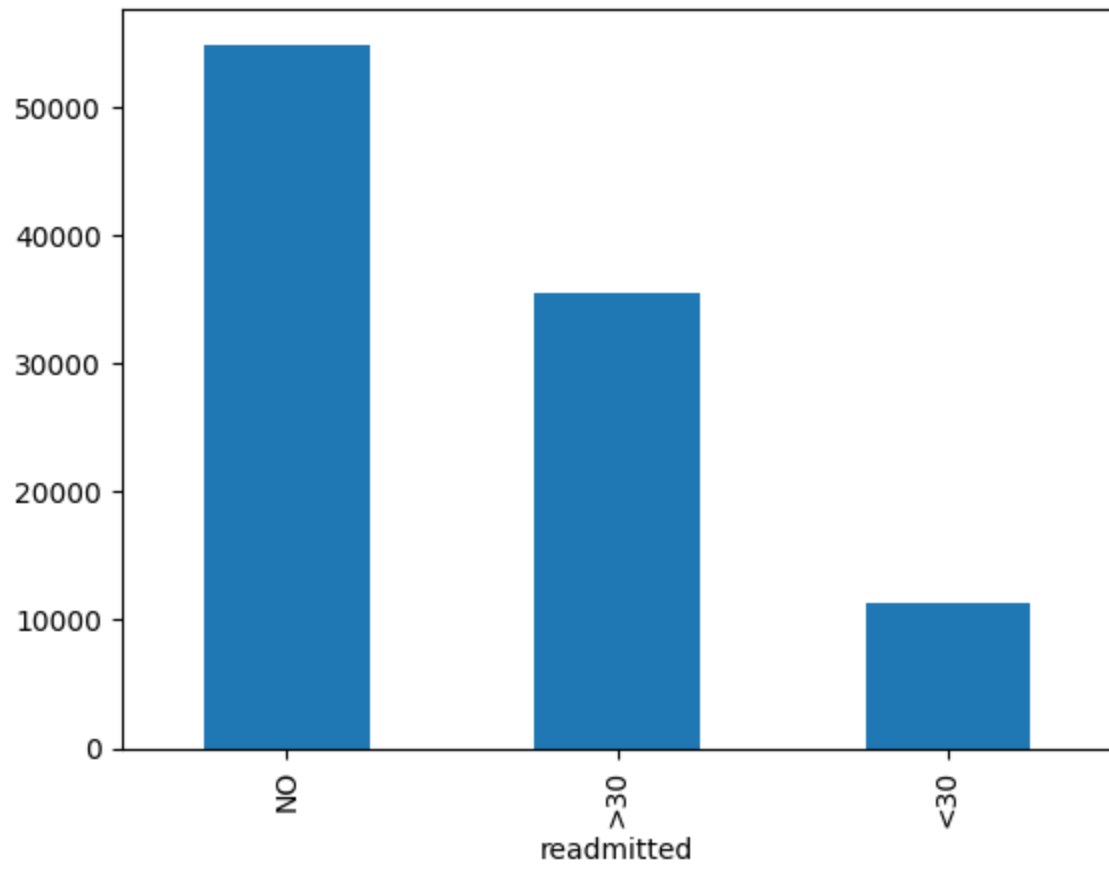
Results and Evaluation

- **Random Forest** showed superior performance in terms of recall and balance between precision and F1 Score.
- **SMOTE** effectively improved model performance by addressing class imbalance.
- Model evaluations were thoroughly conducted with detailed performance metrics and visualizations (confusion matrices, ROC curves).

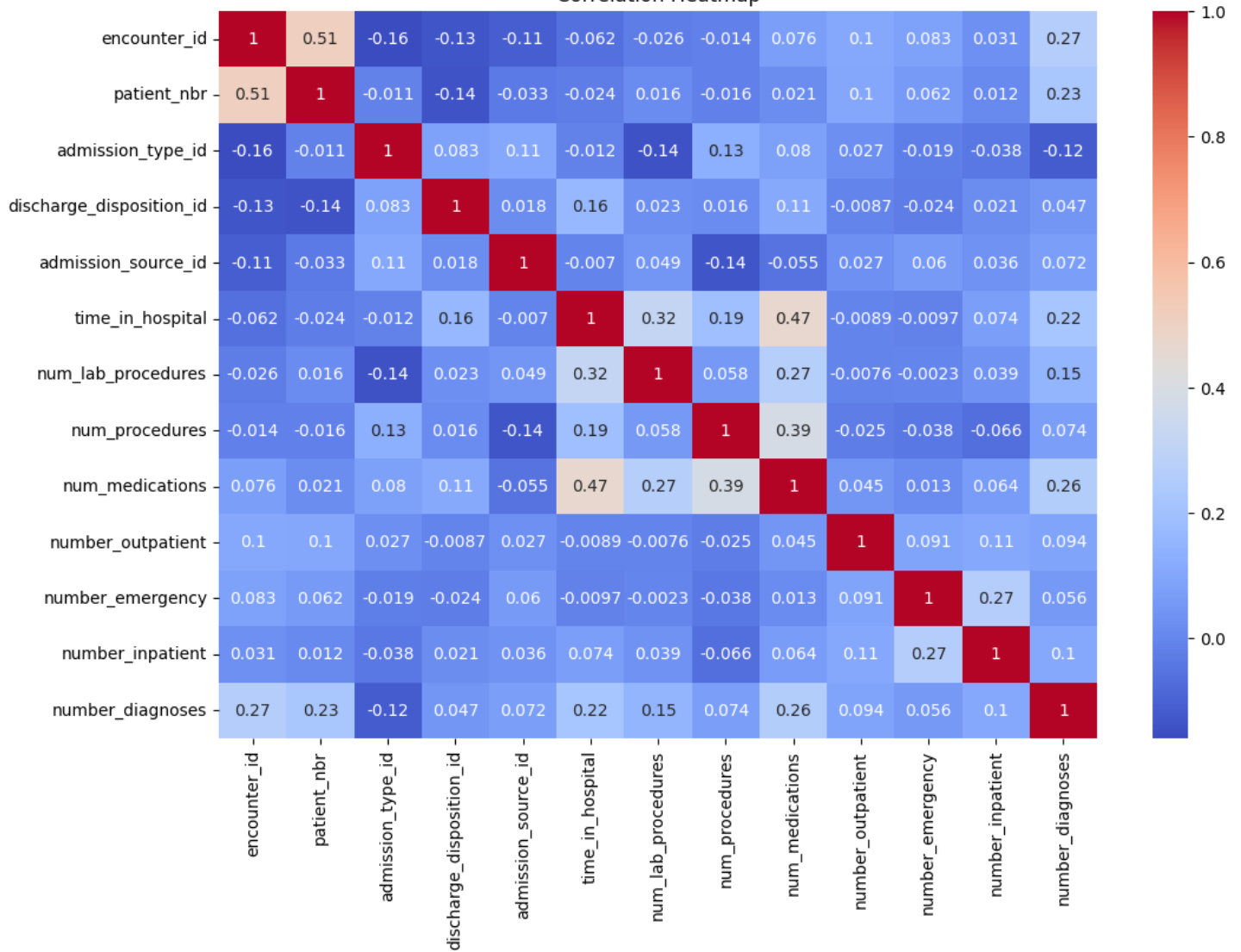
Graphs



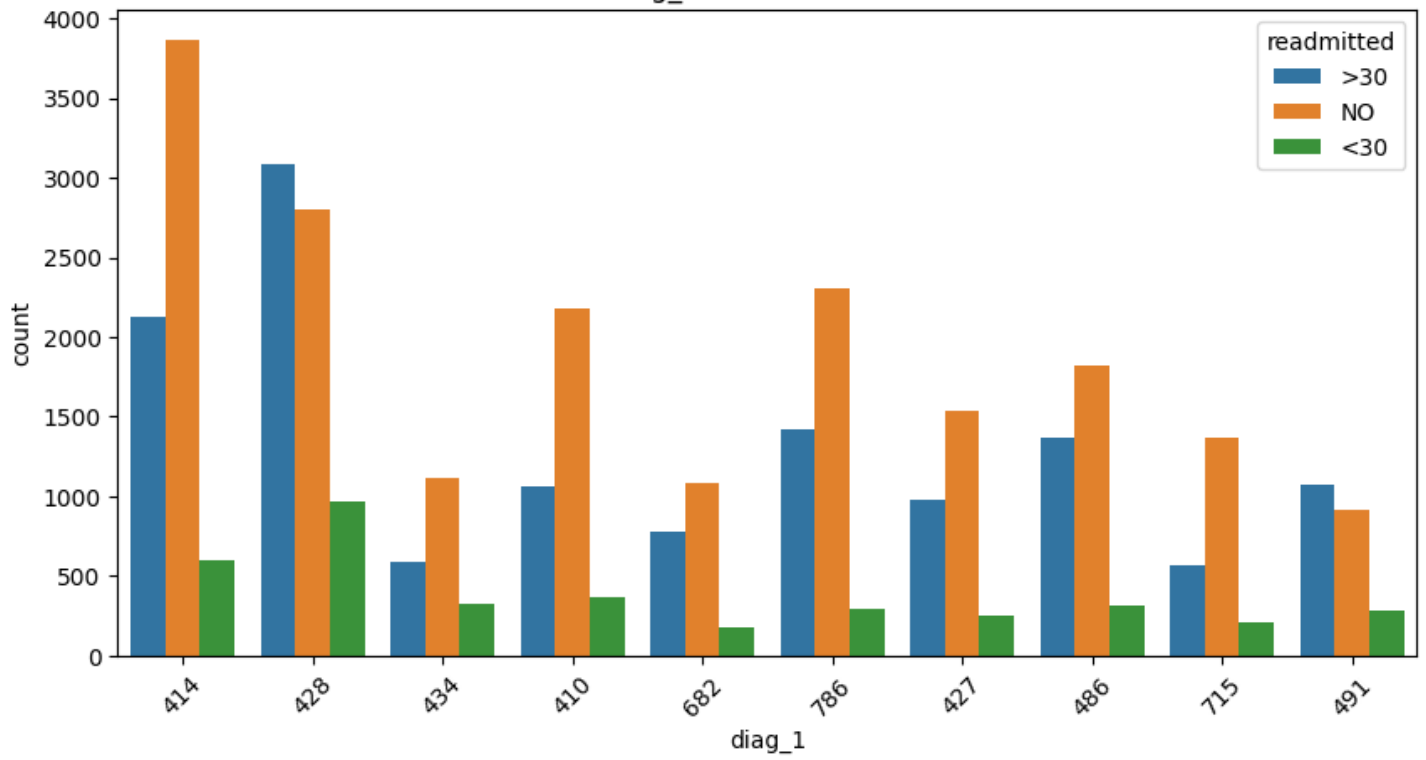
Readmission Count



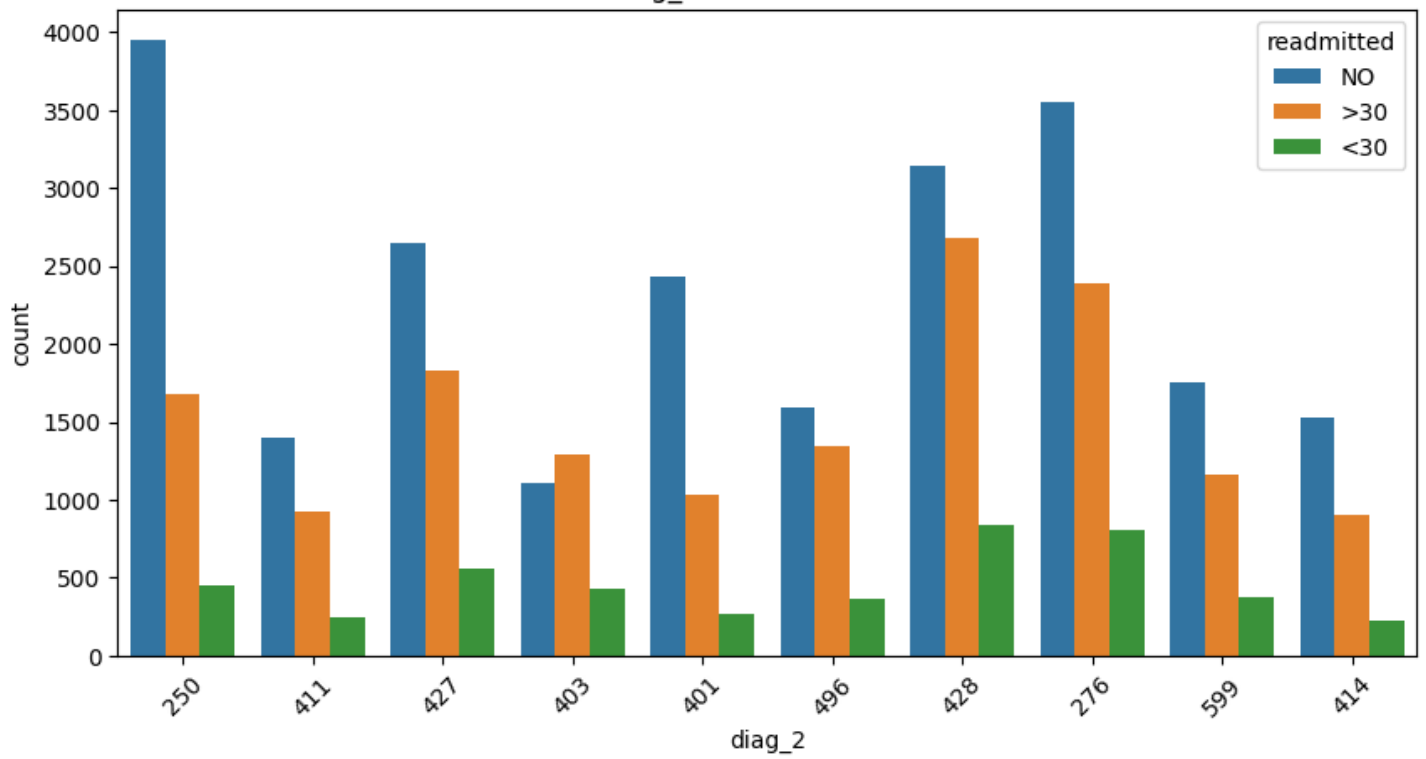
Correlation Heatmap



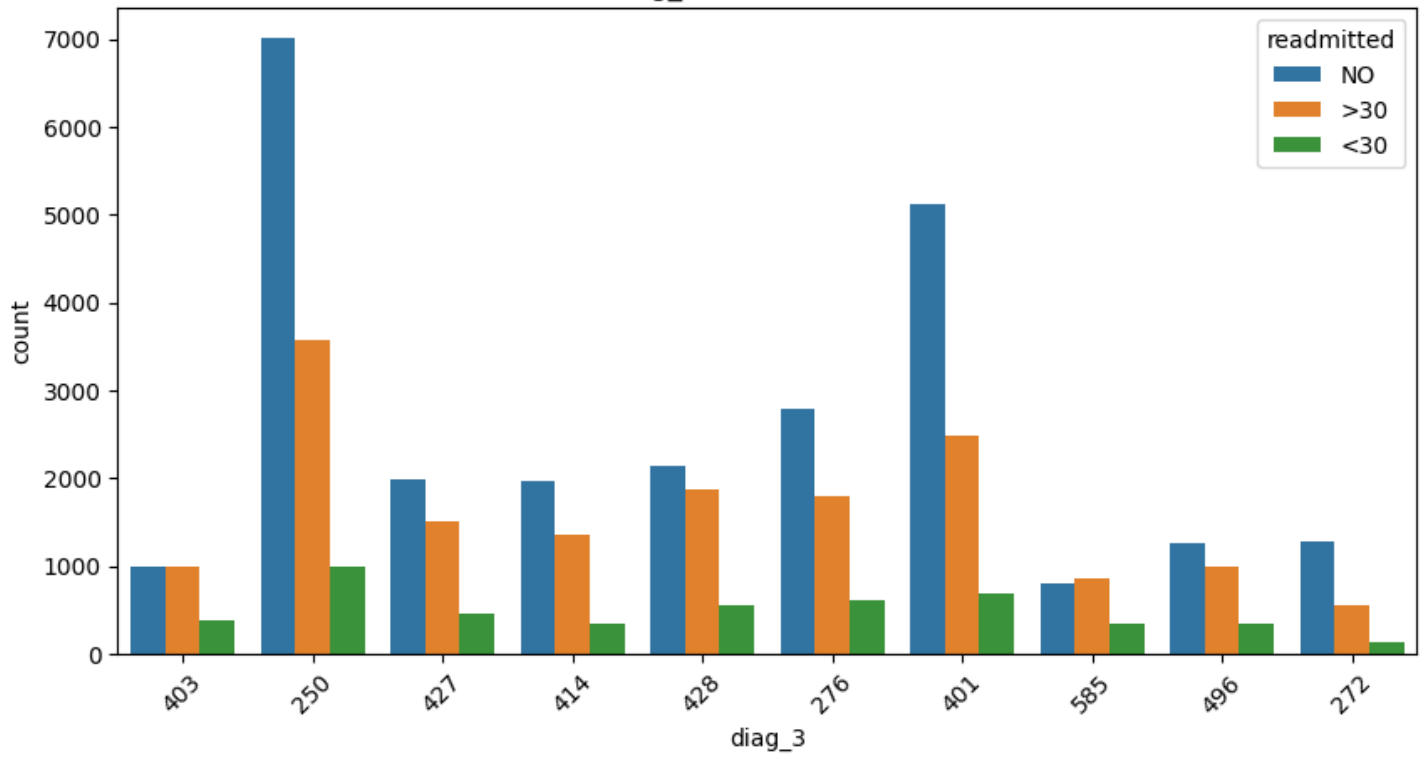
diag_1 vs Readmission



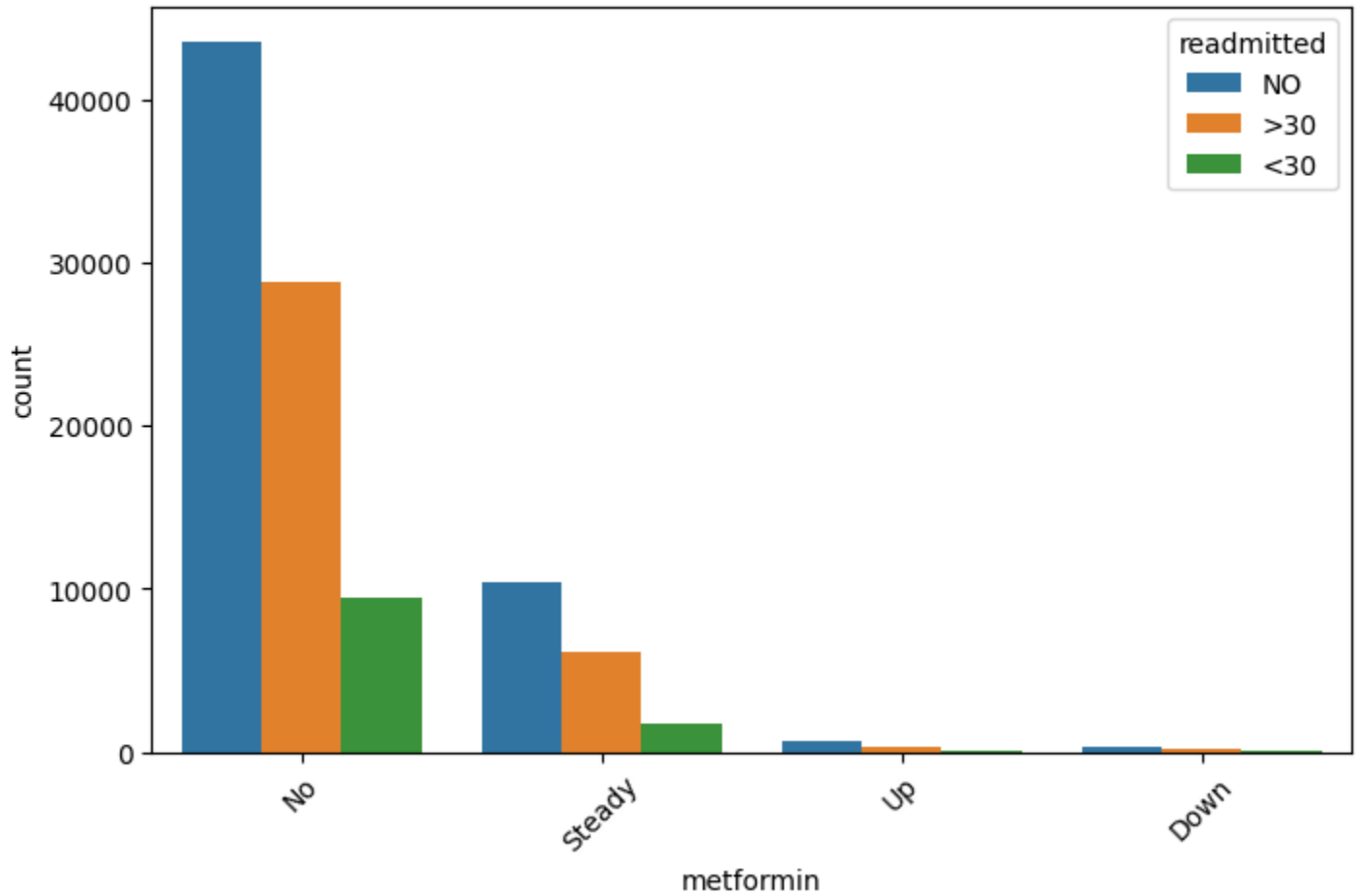
diag_2 vs Readmission



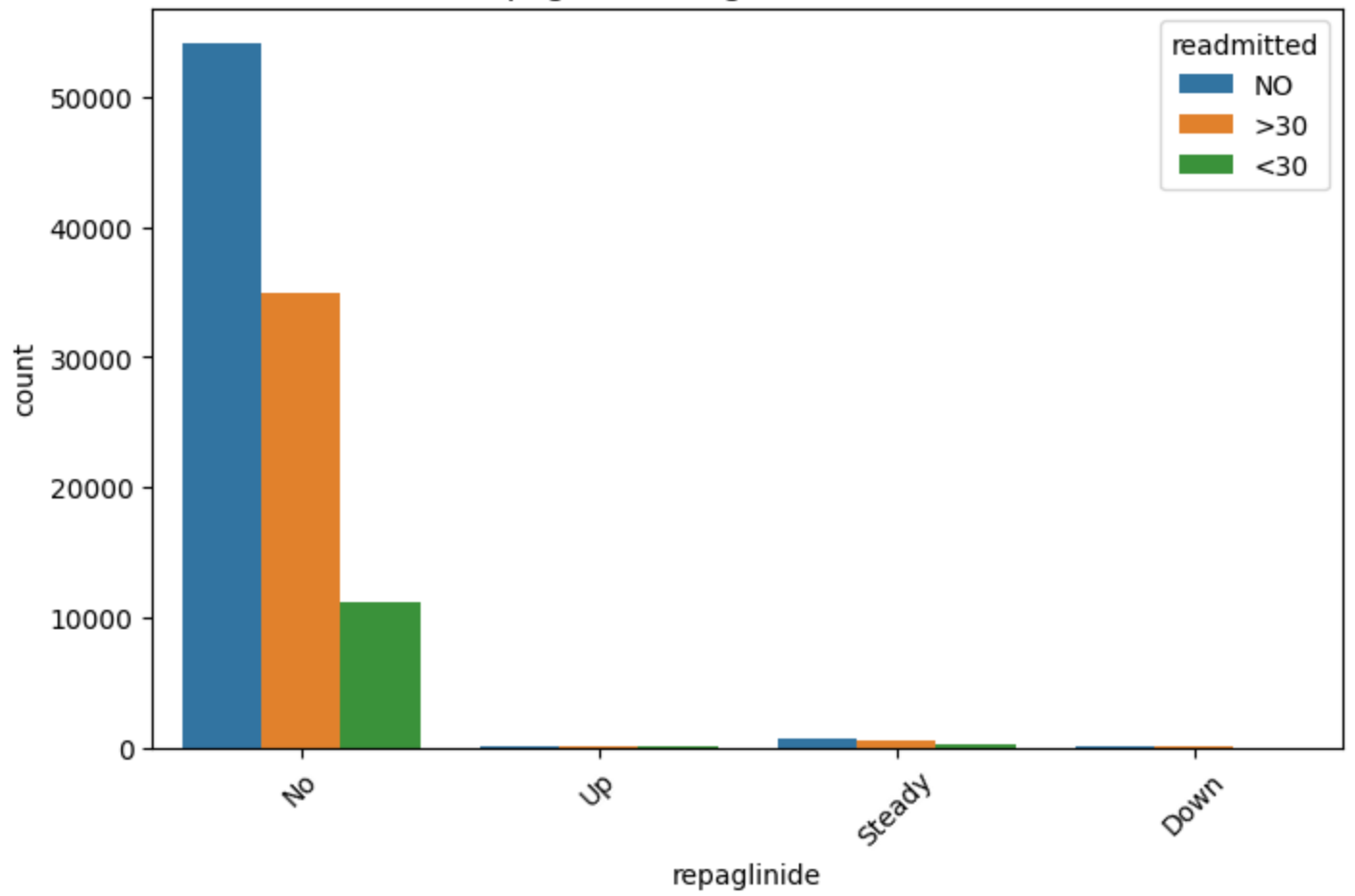
diag_3 vs Readmission



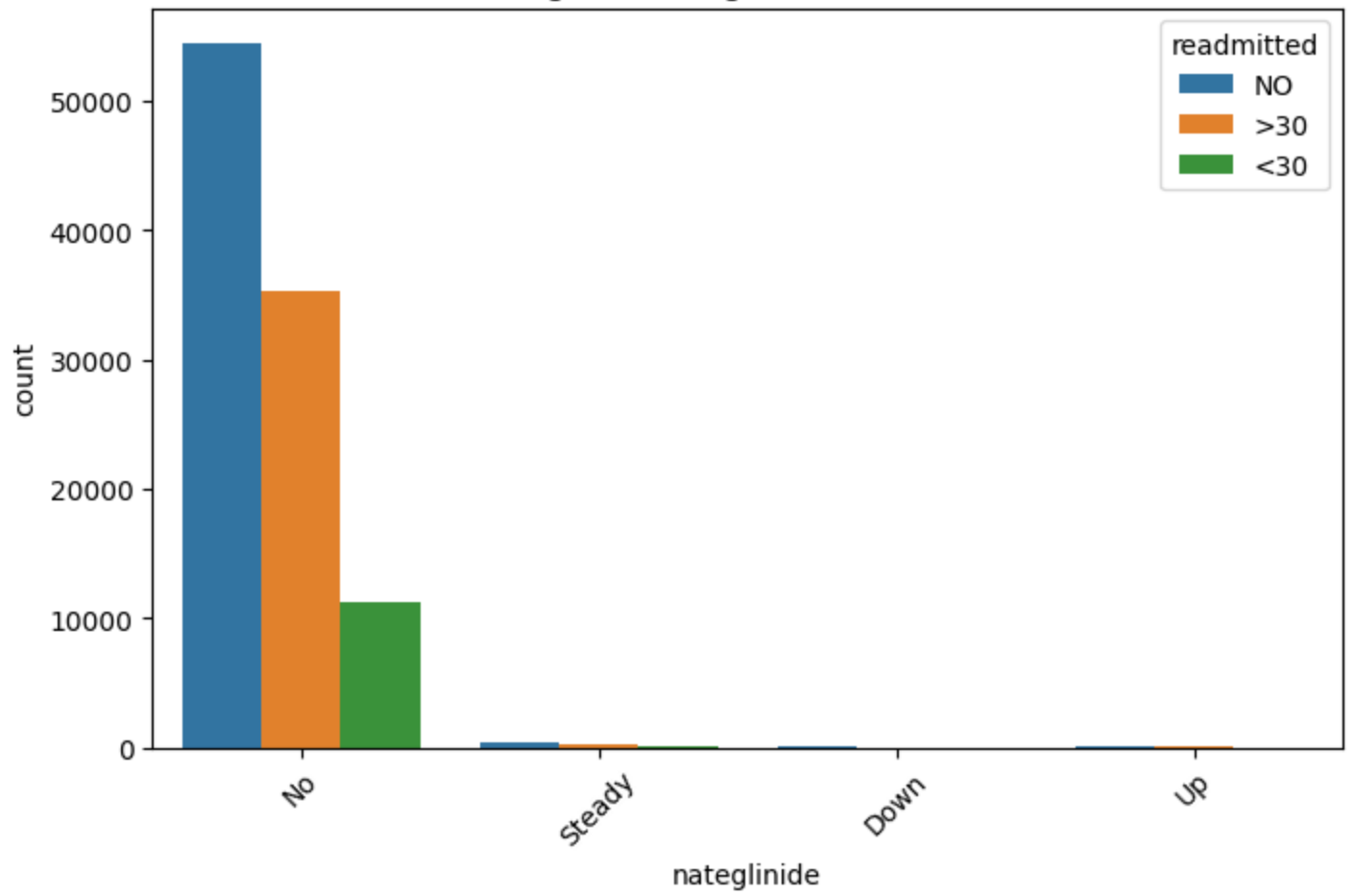
metformin Usage vs Readmission



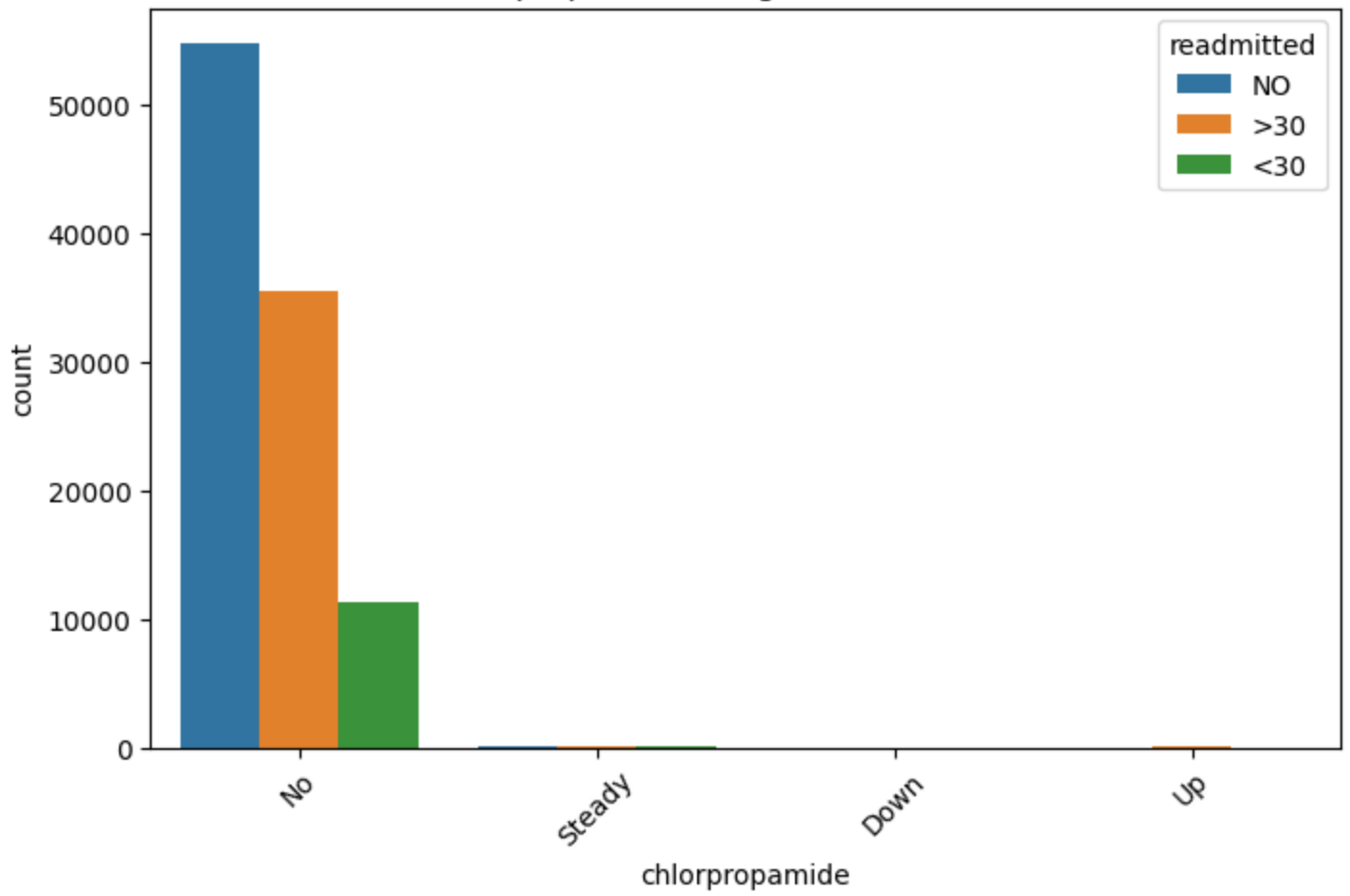
repaglinide Usage vs Readmission



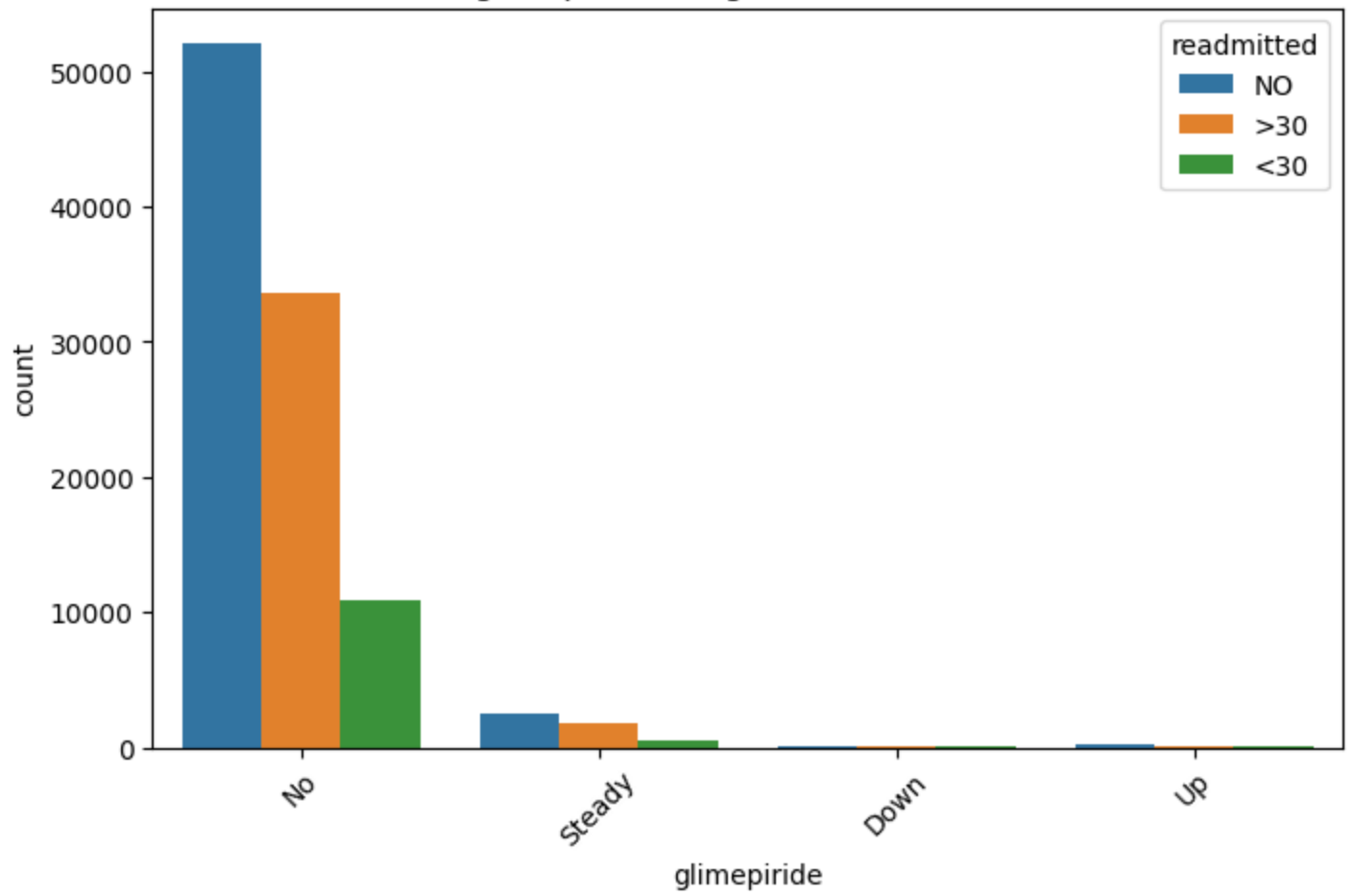
nateglinide Usage vs Readmission



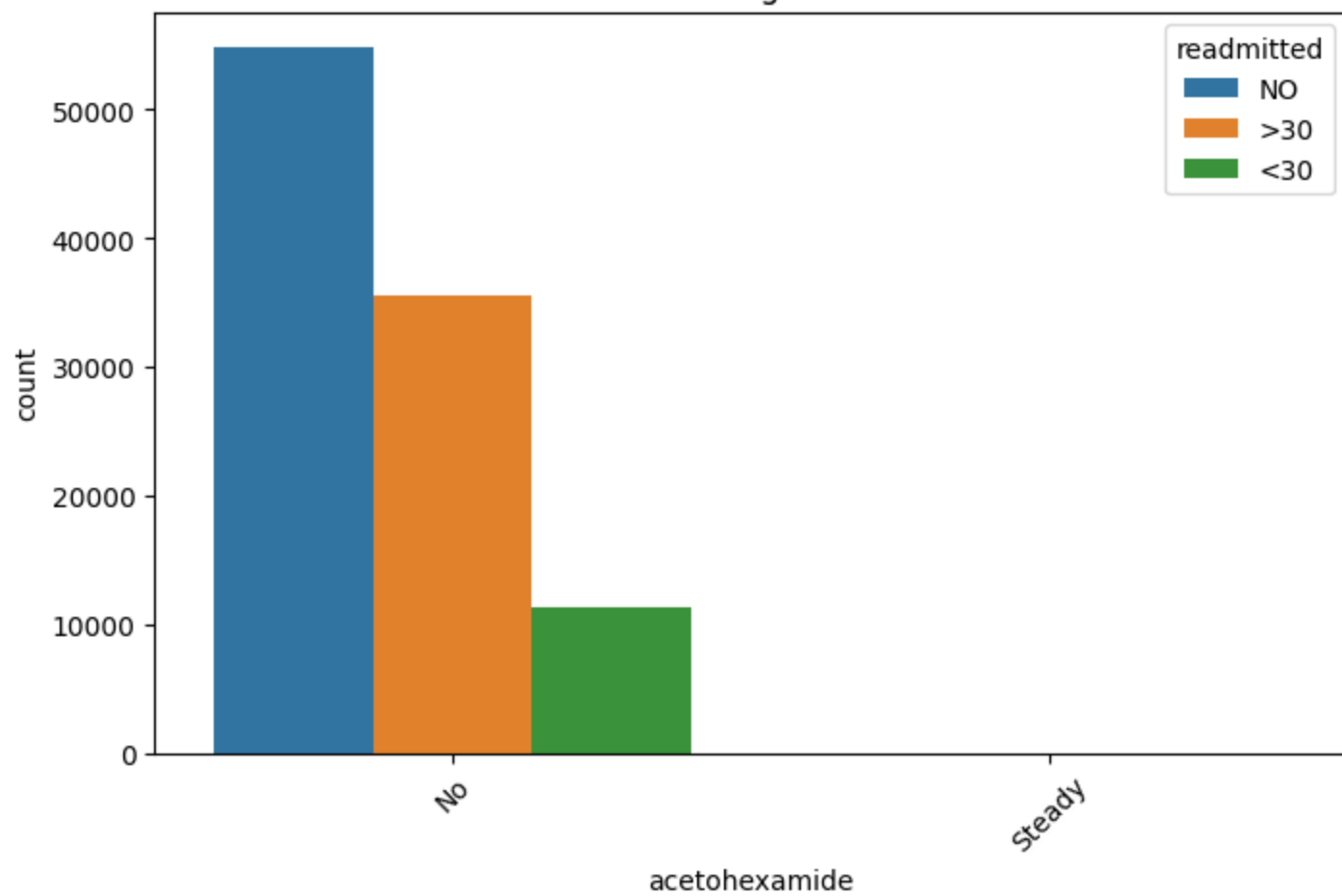
chlorpropamide Usage vs Readmission



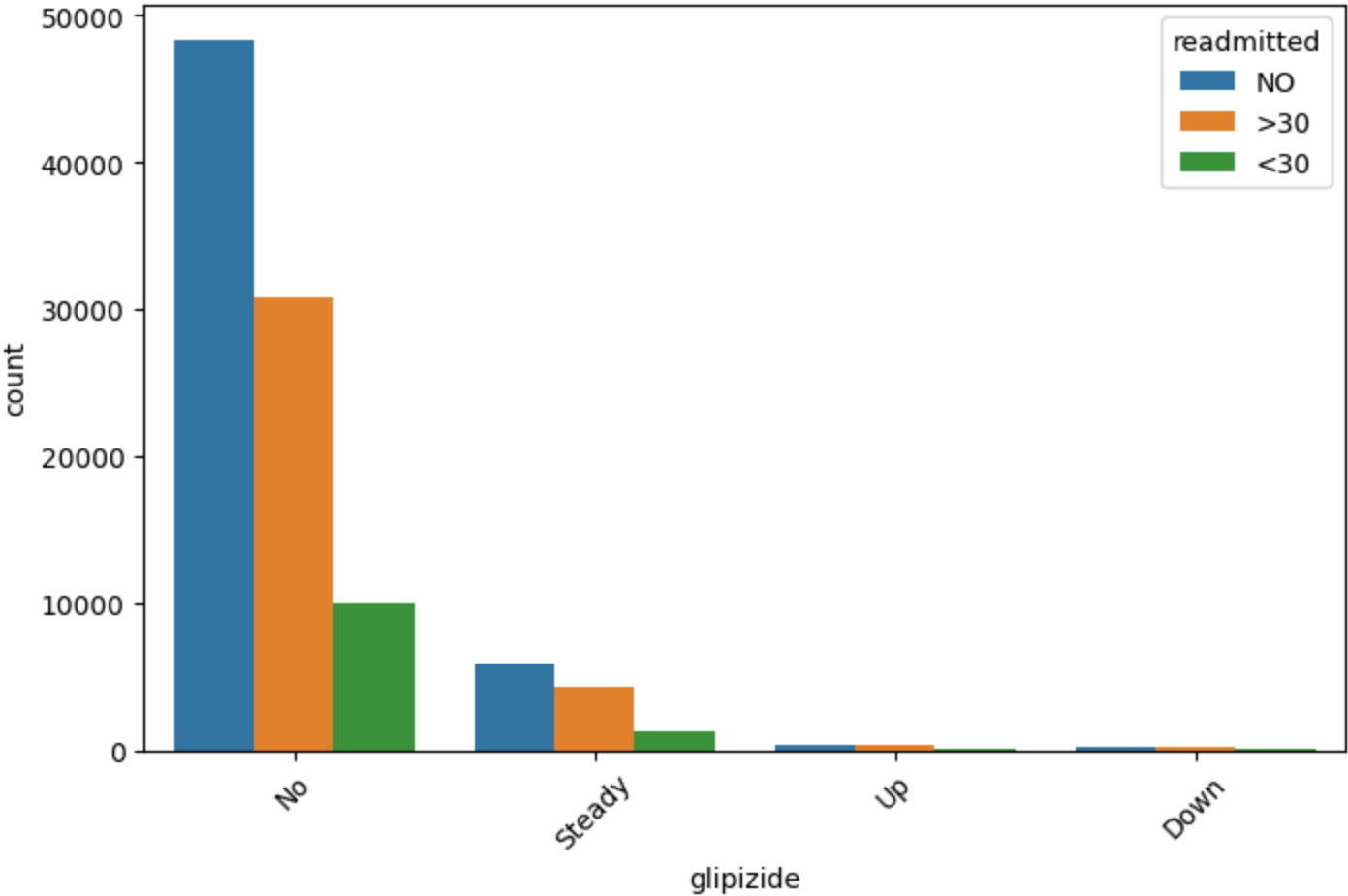
glimepiride Usage vs Readmission



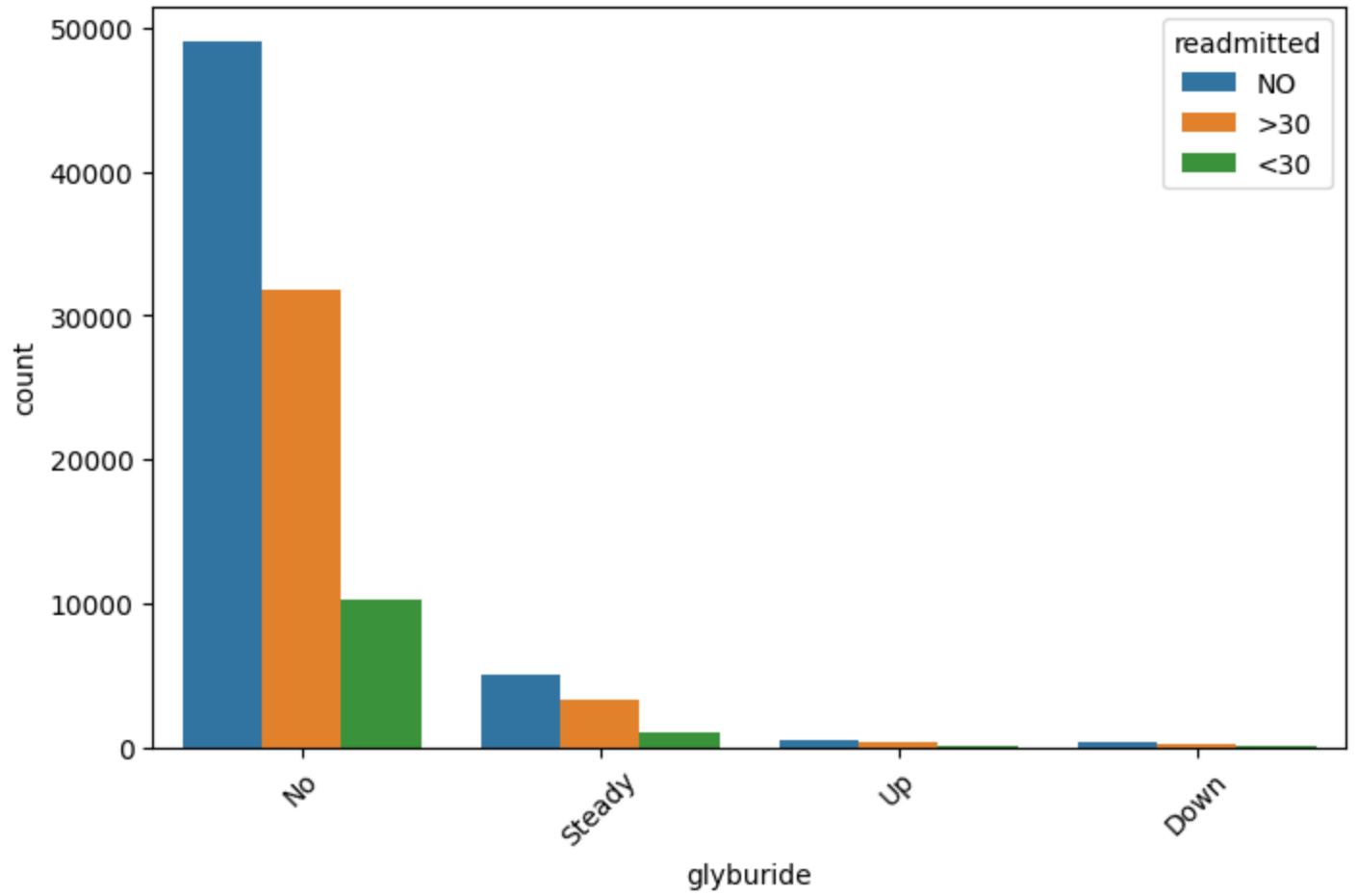
acetohexamide Usage vs Readmission



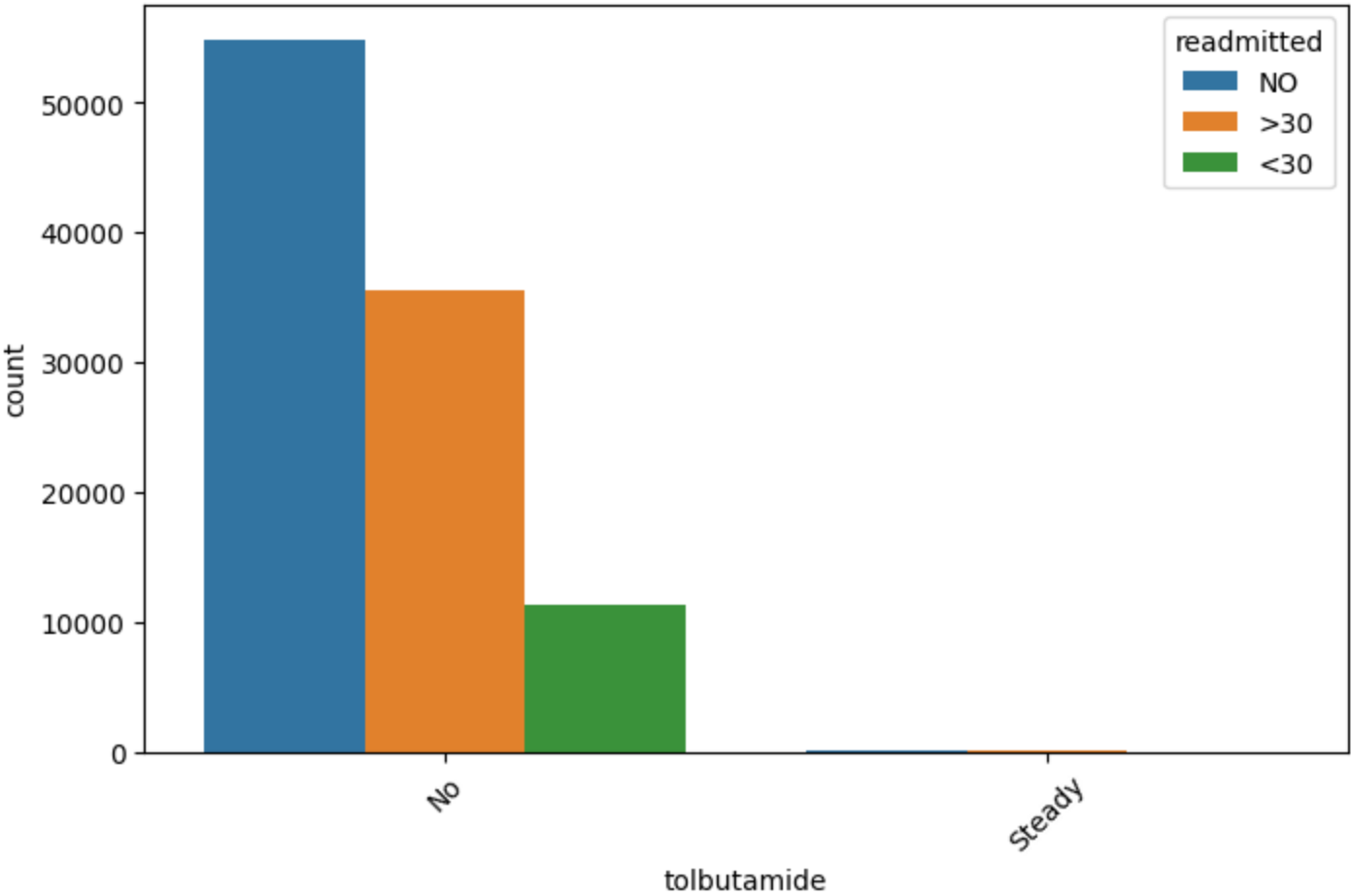
glipizide Usage vs Readmission



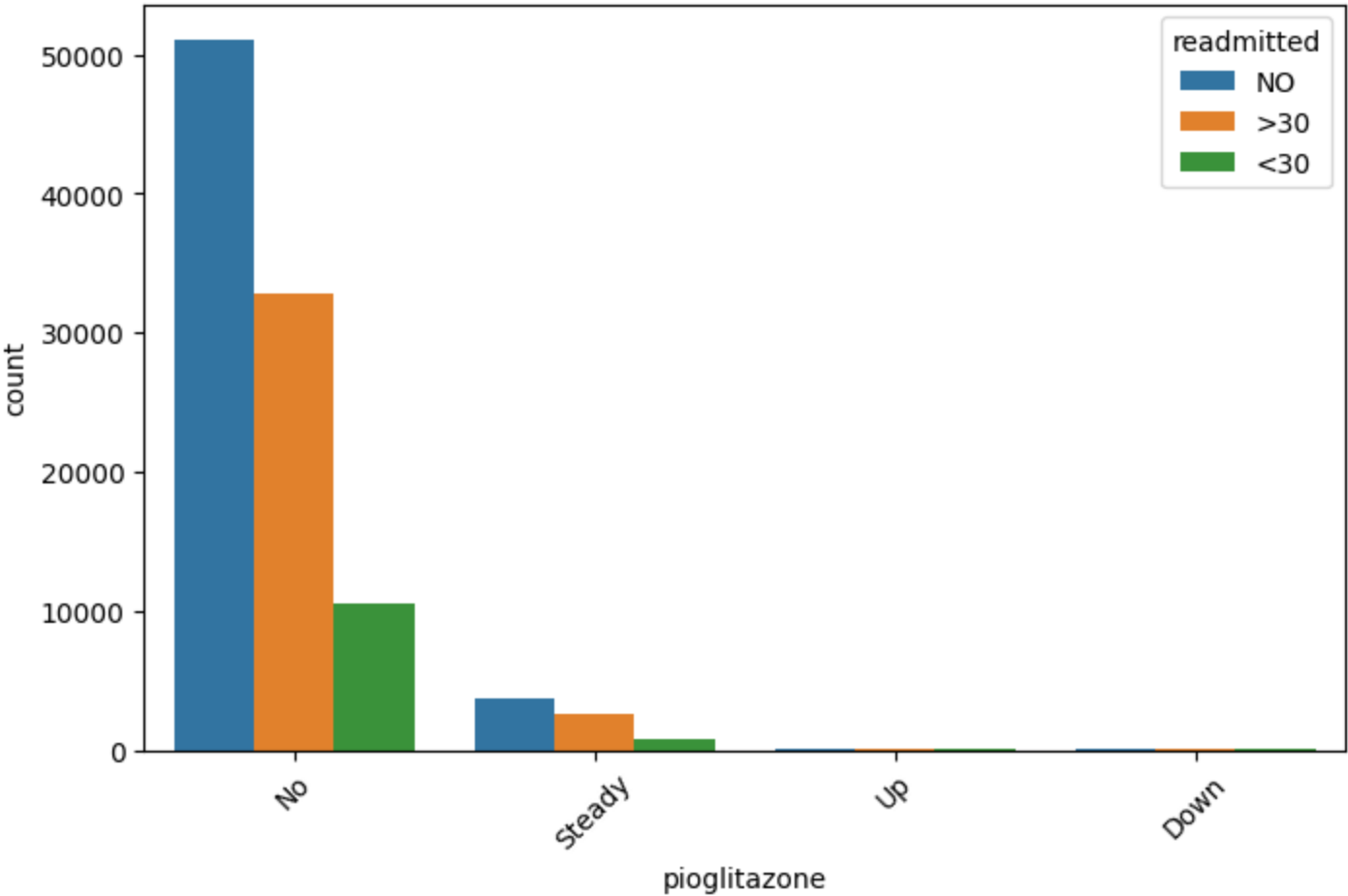
glyburide Usage vs Readmission



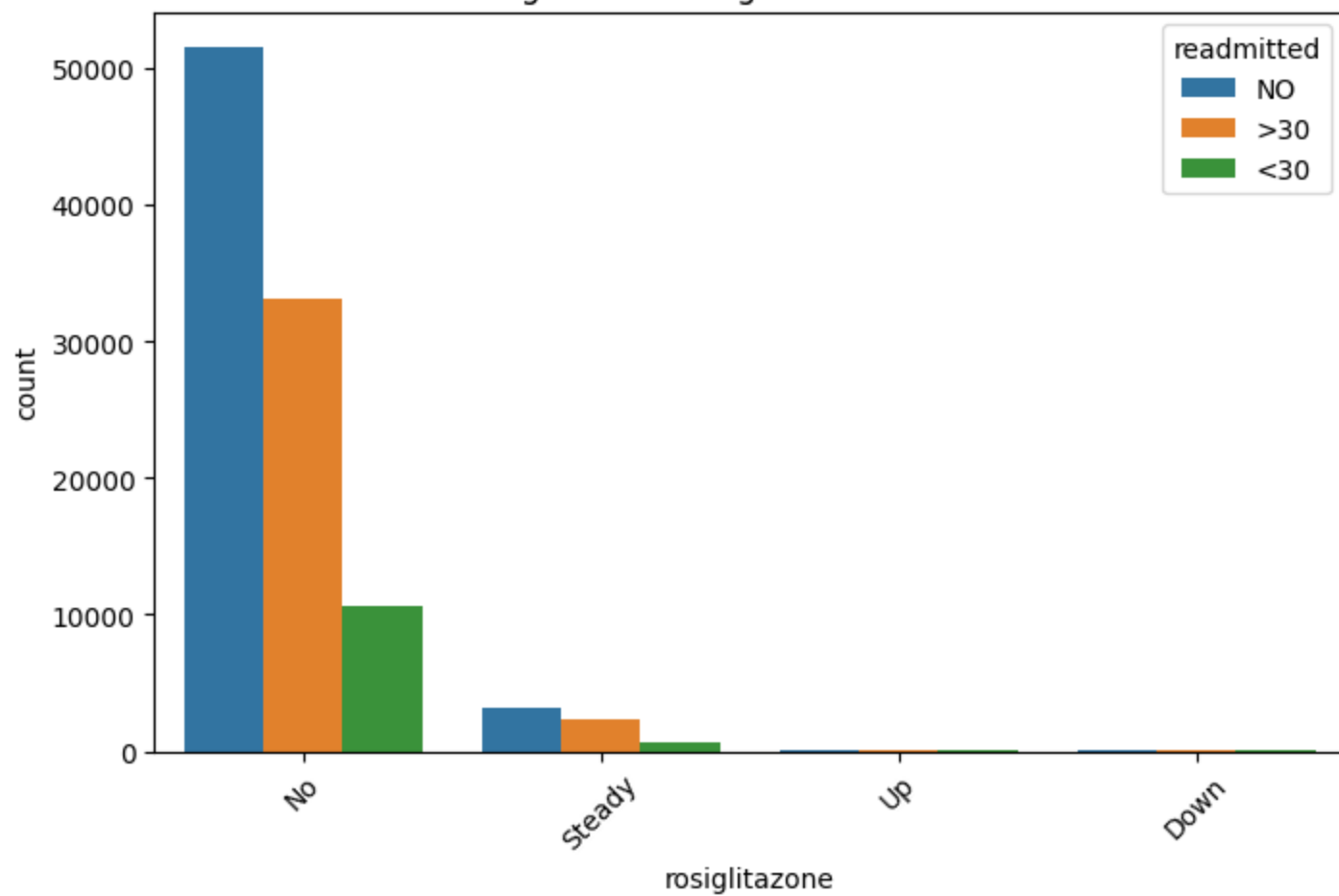
tolbutamide Usage vs Readmission



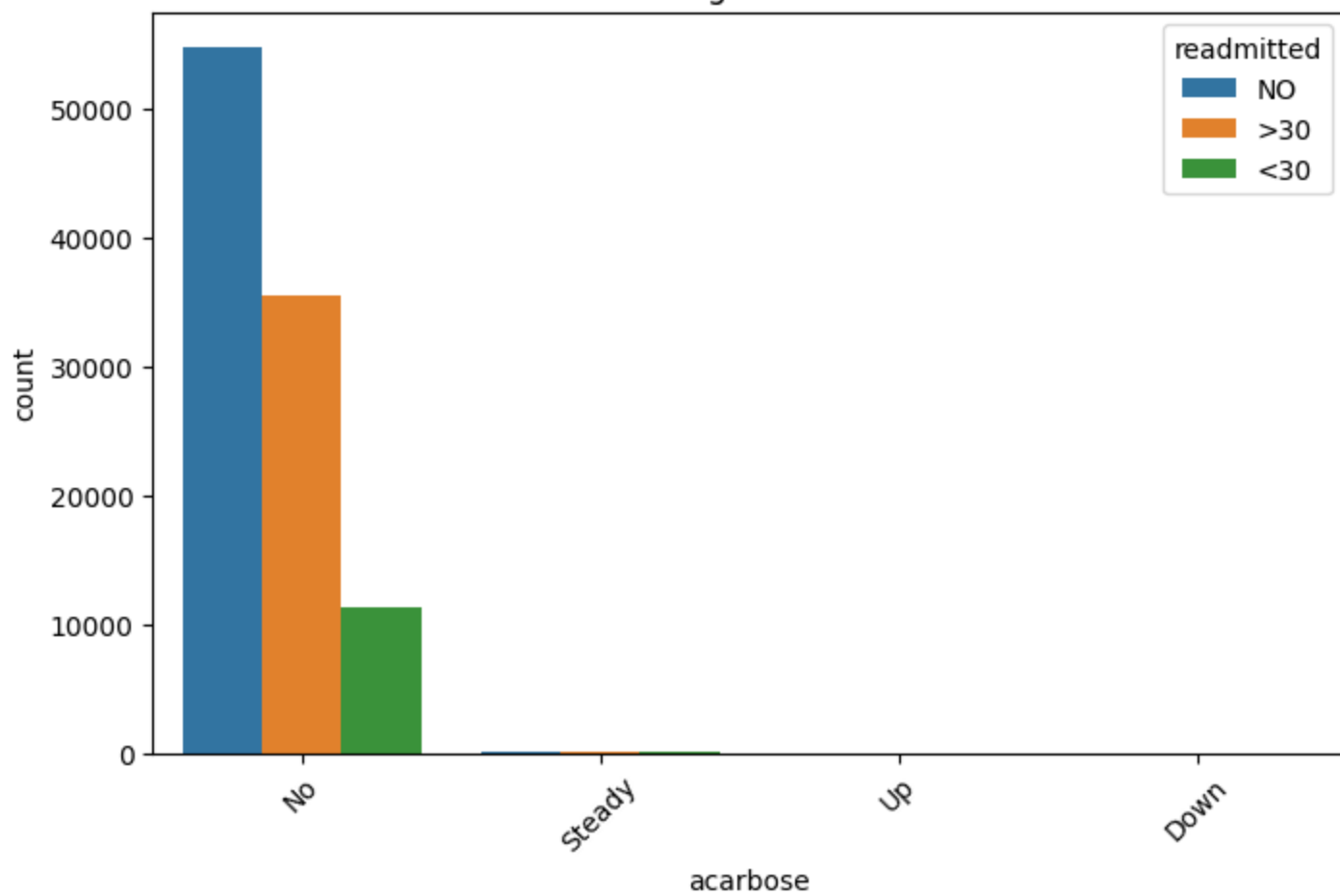
pioglitazone Usage vs Readmission



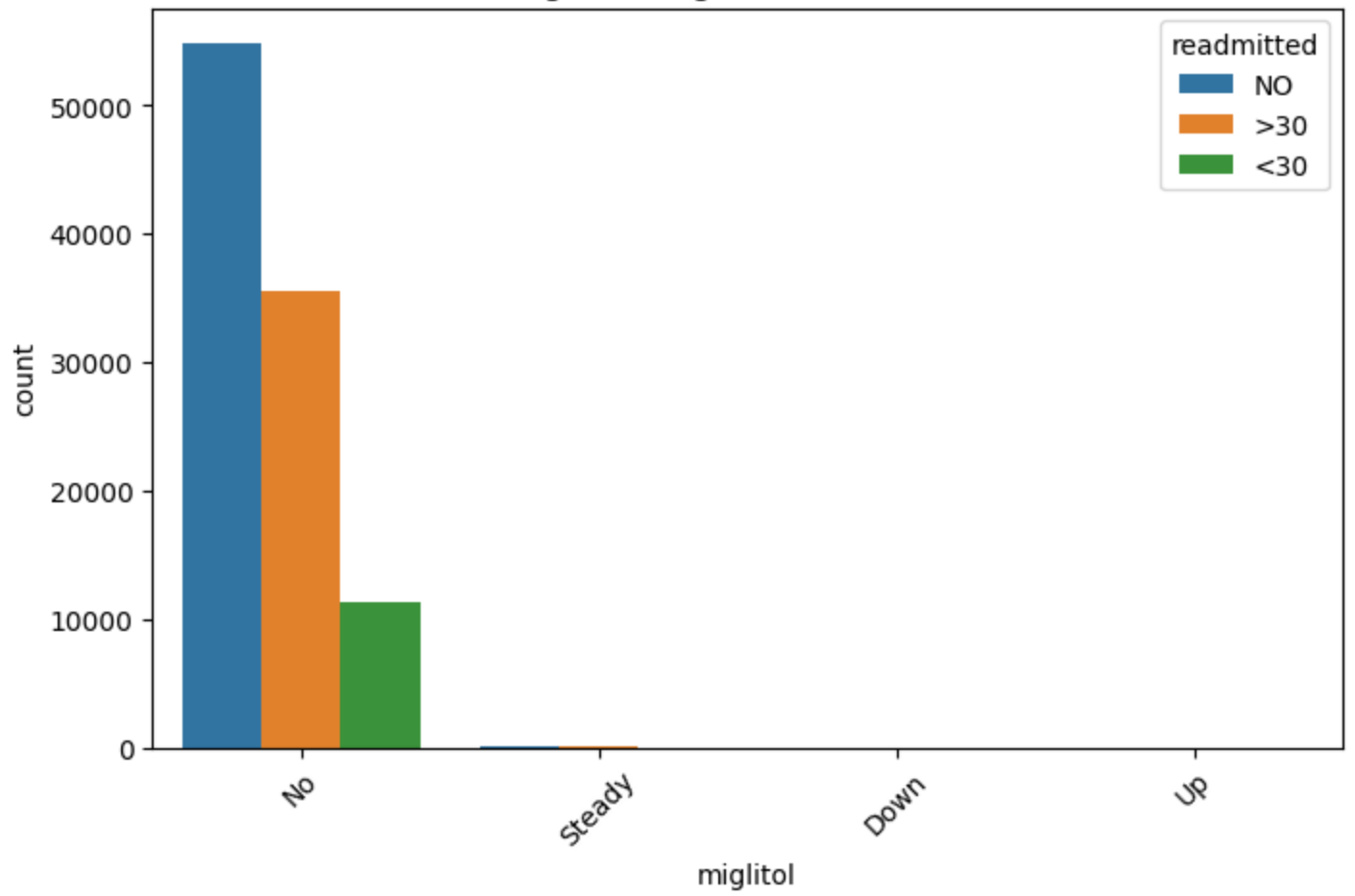
rosiglitazone Usage vs Readmission



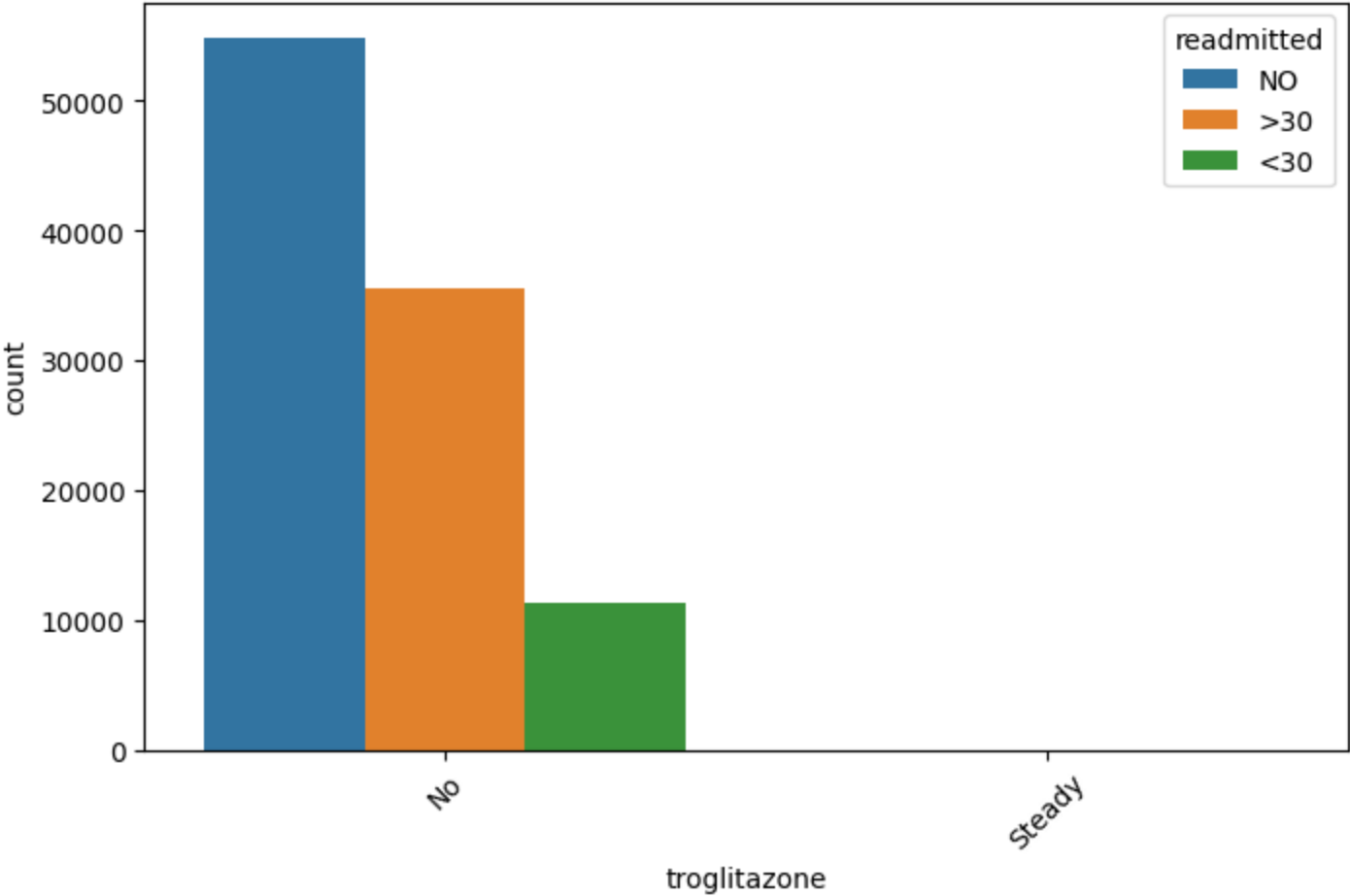
acarbose Usage vs Readmission



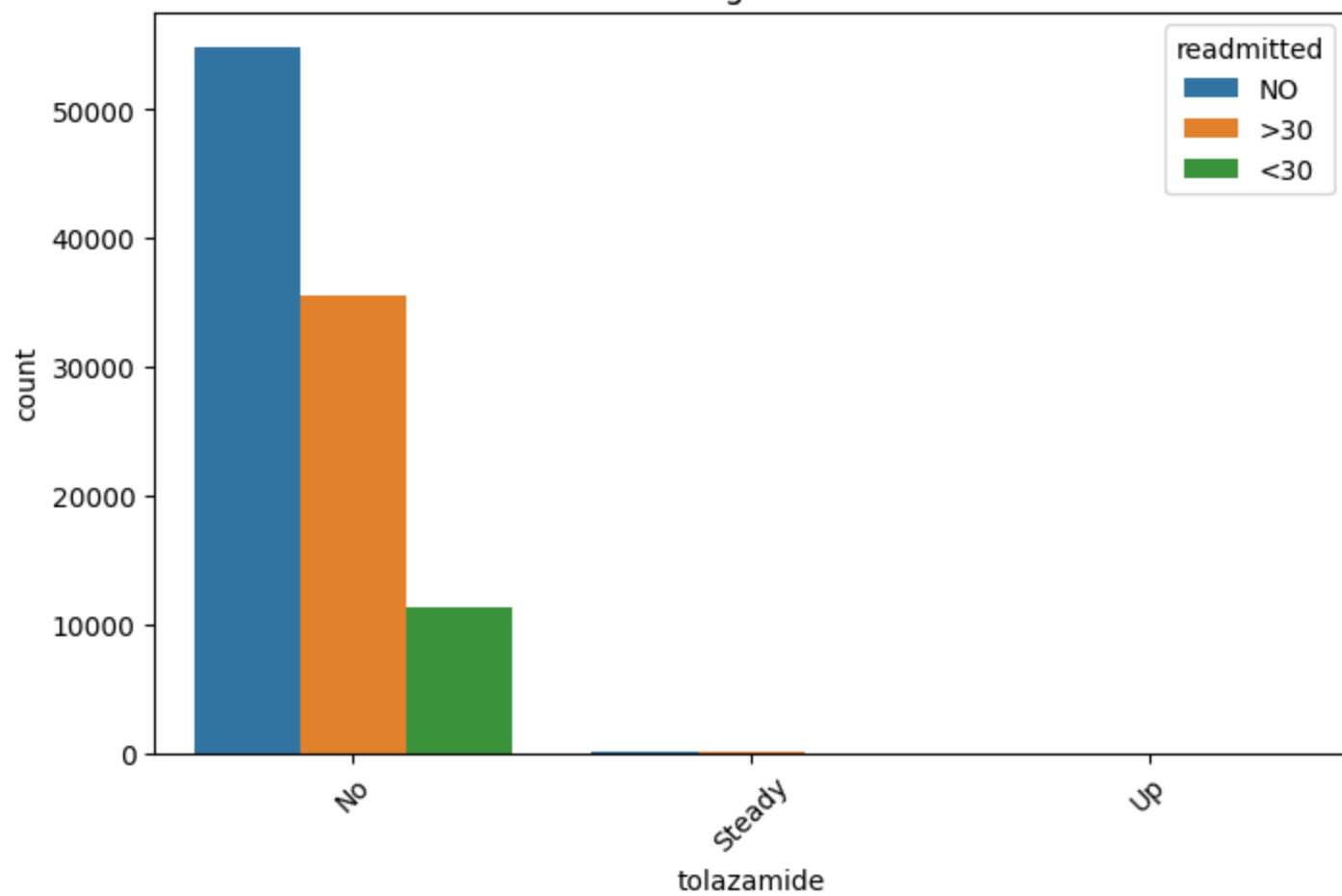
miglitol Usage vs Readmission



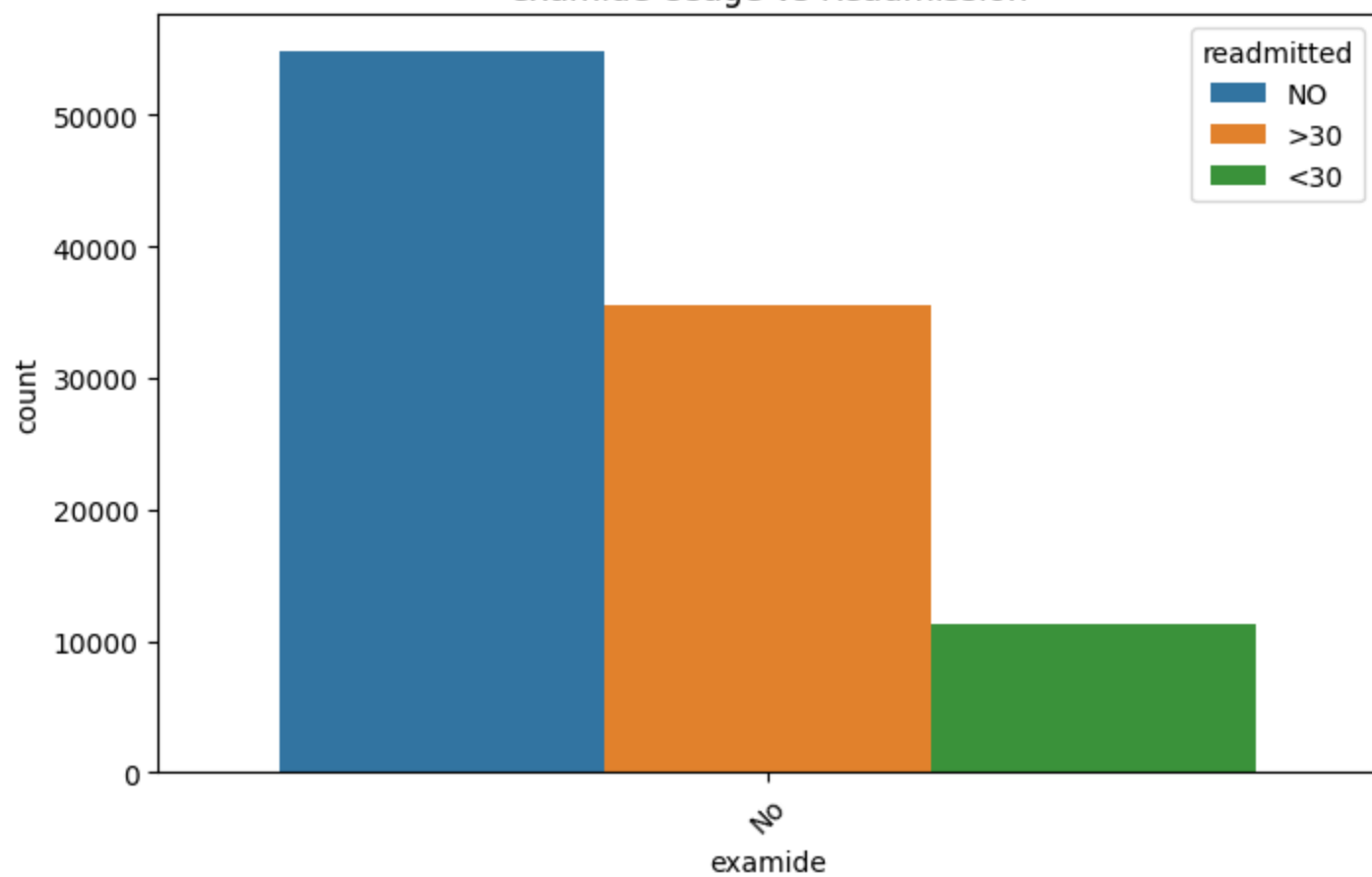
troglitazone Usage vs Readmission



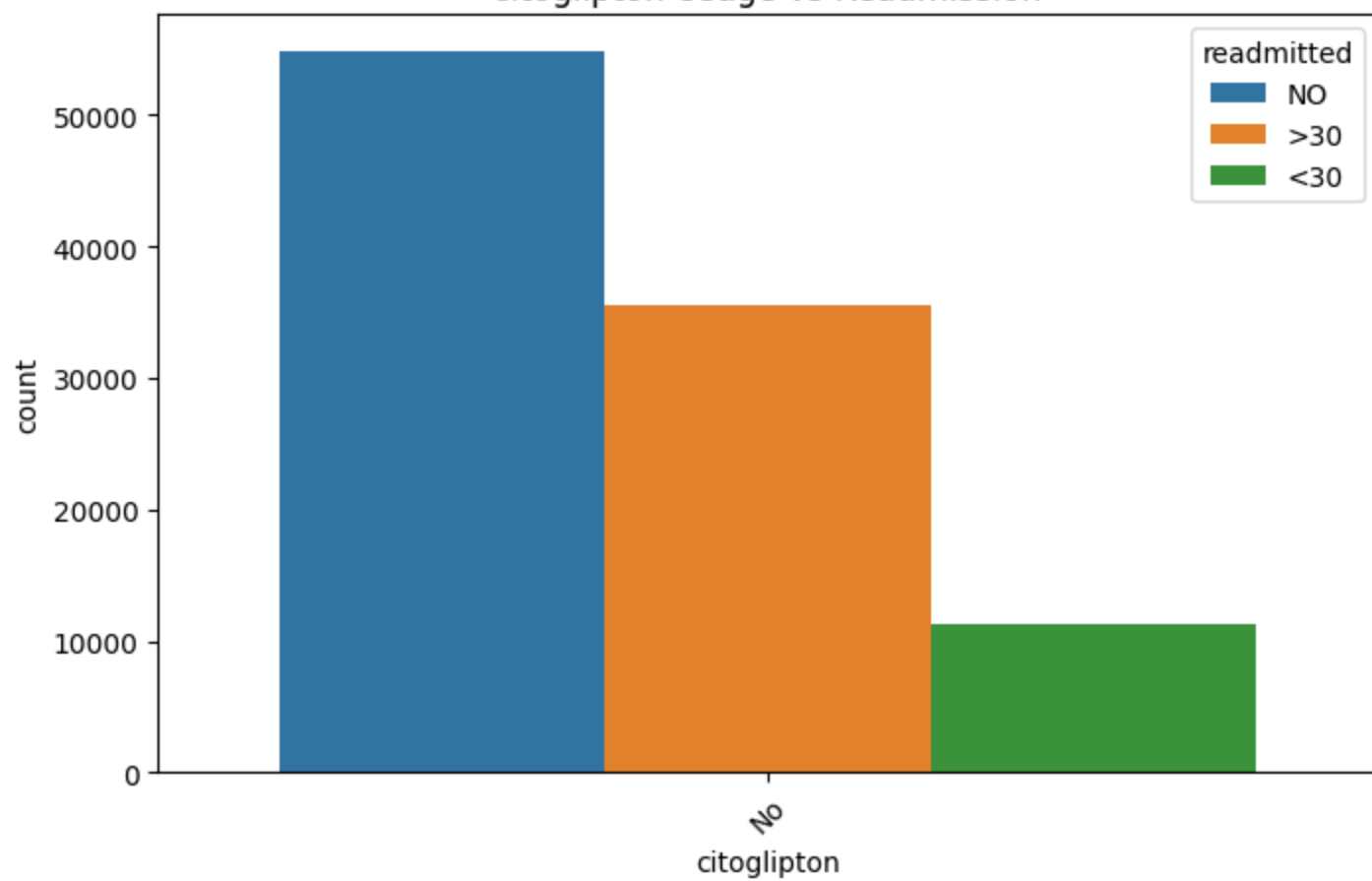
tolazamide Usage vs Readmission



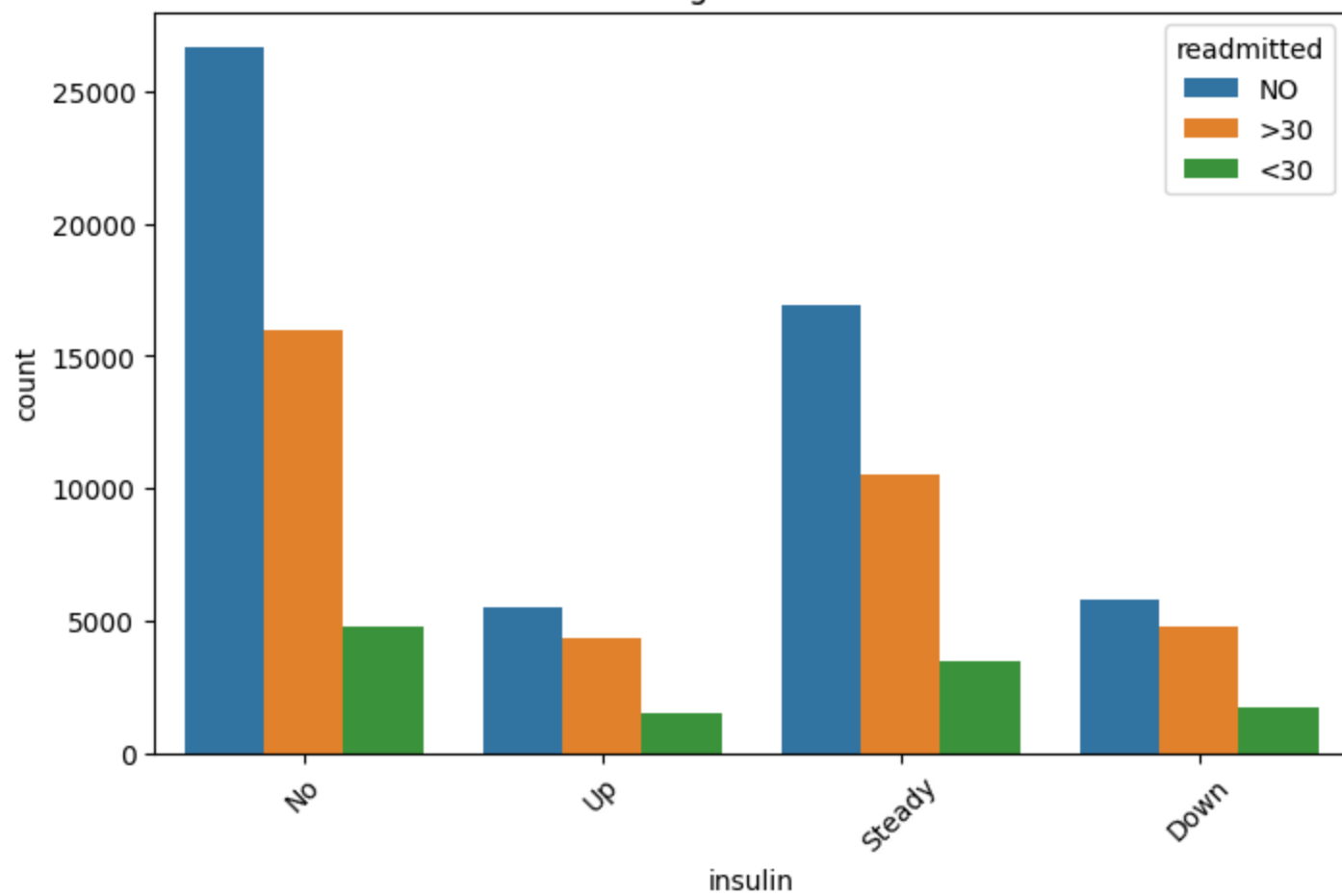
examide Usage vs Readmission



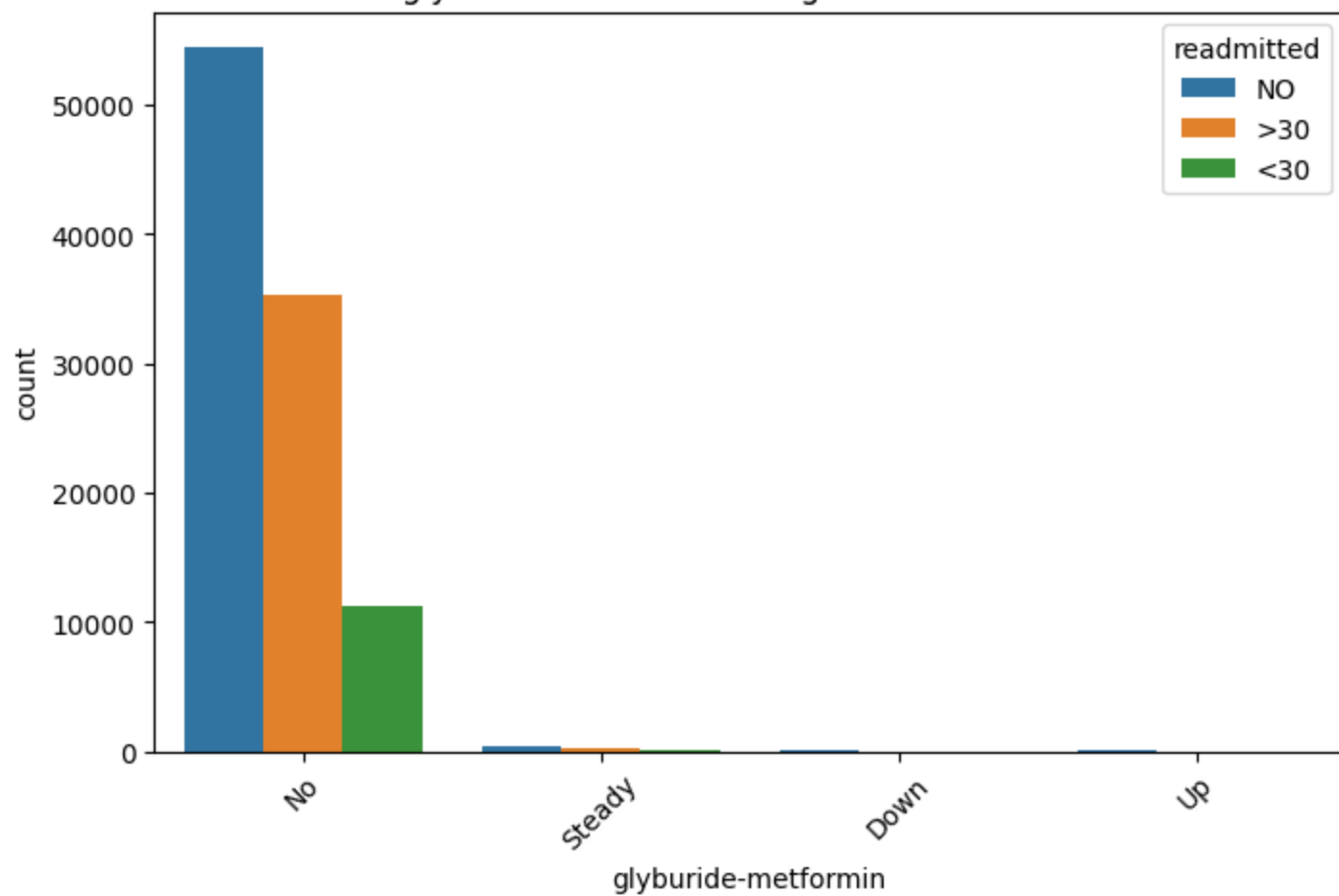
citoglipton Usage vs Readmission



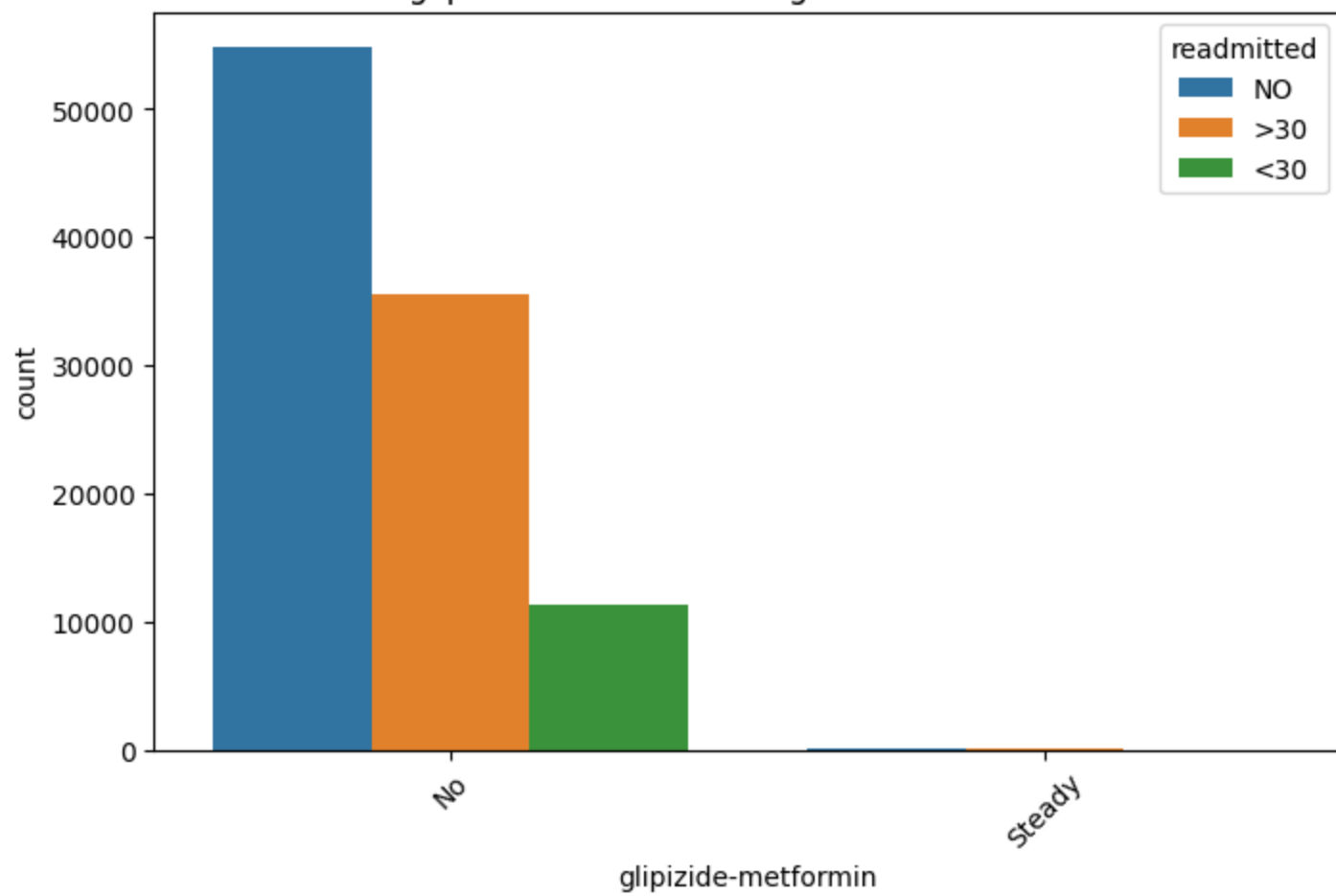
insulin Usage vs Readmission



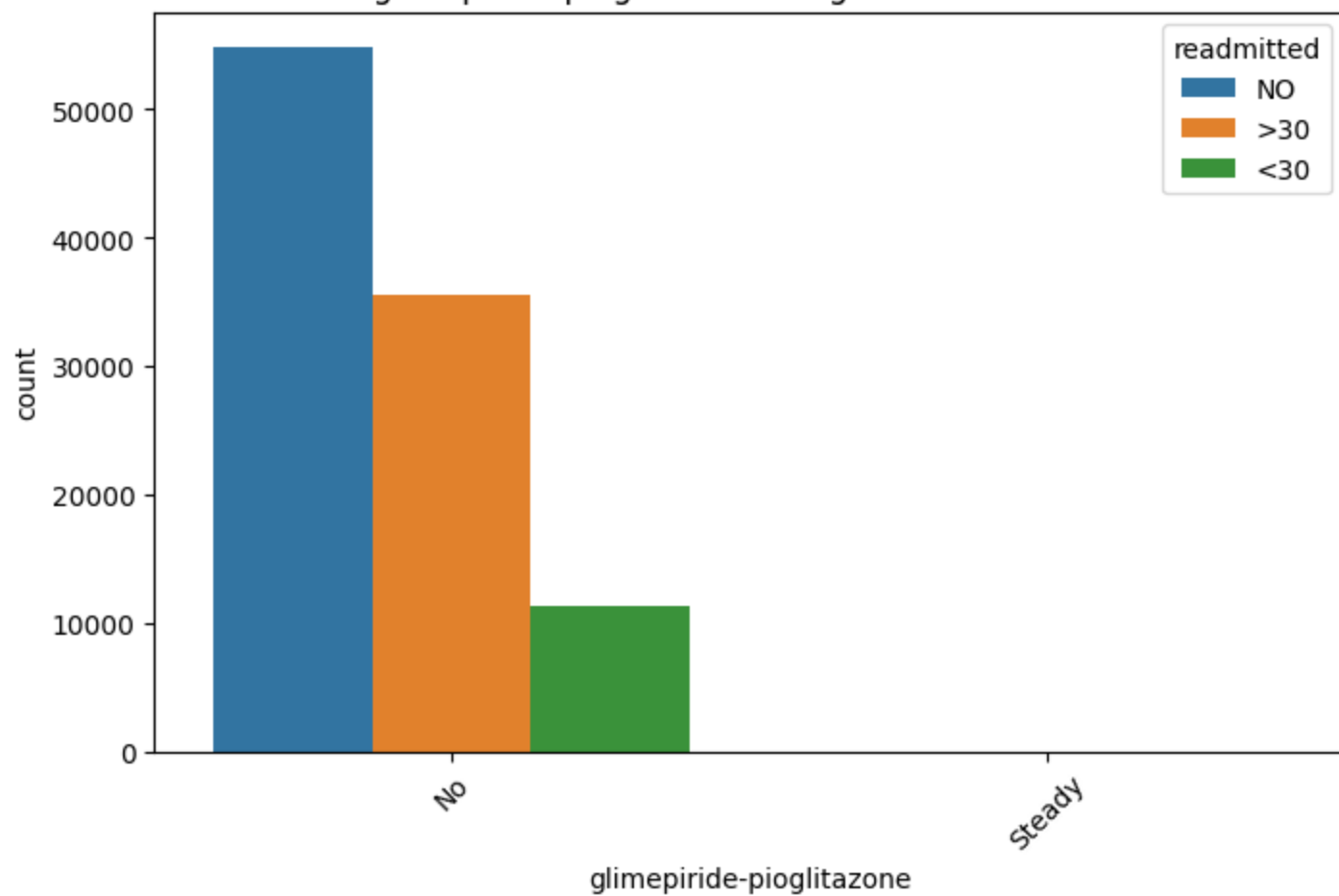
glyburide-metformin Usage vs Readmission



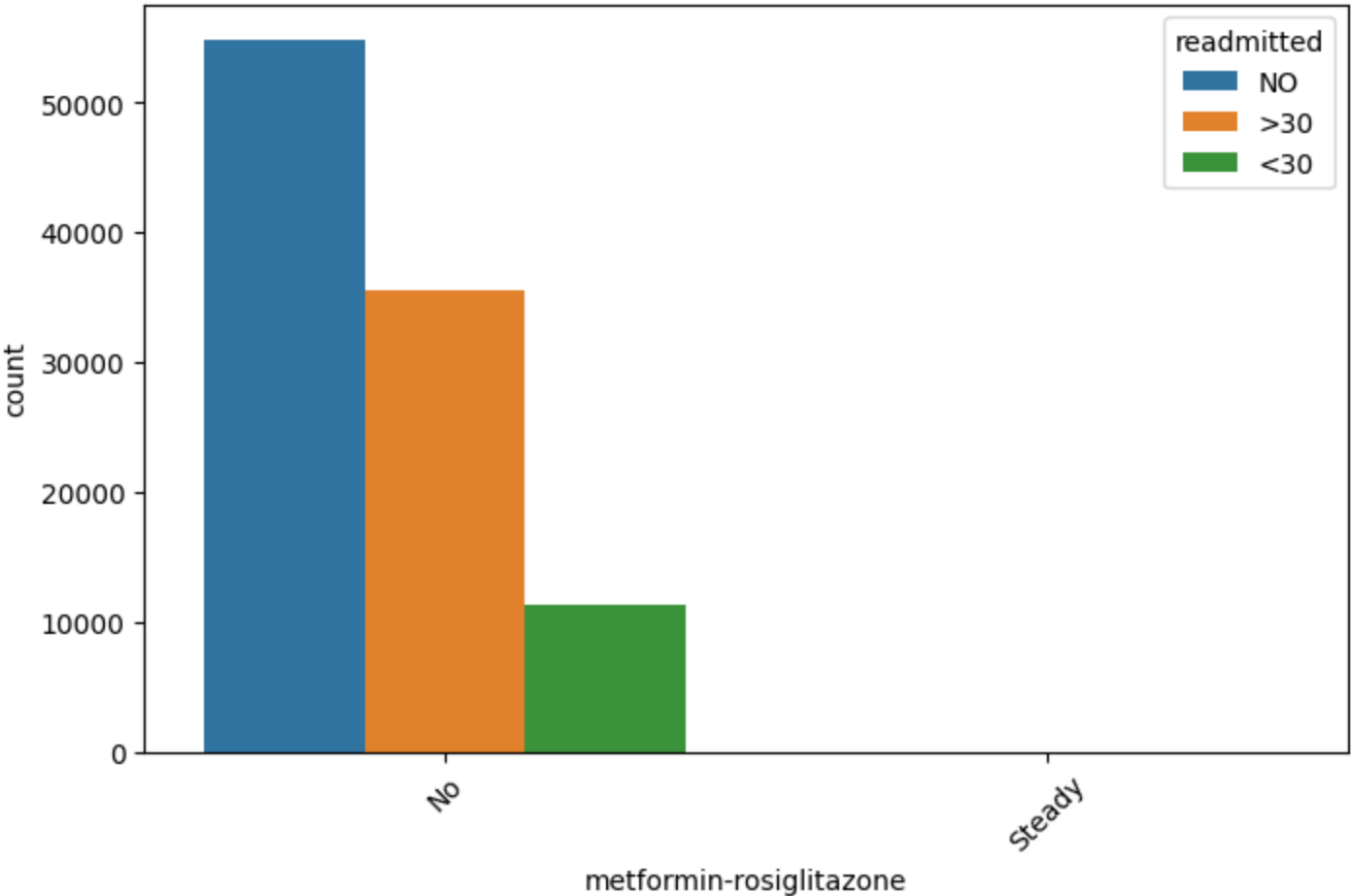
glipizide-metformin Usage vs Readmission



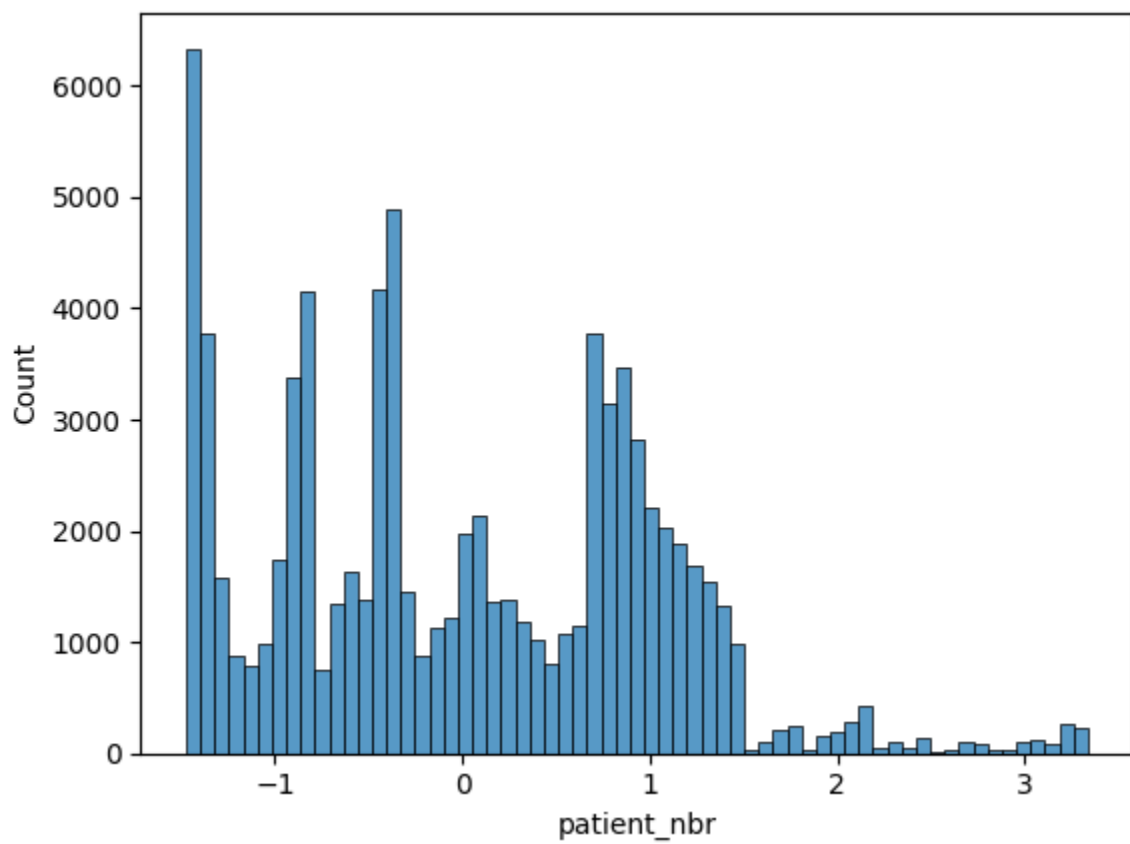
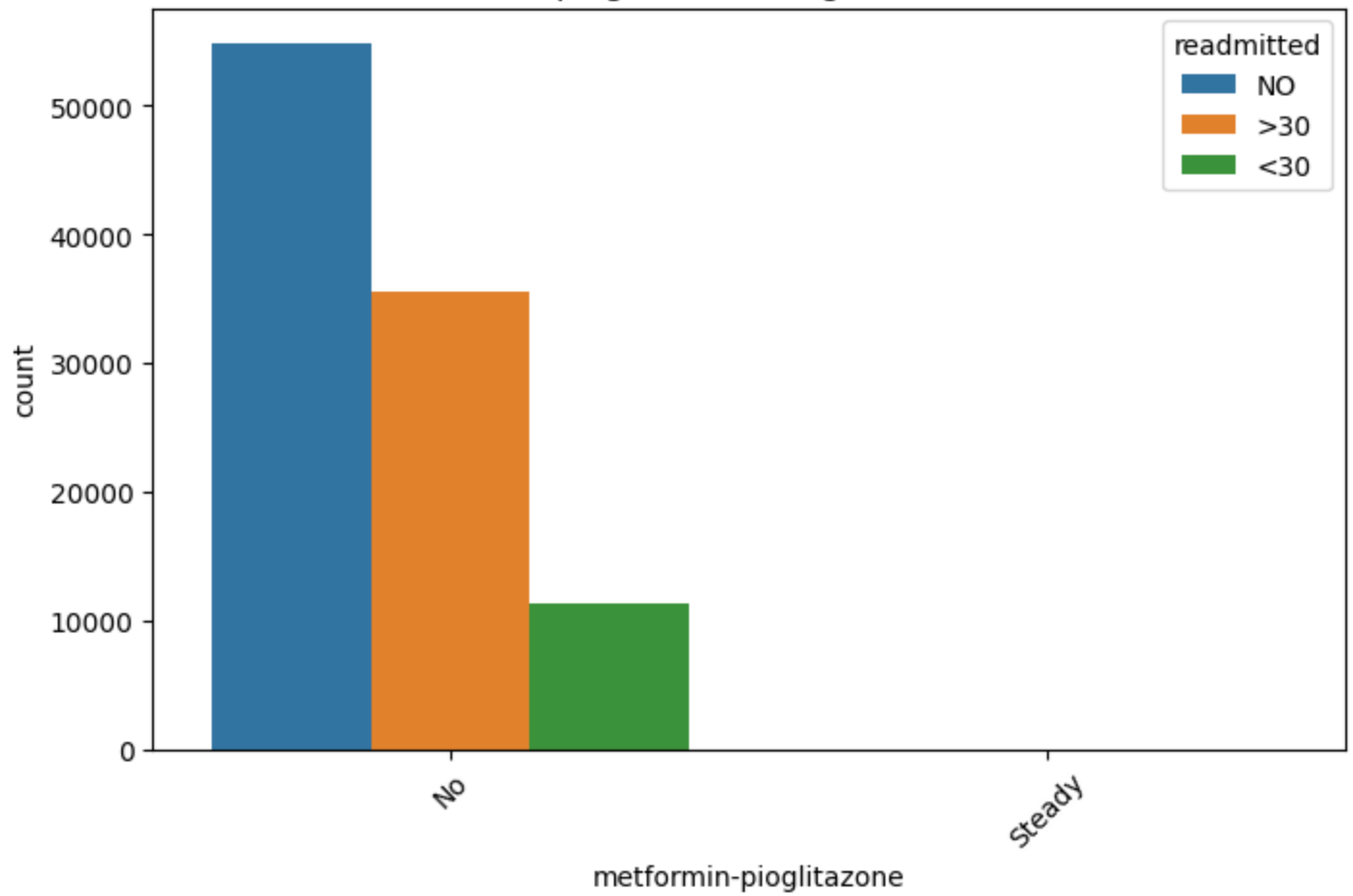
glimepiride-pioglitazone Usage vs Readmission

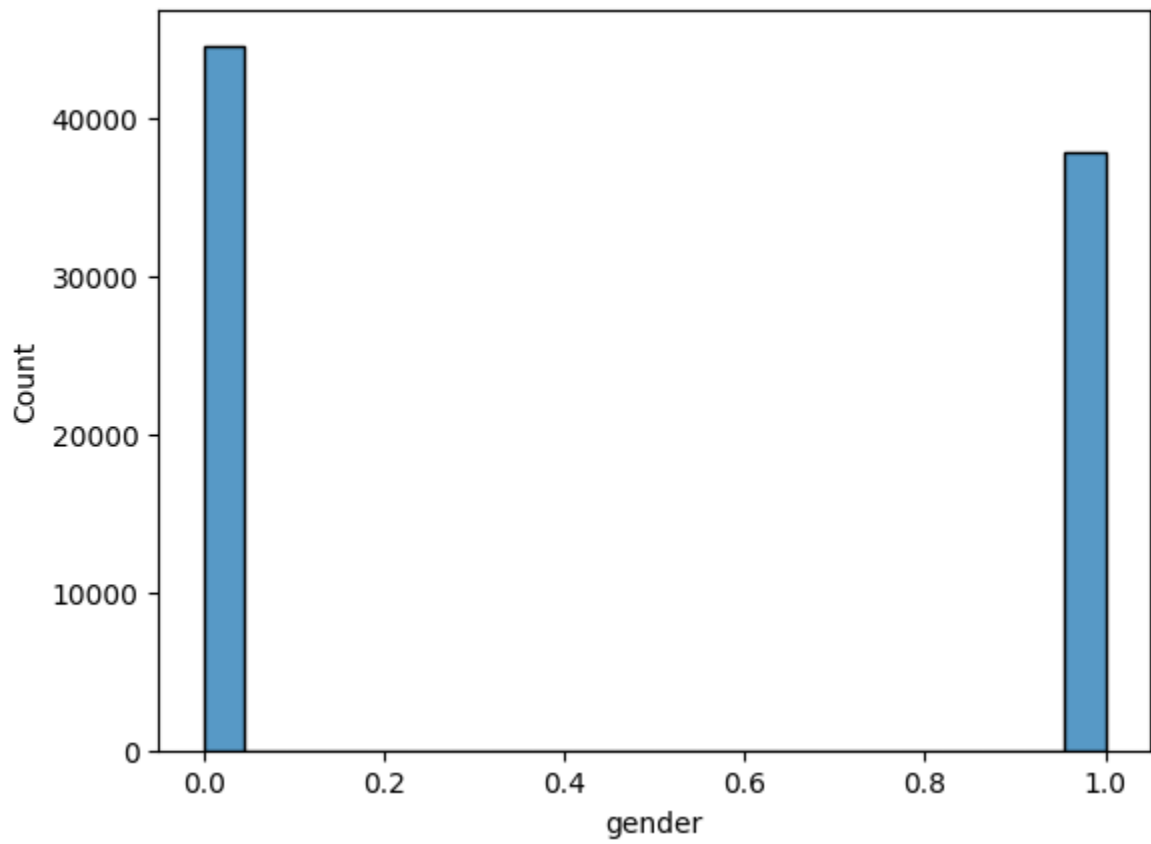
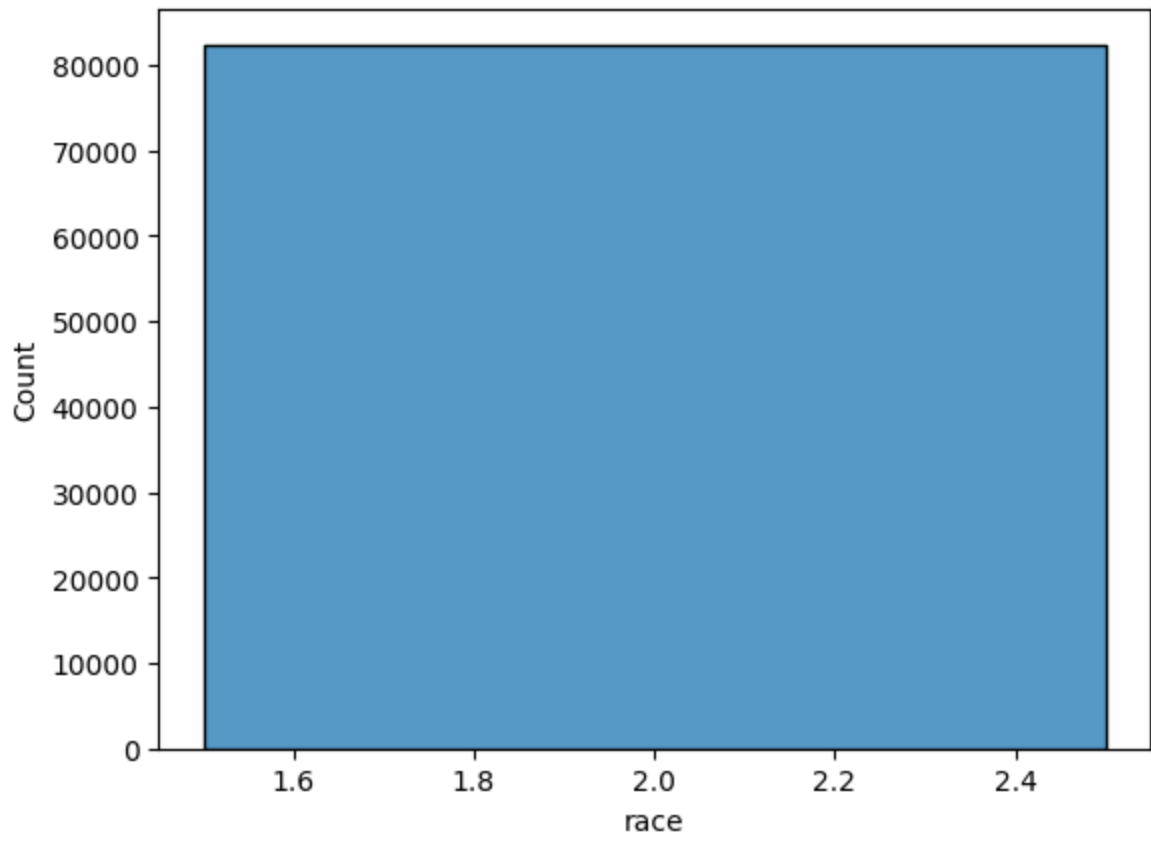


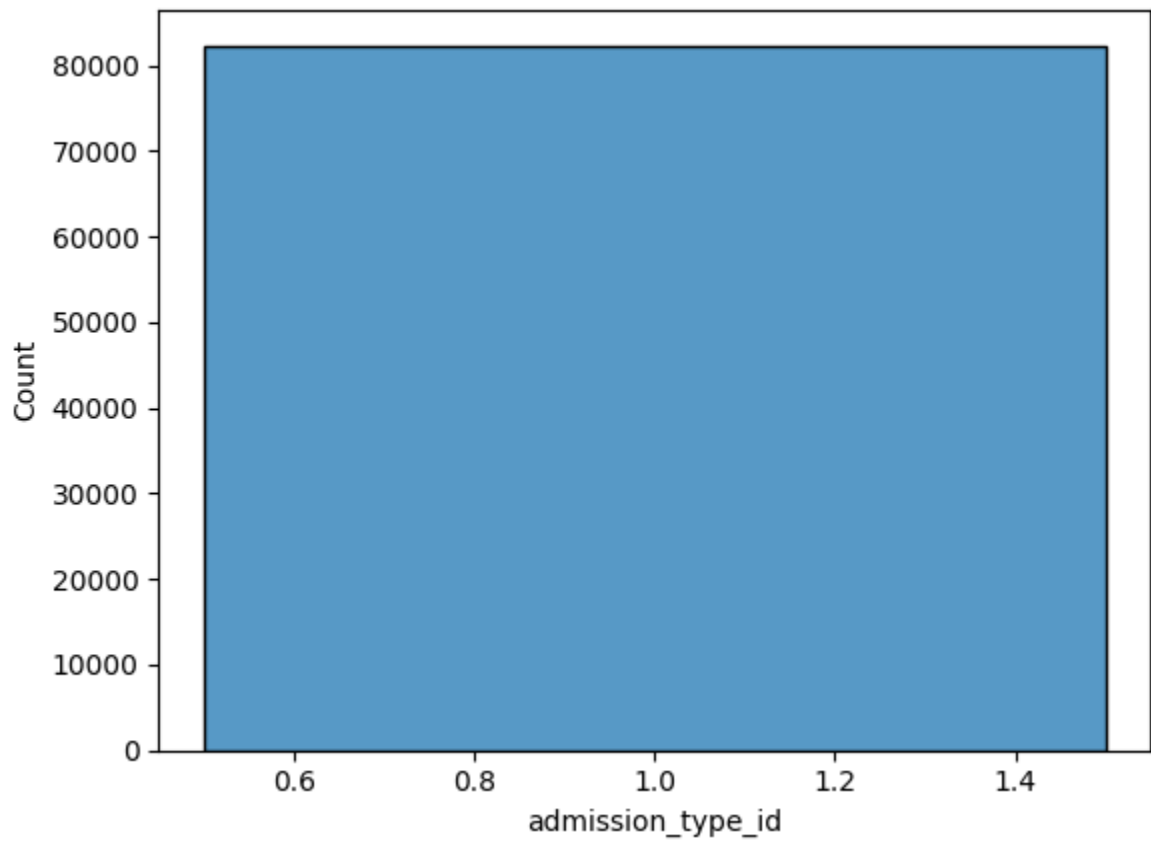
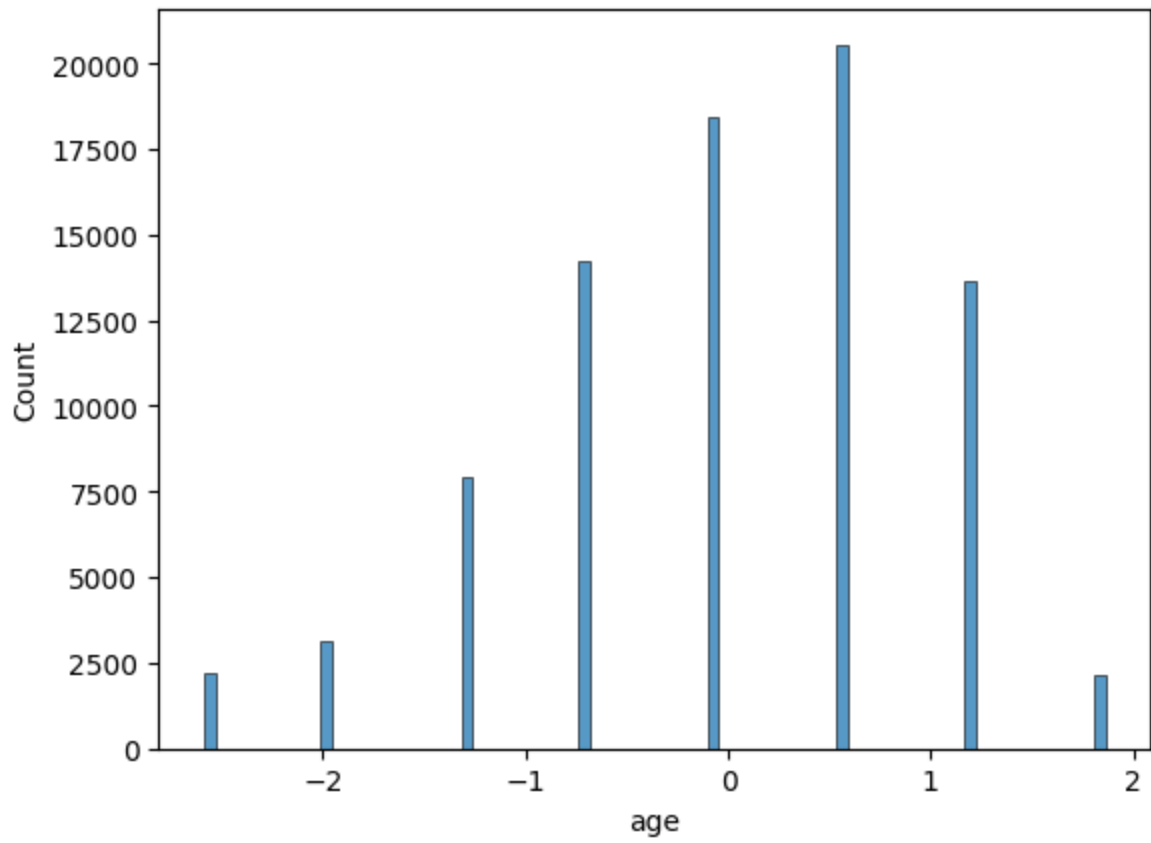
metformin-rosiglitazone Usage vs Readmission

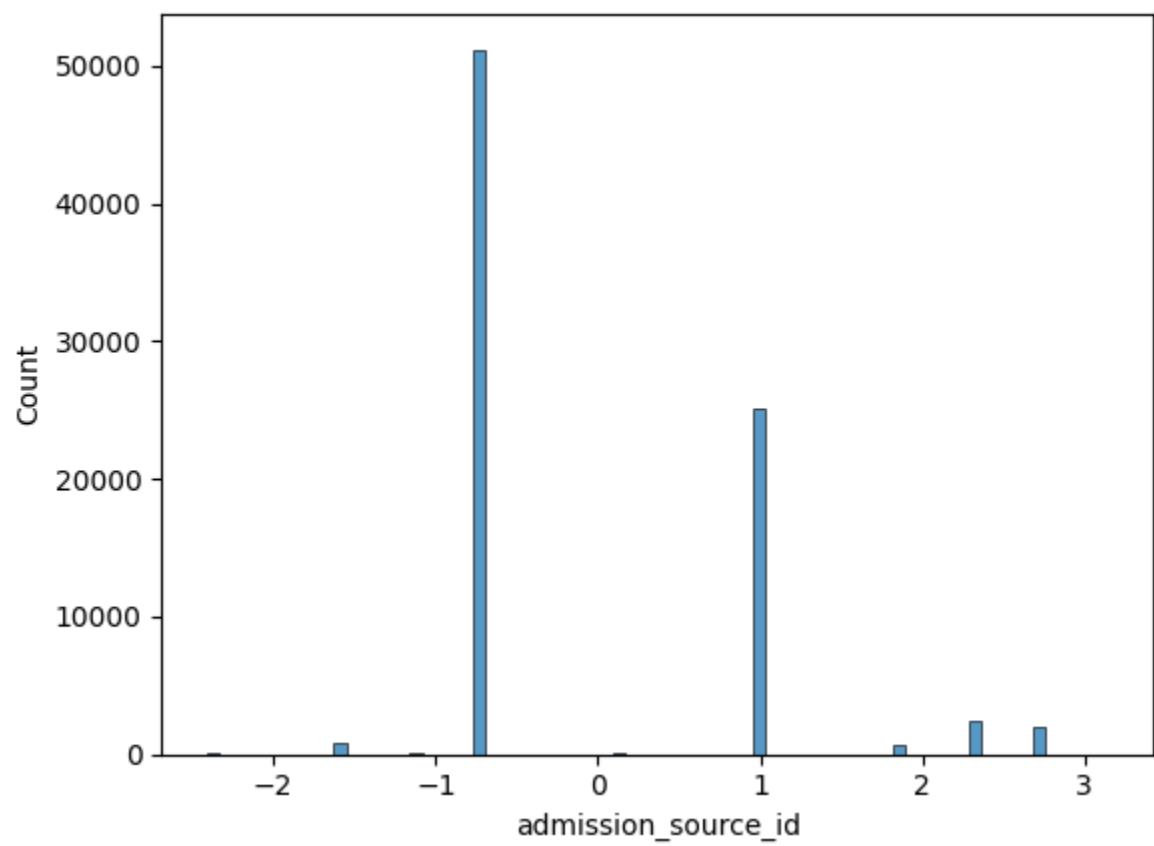
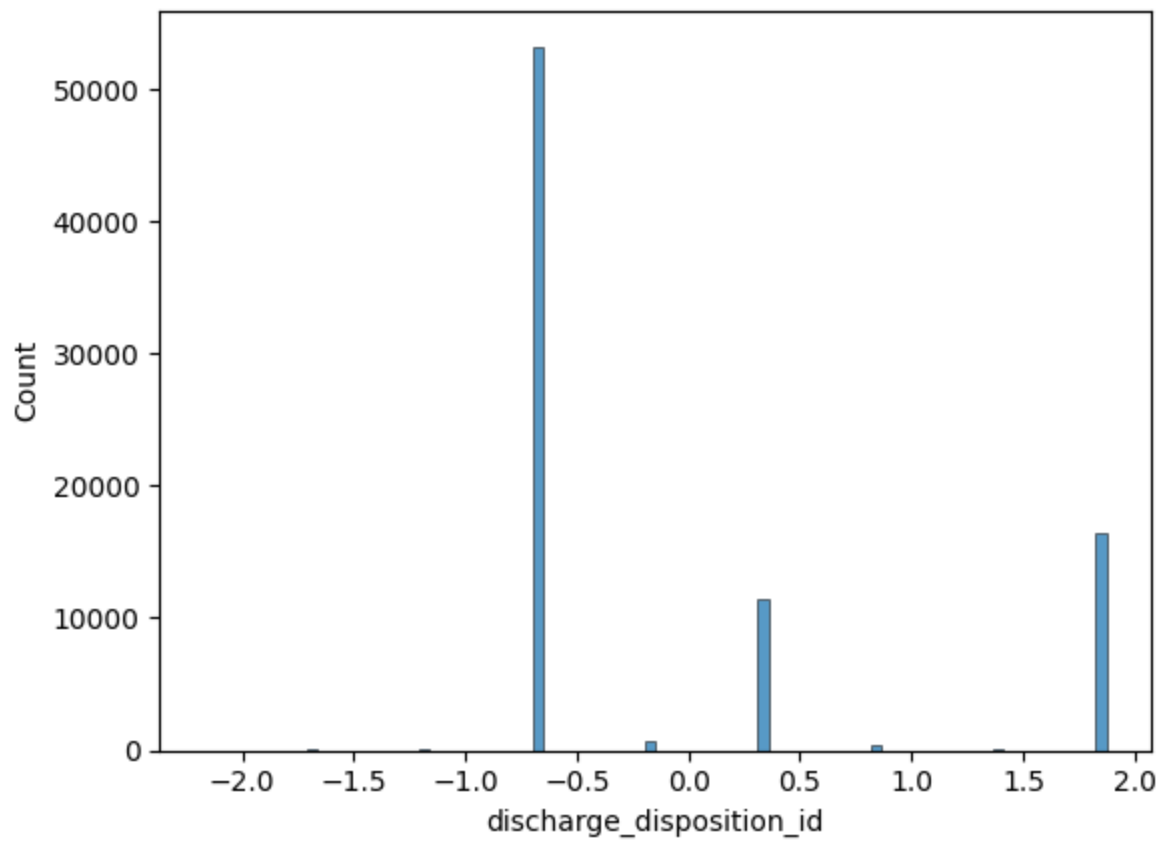


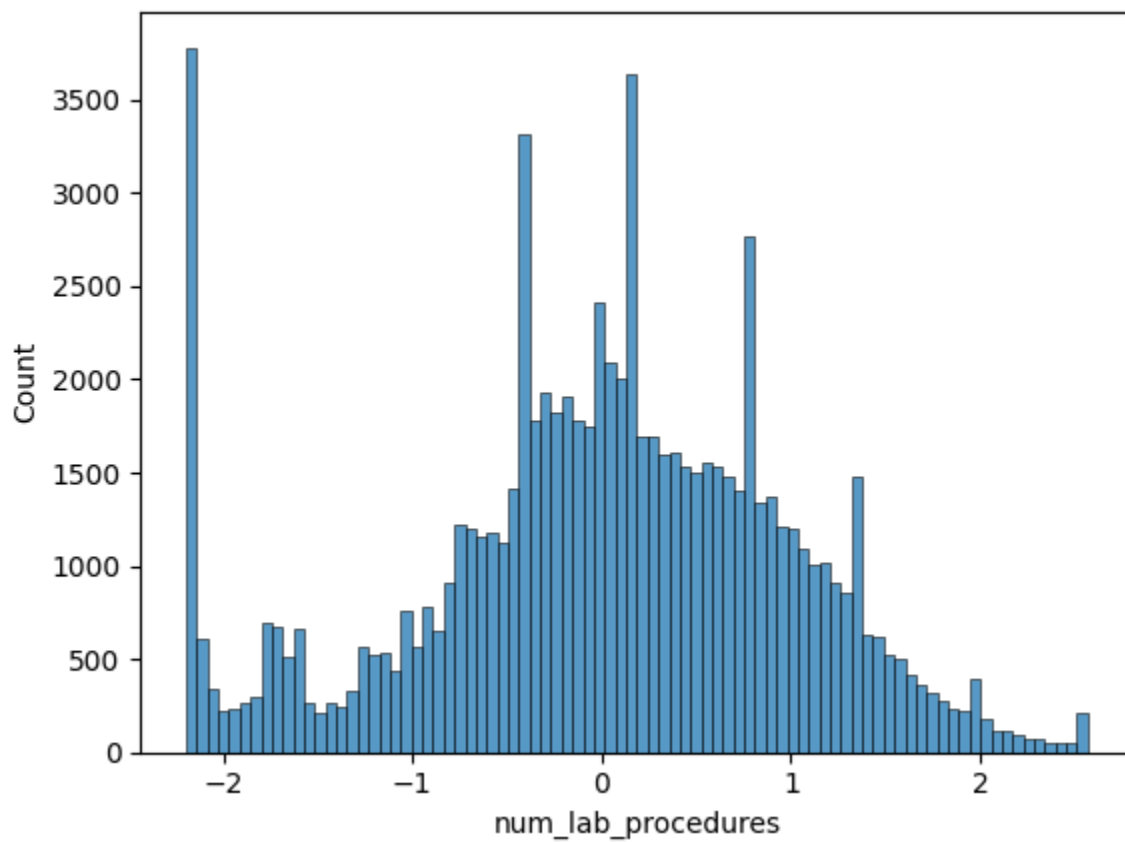
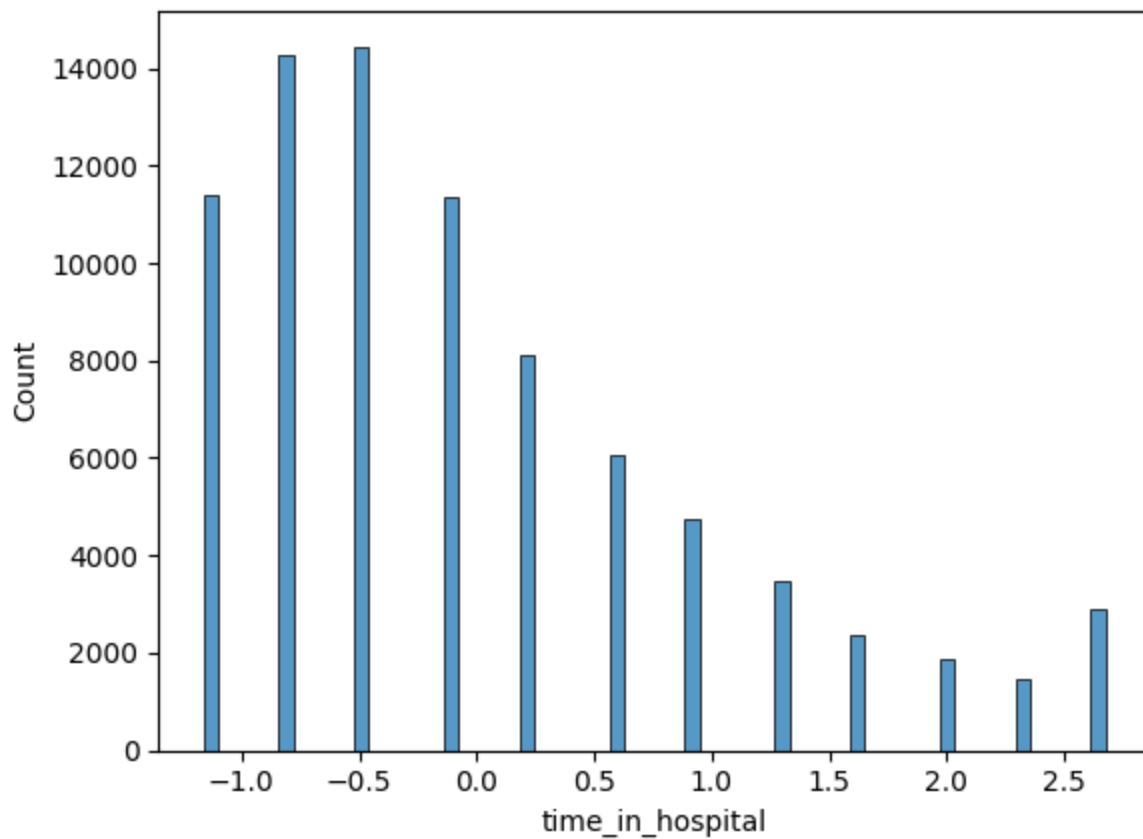
metformin-pioglitazone Usage vs Readmission

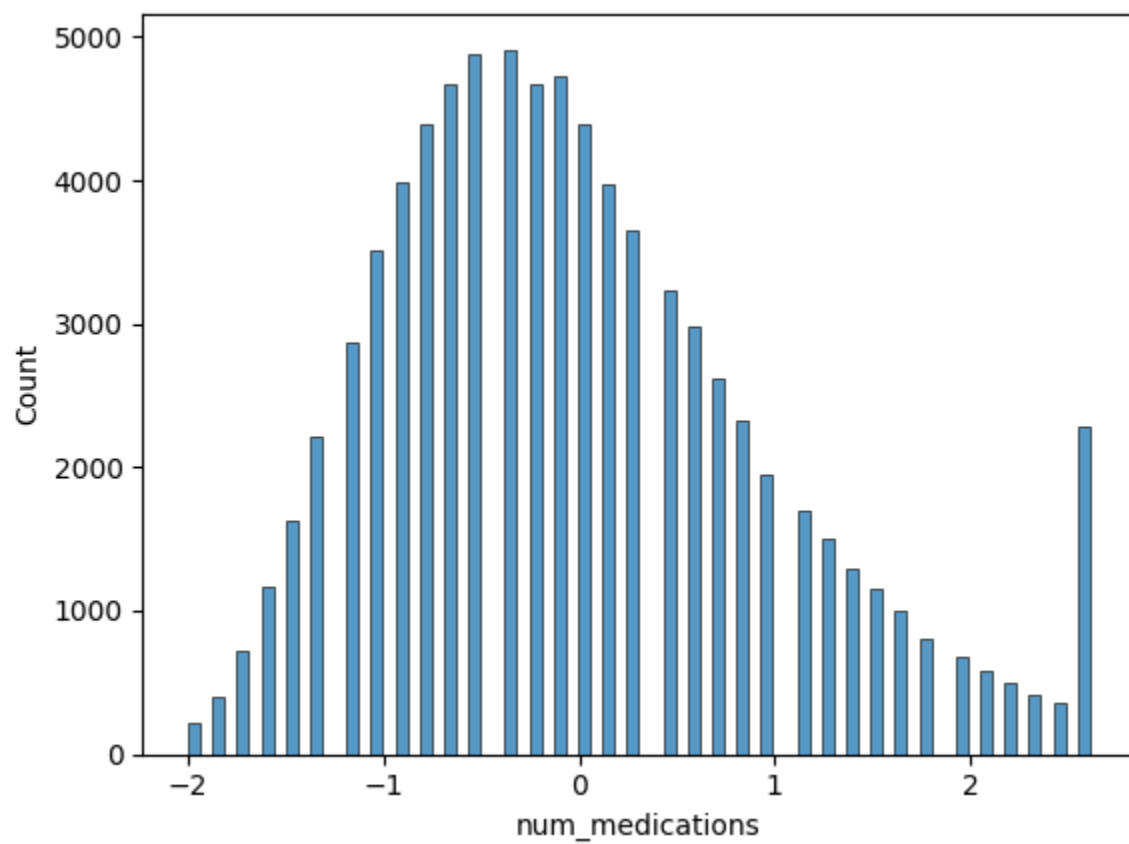
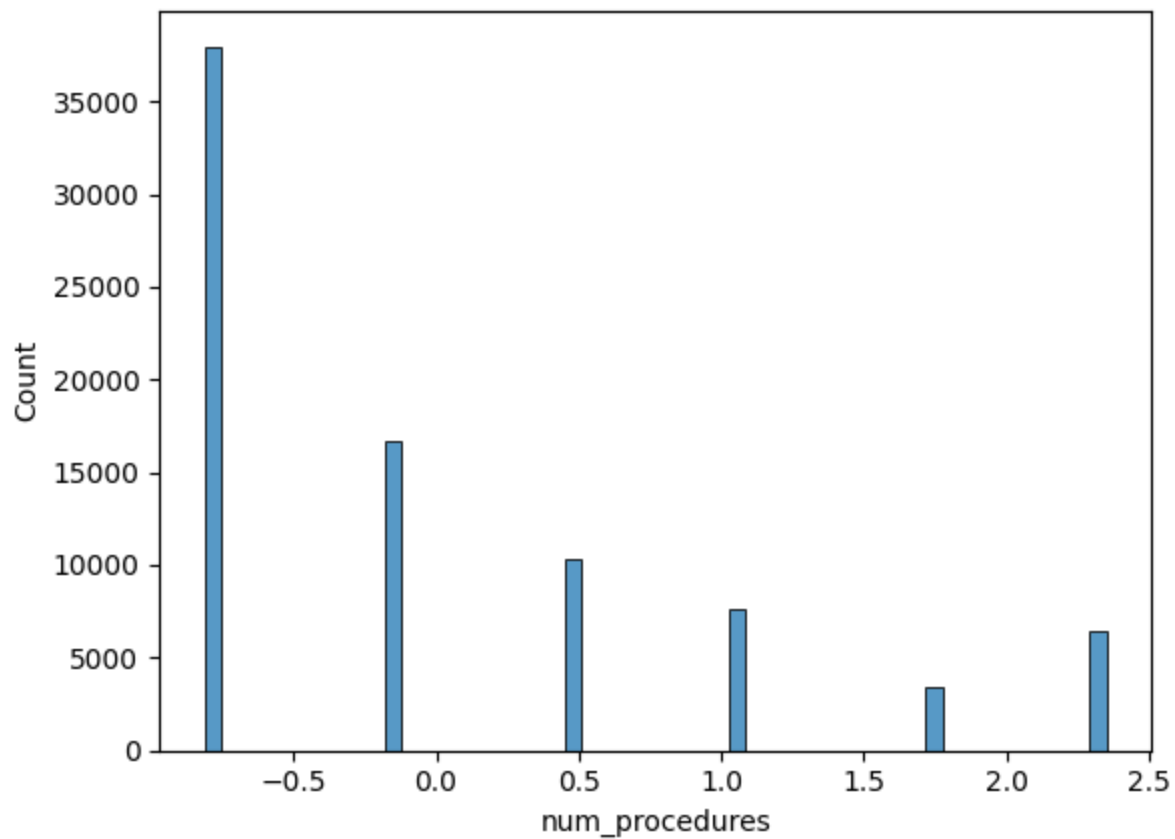


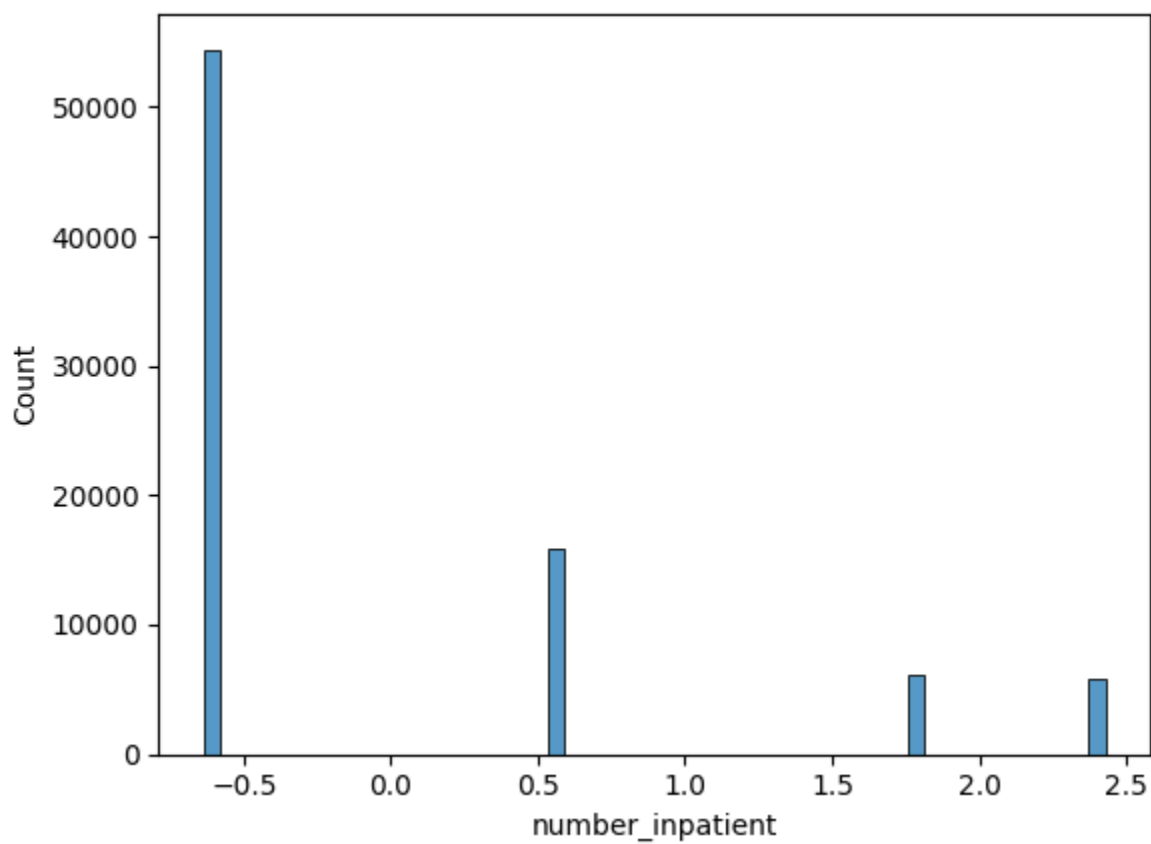
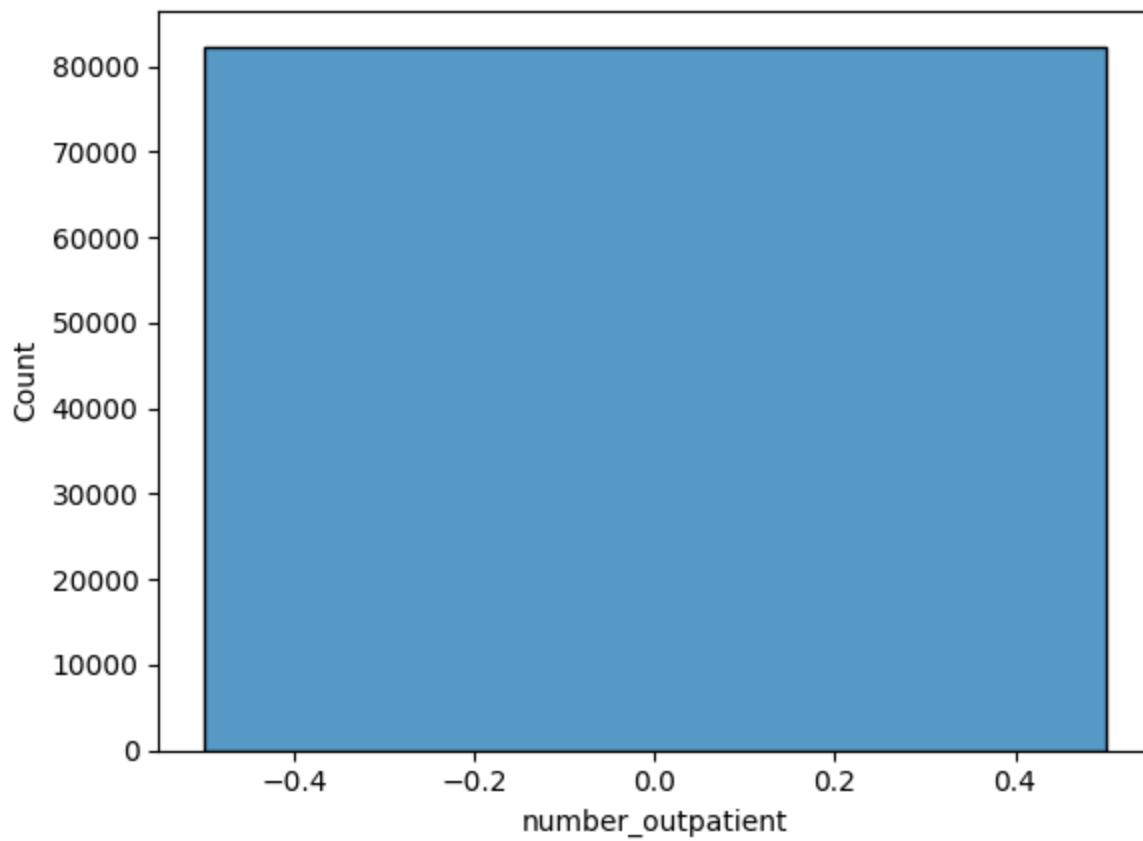


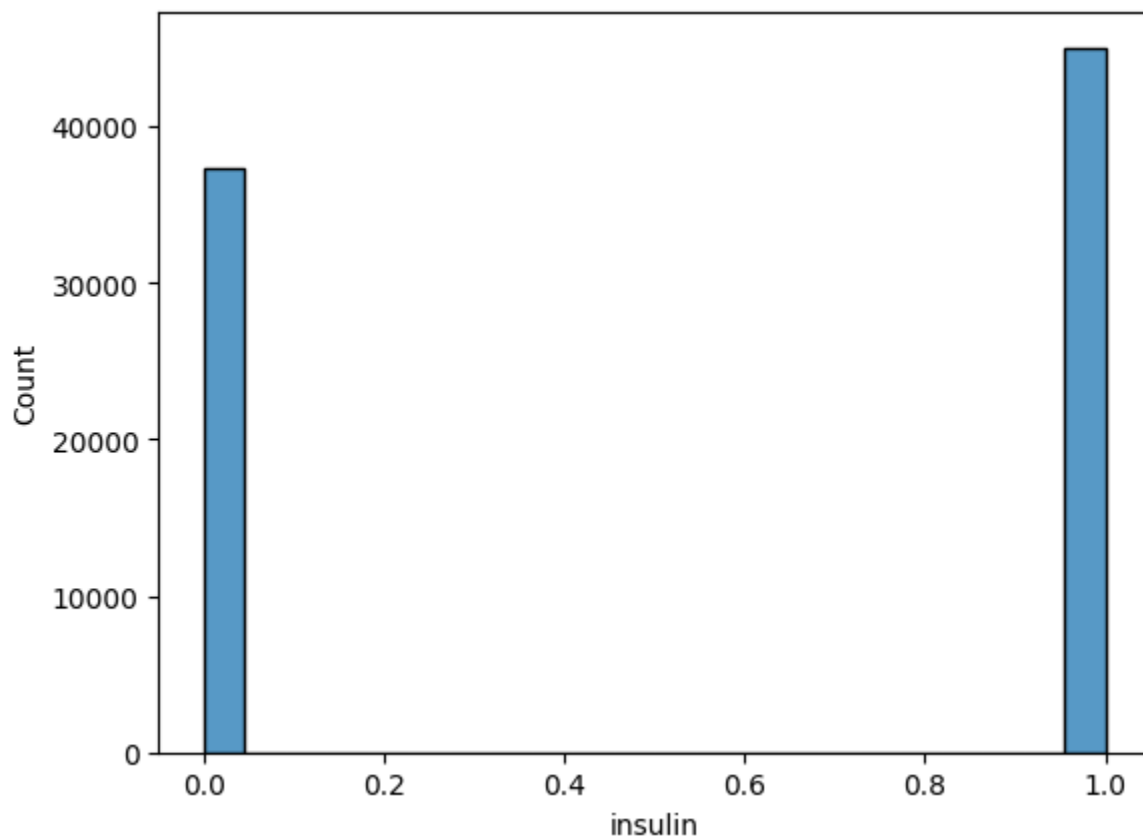
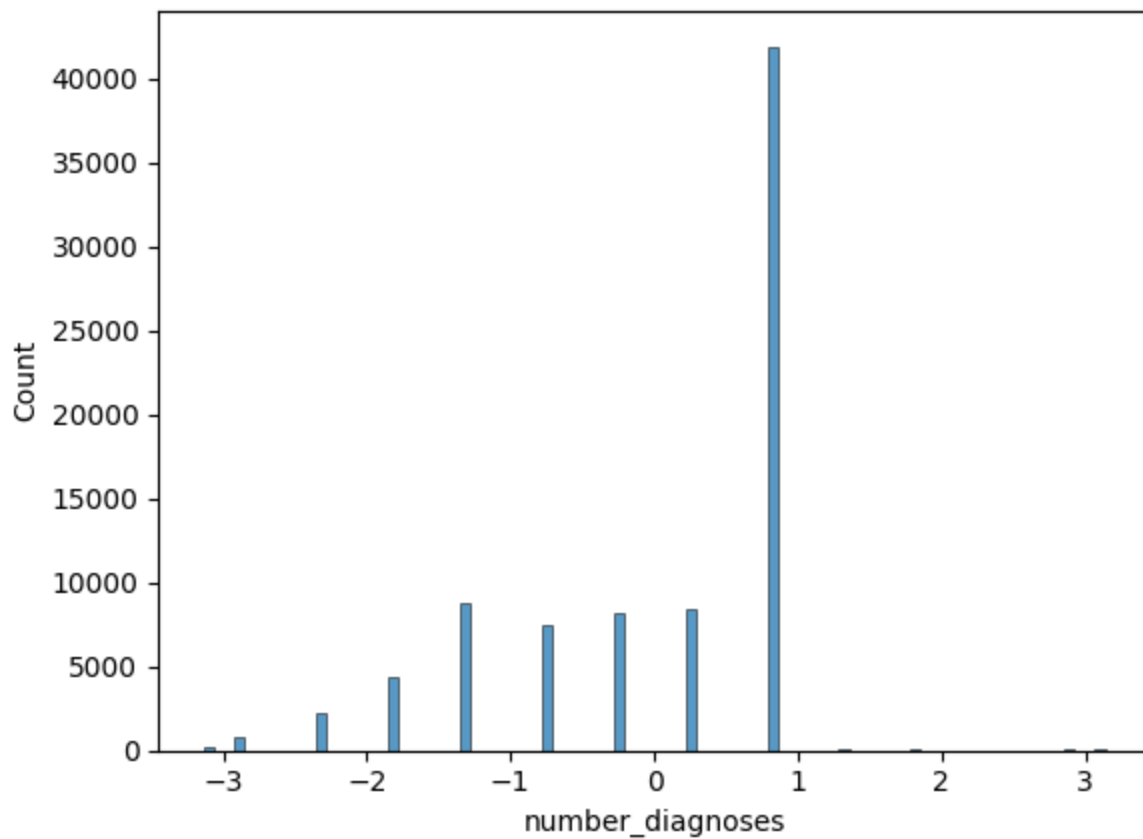


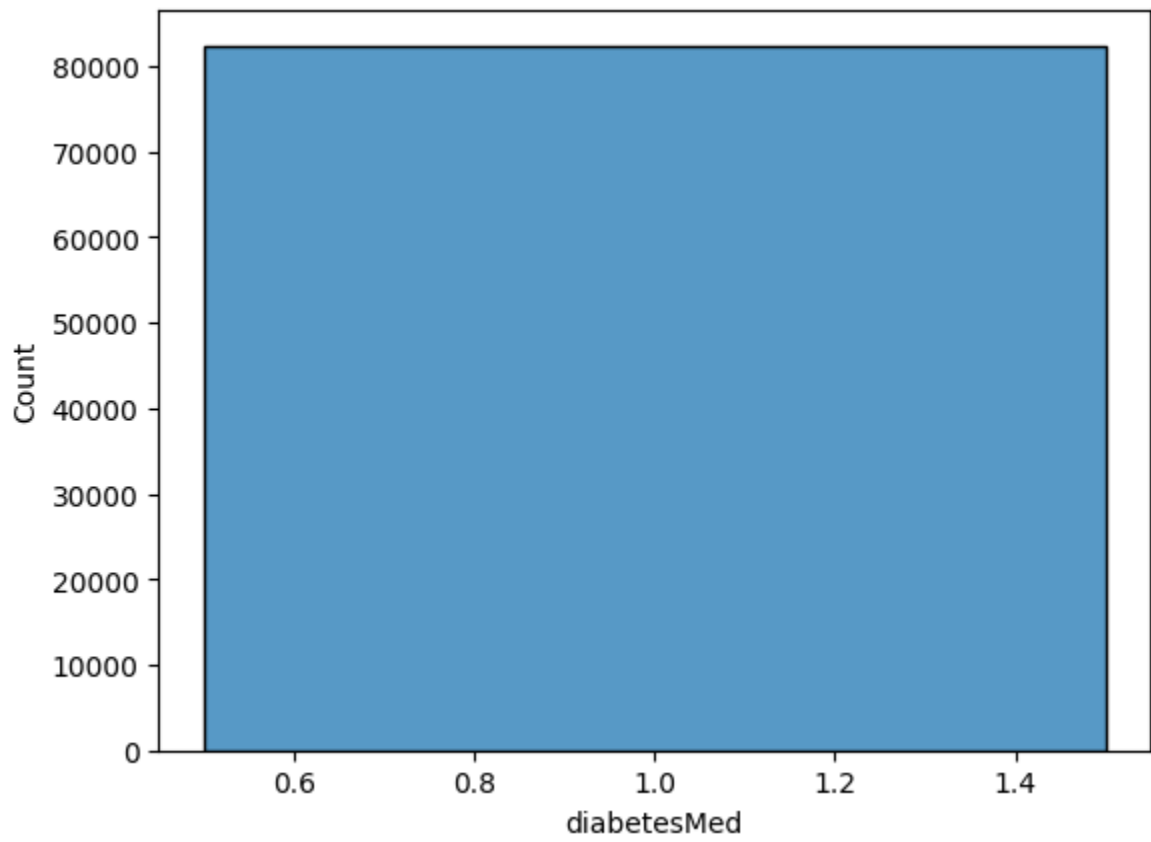
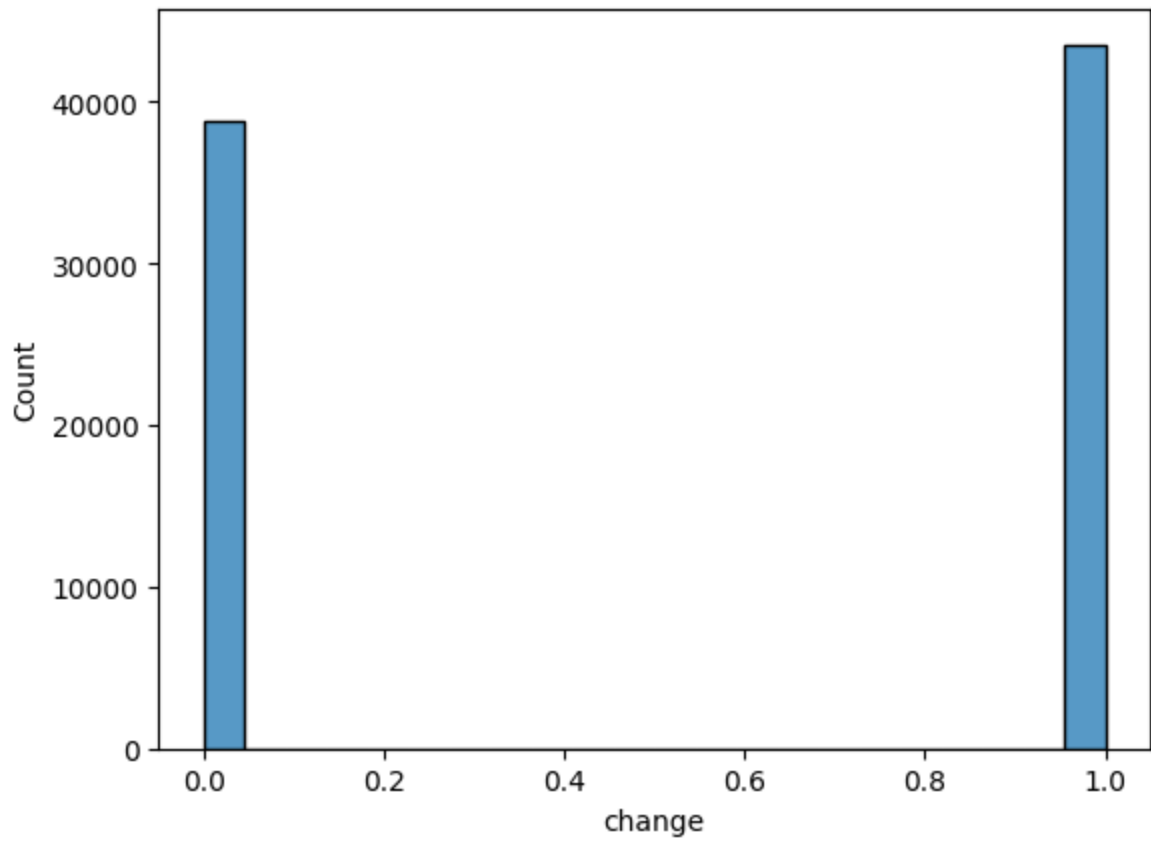


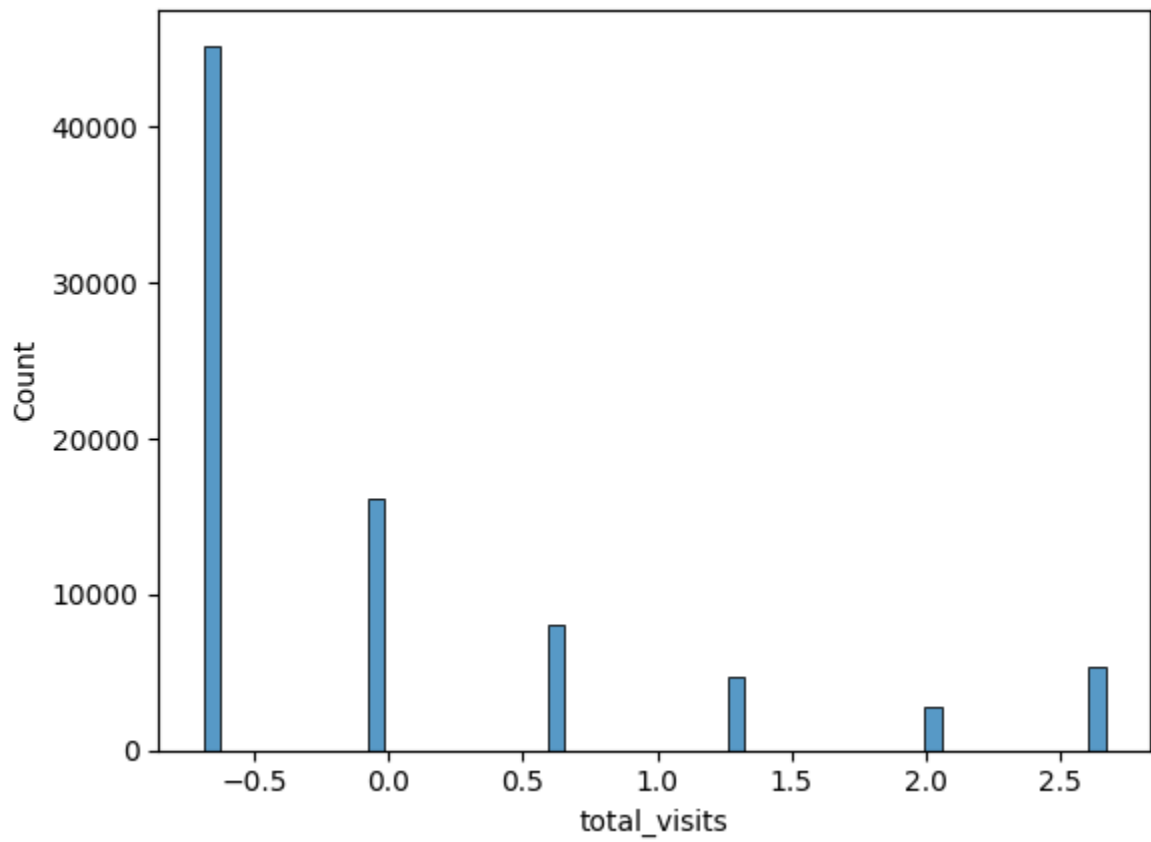
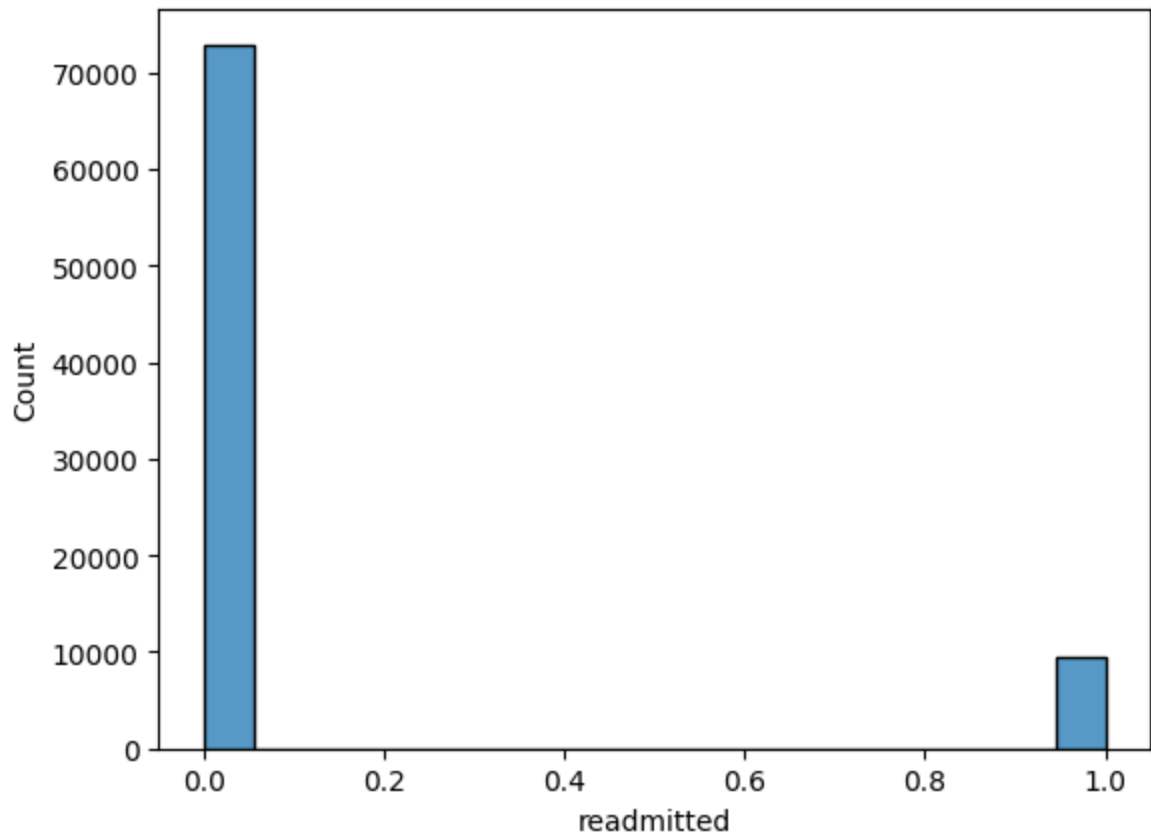


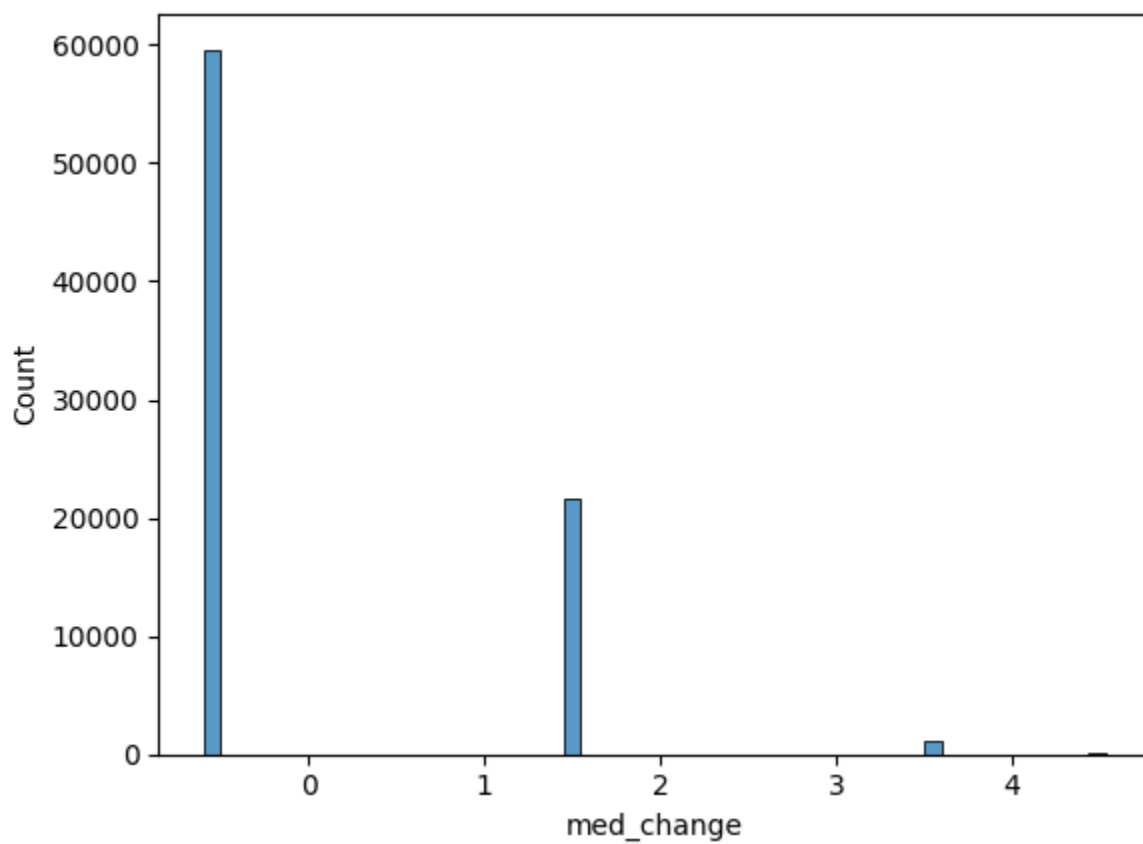
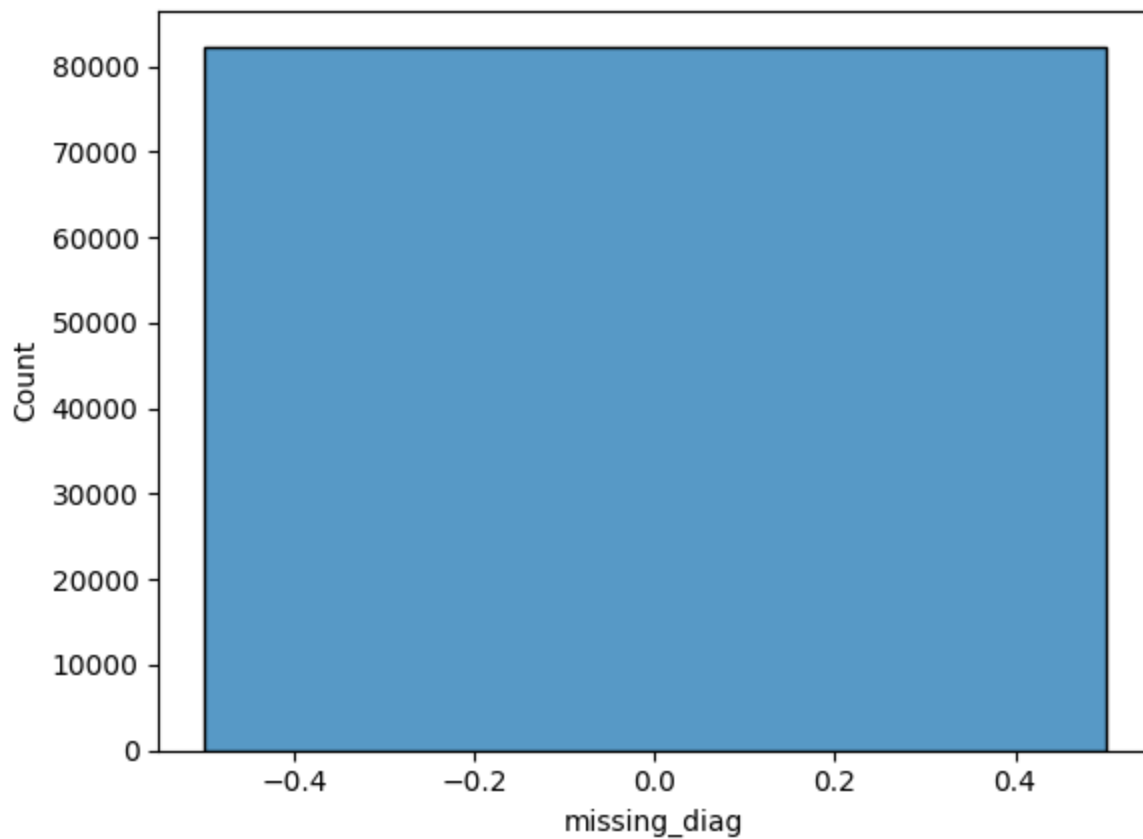


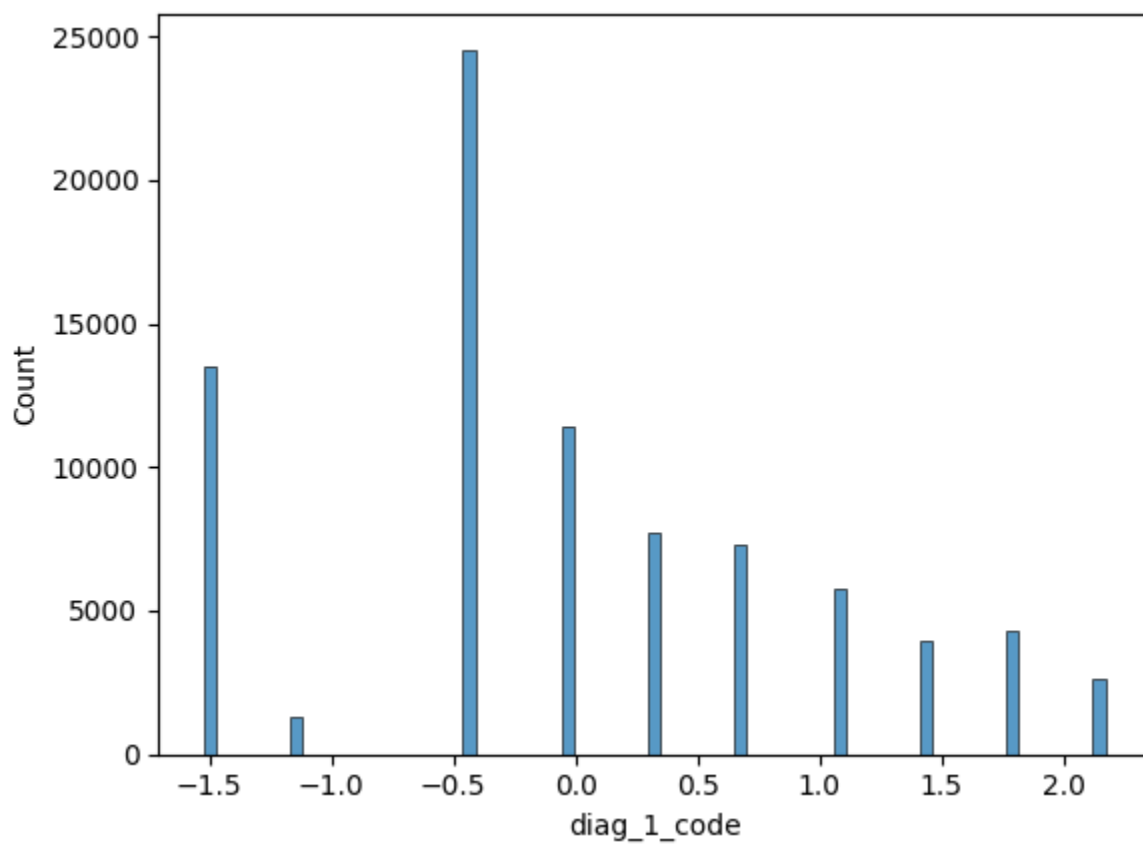
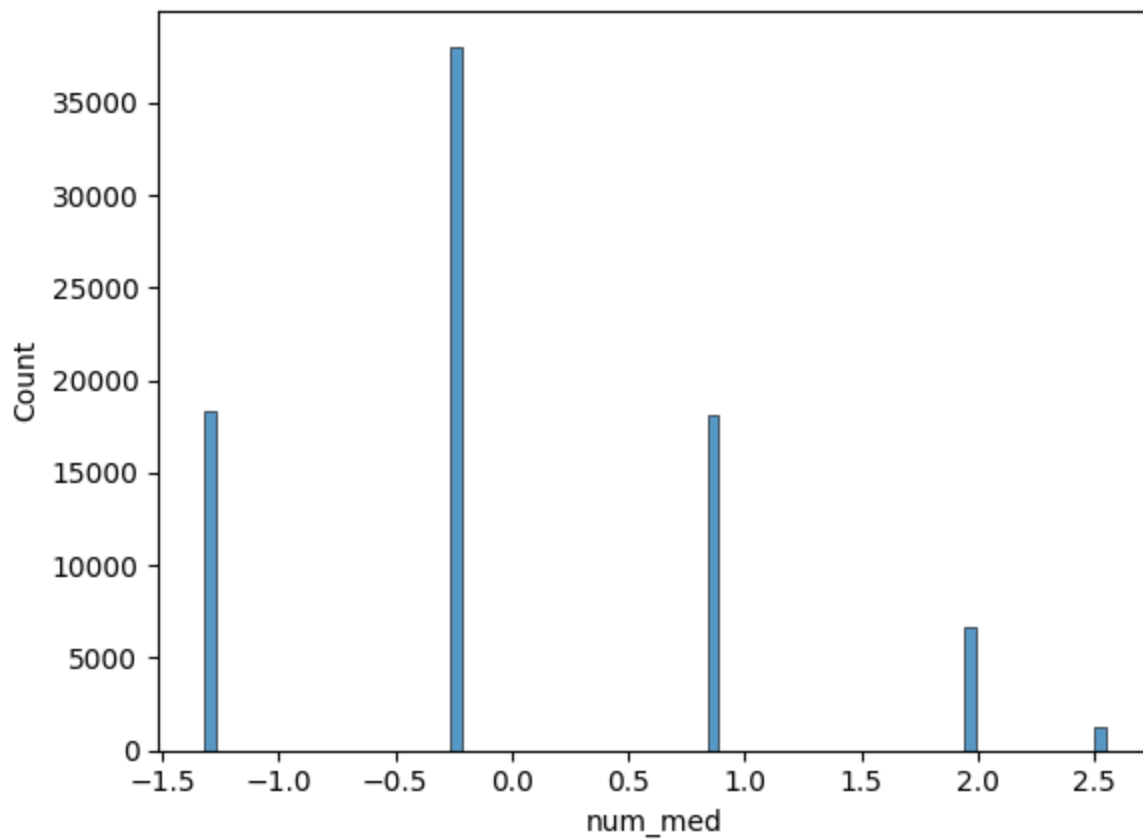


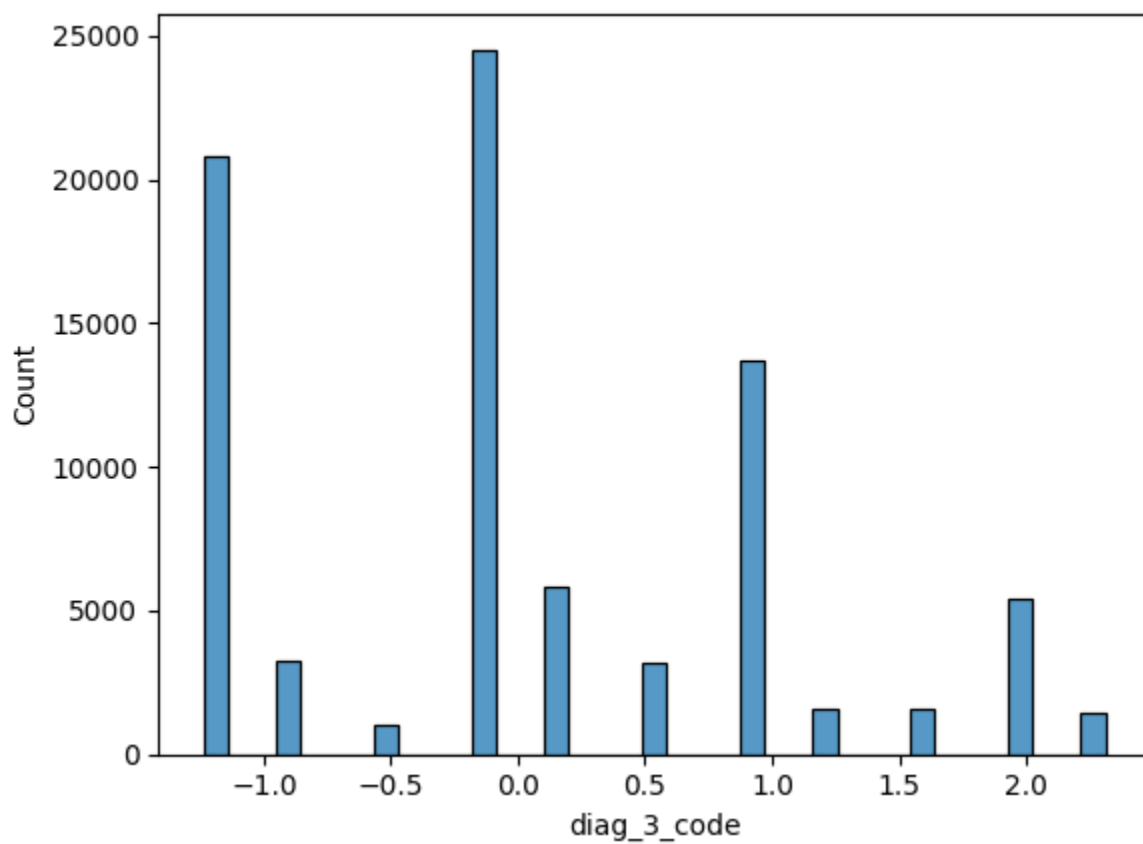
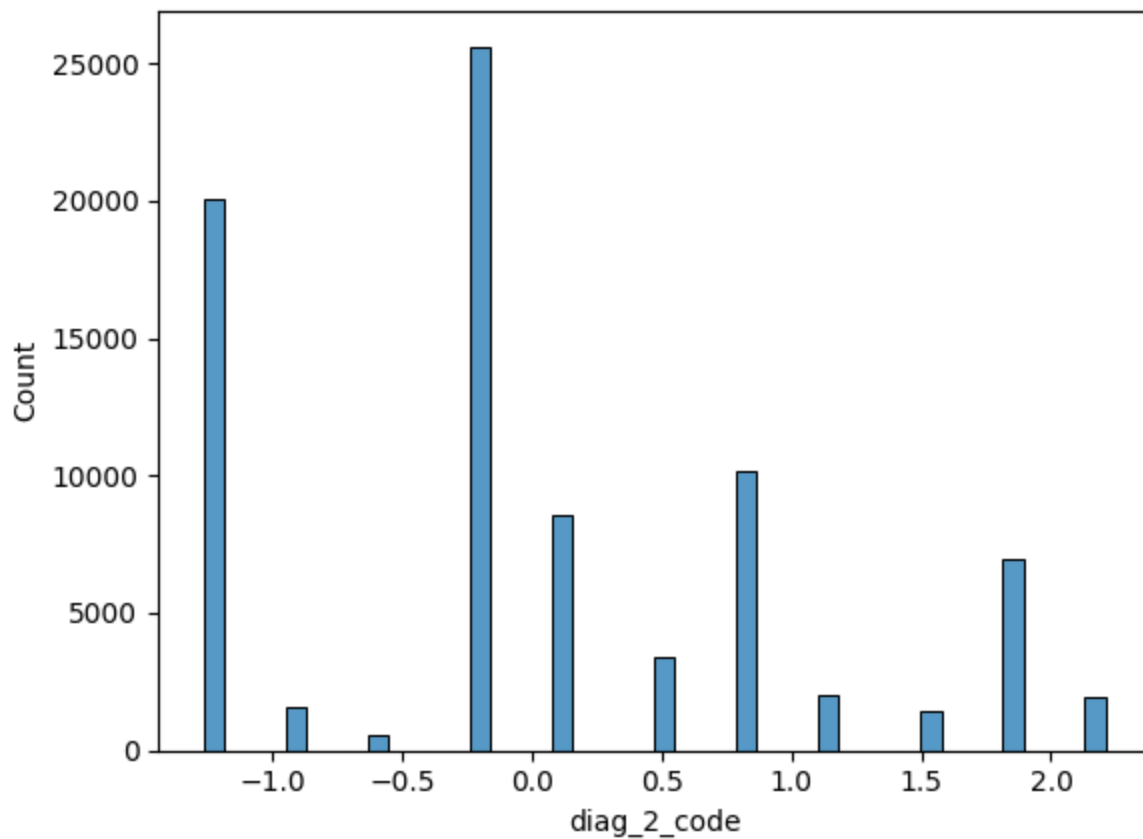


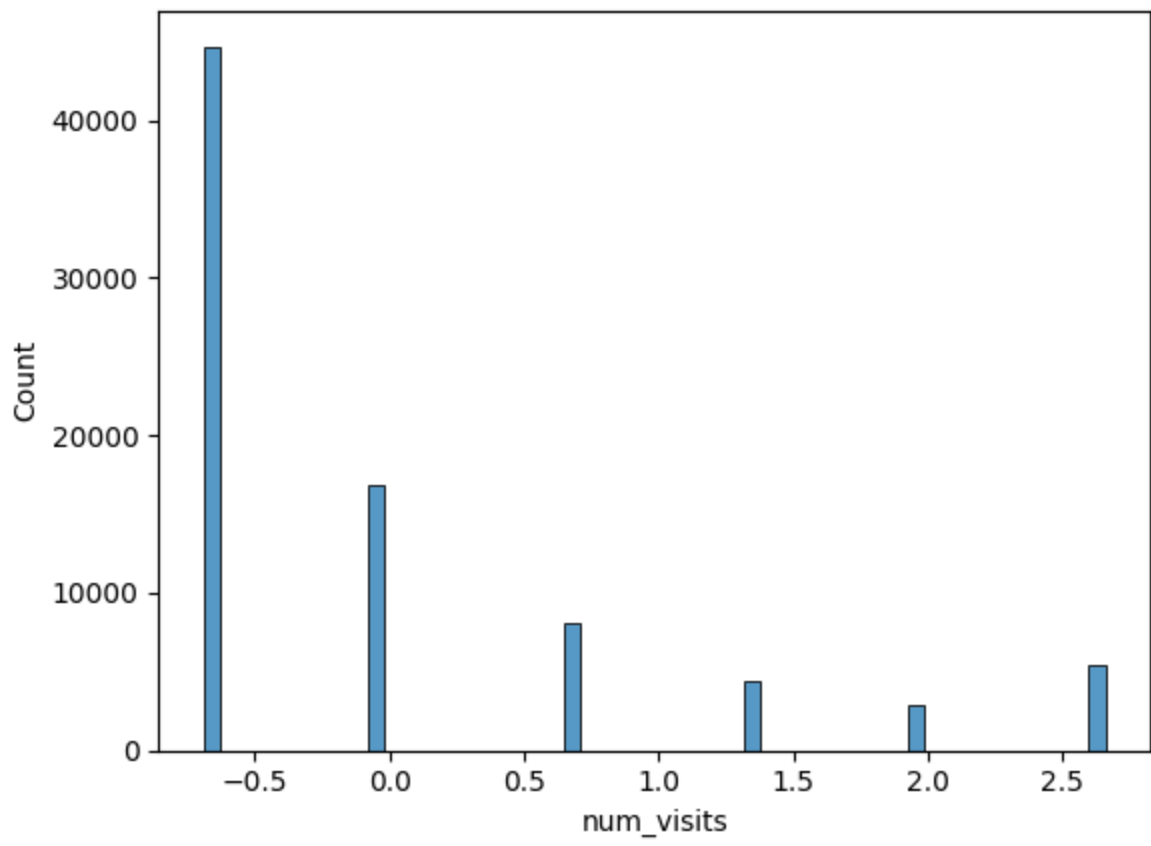


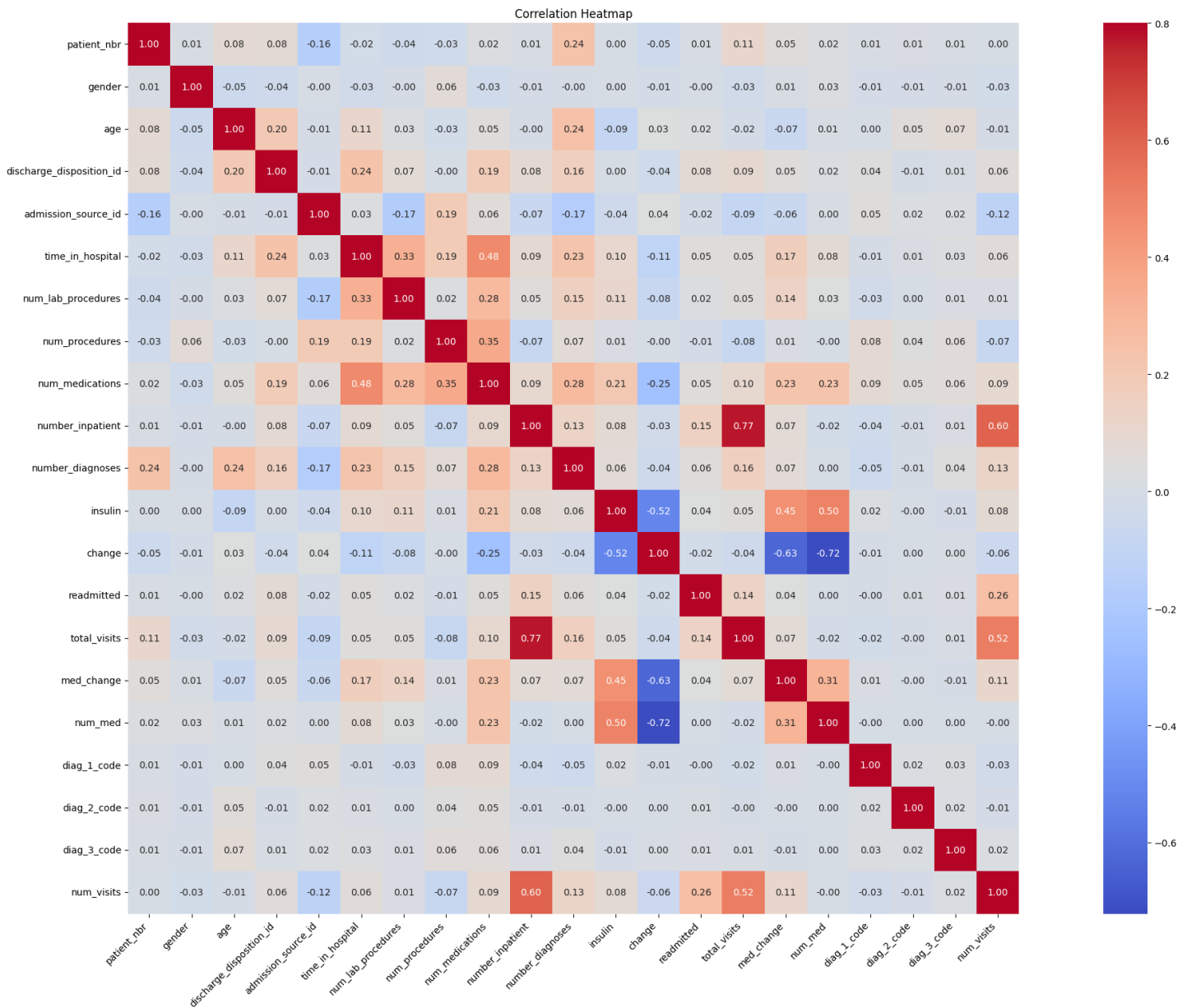




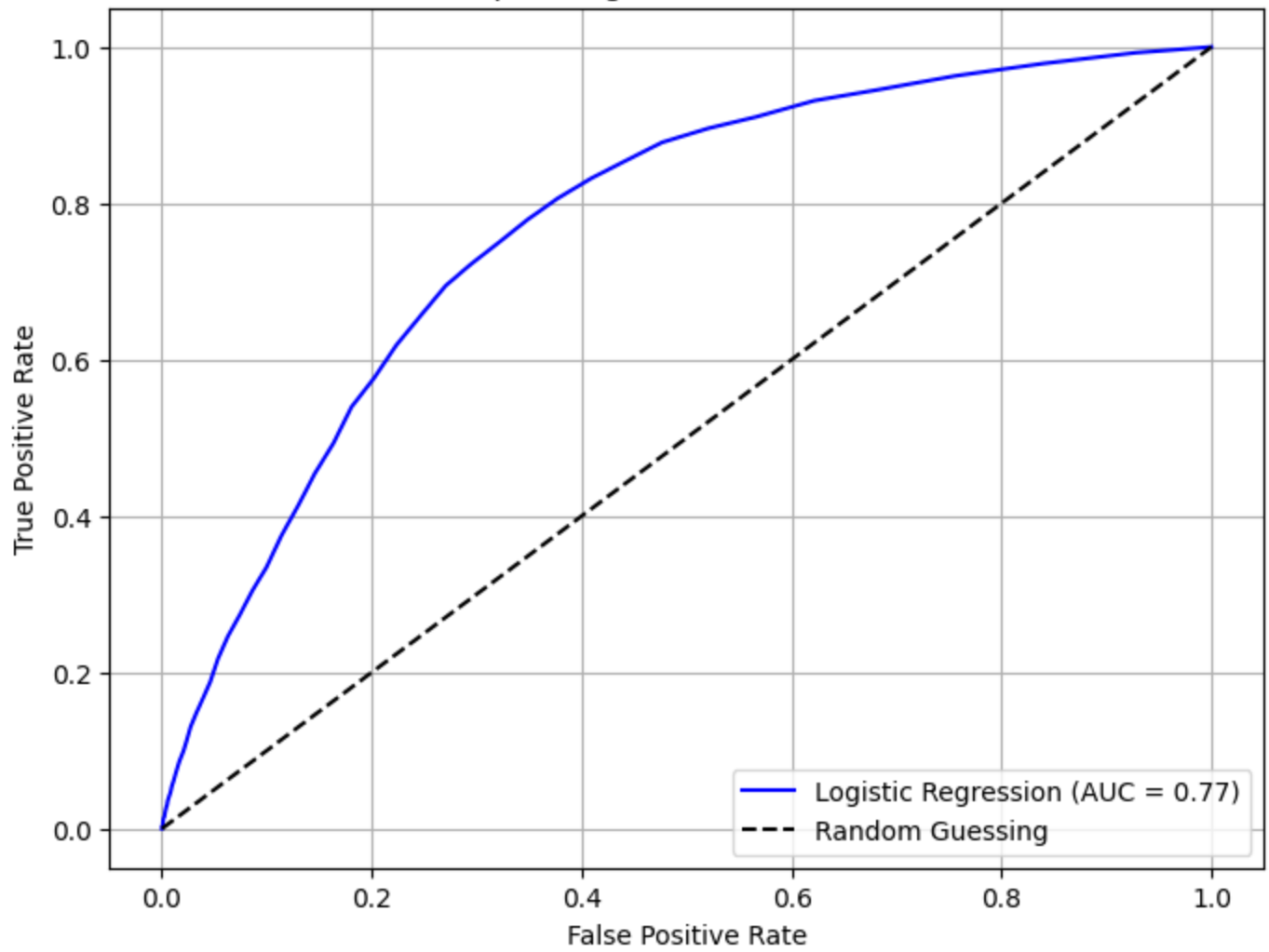




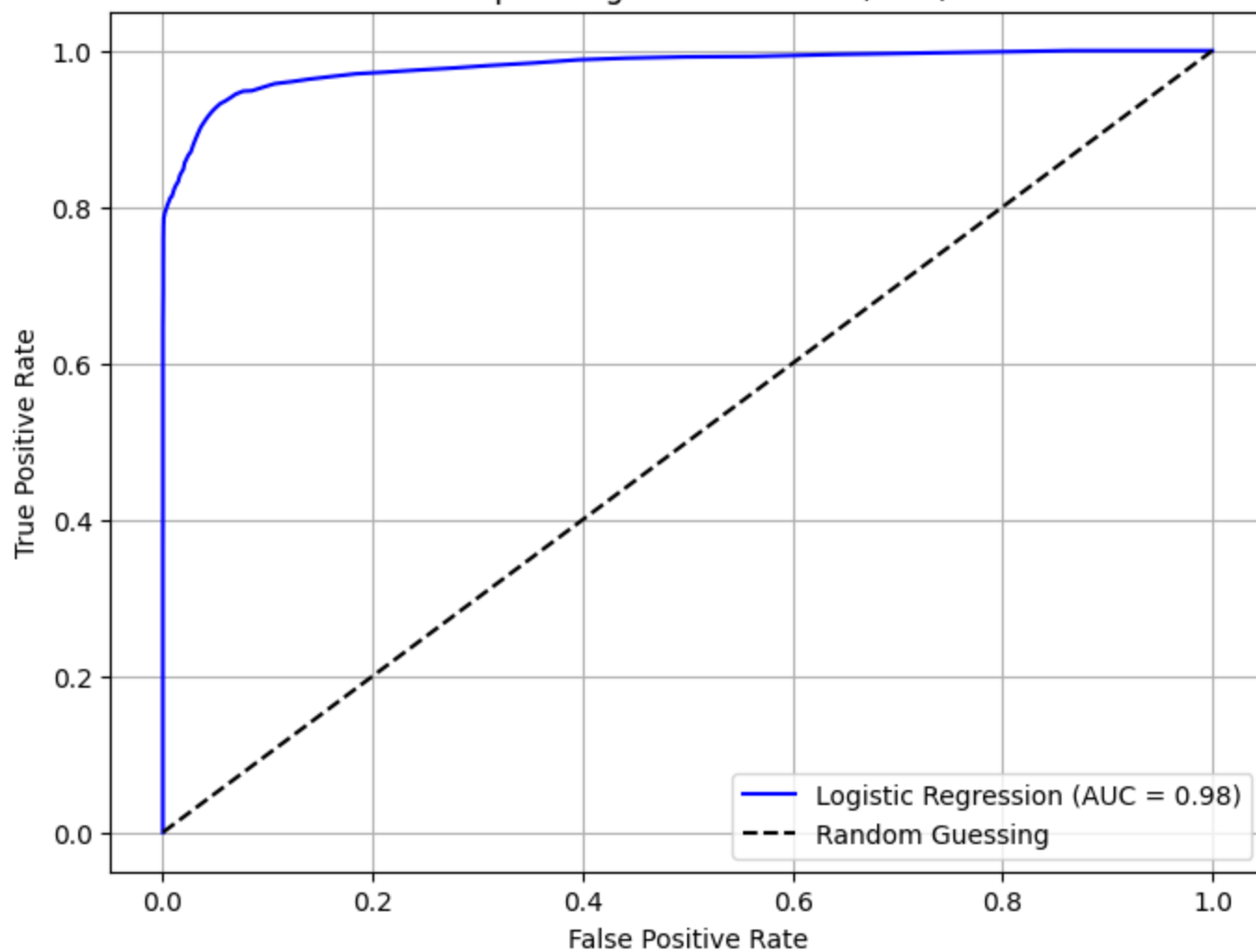




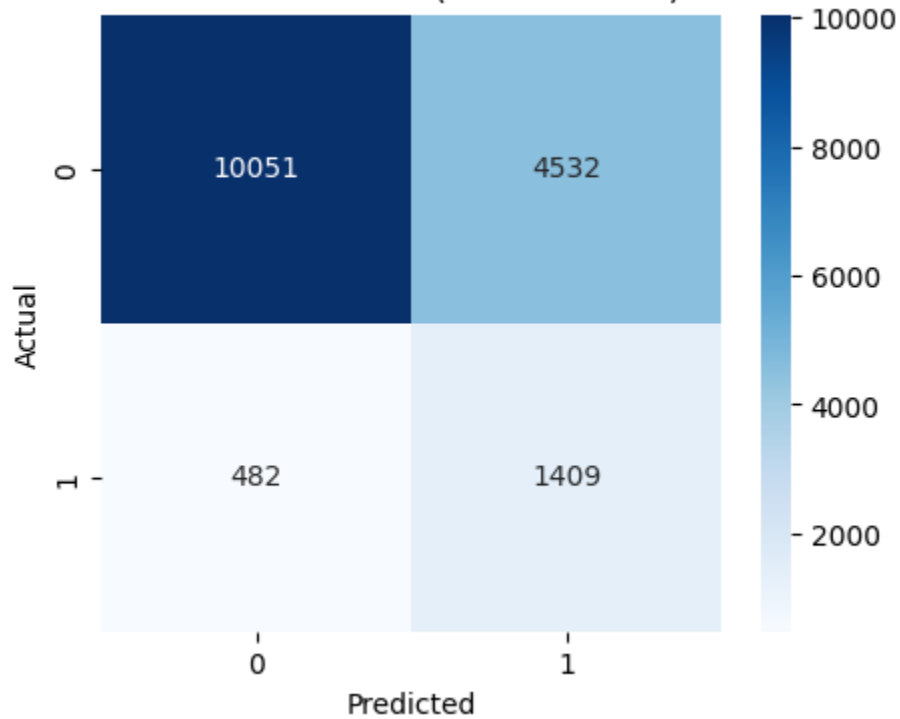
Receiver Operating Characteristic (ROC) Curve



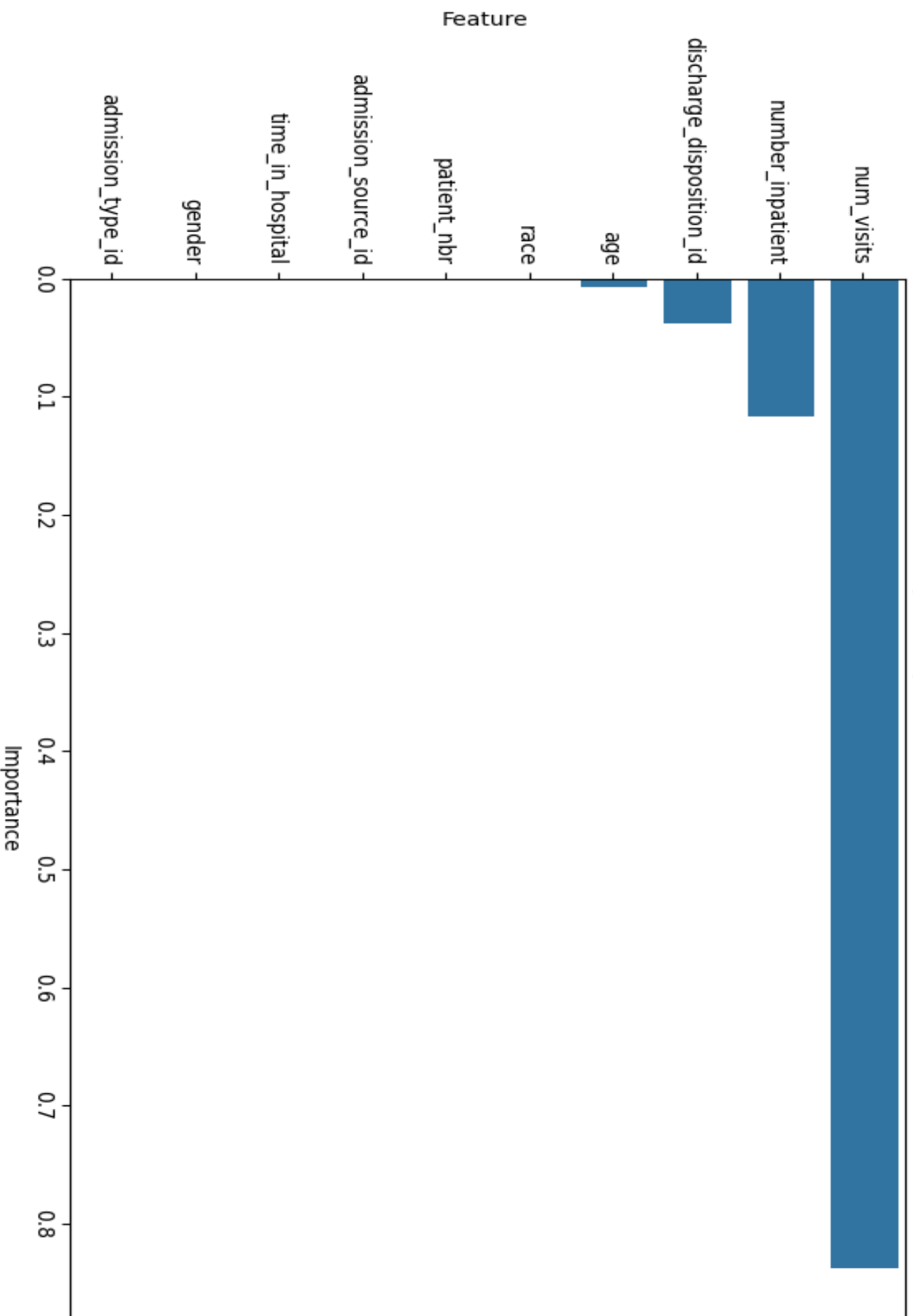
Receiver Operating Characteristic (ROC) Curve



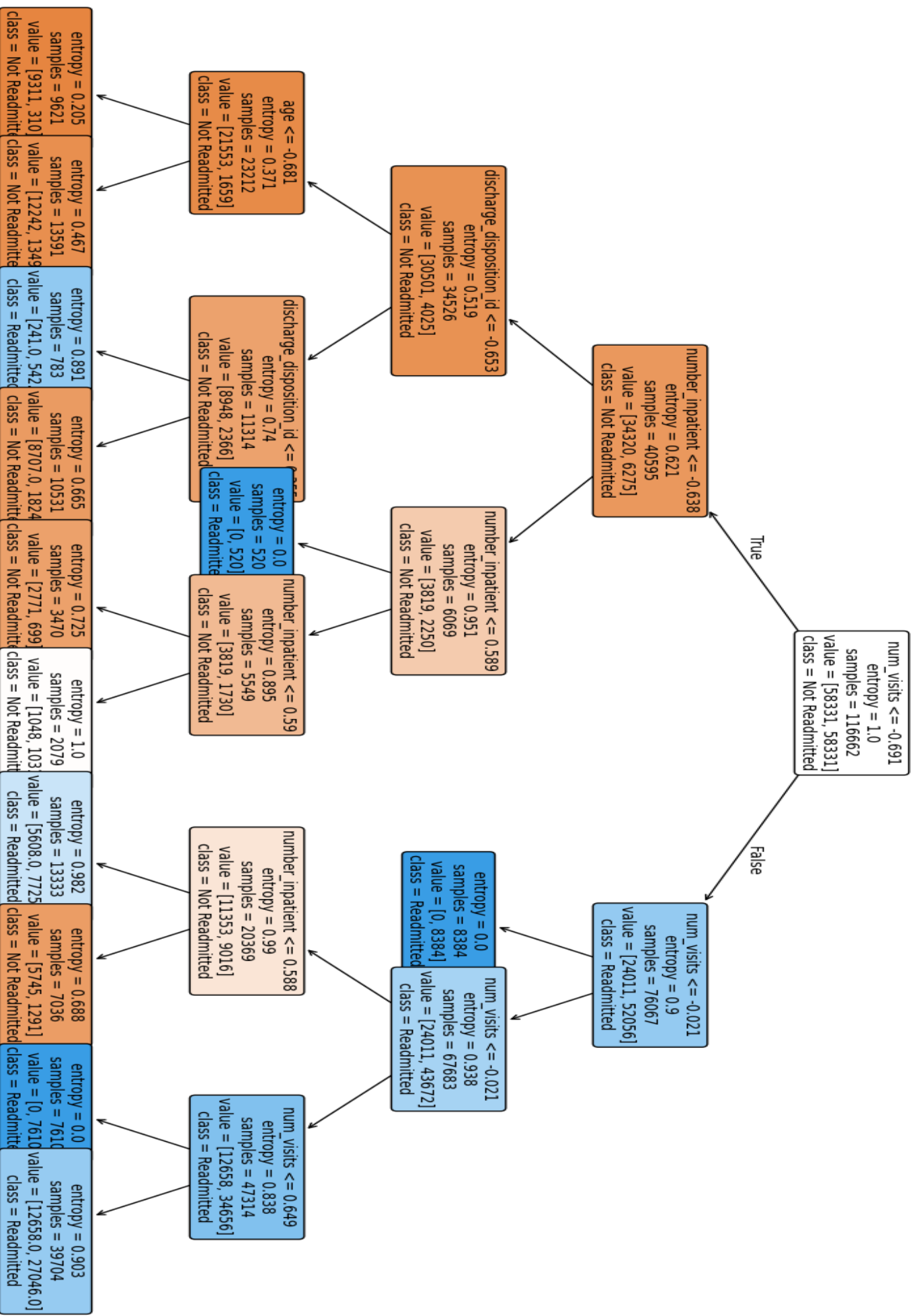
Confusion Matrix (Decision Tree)

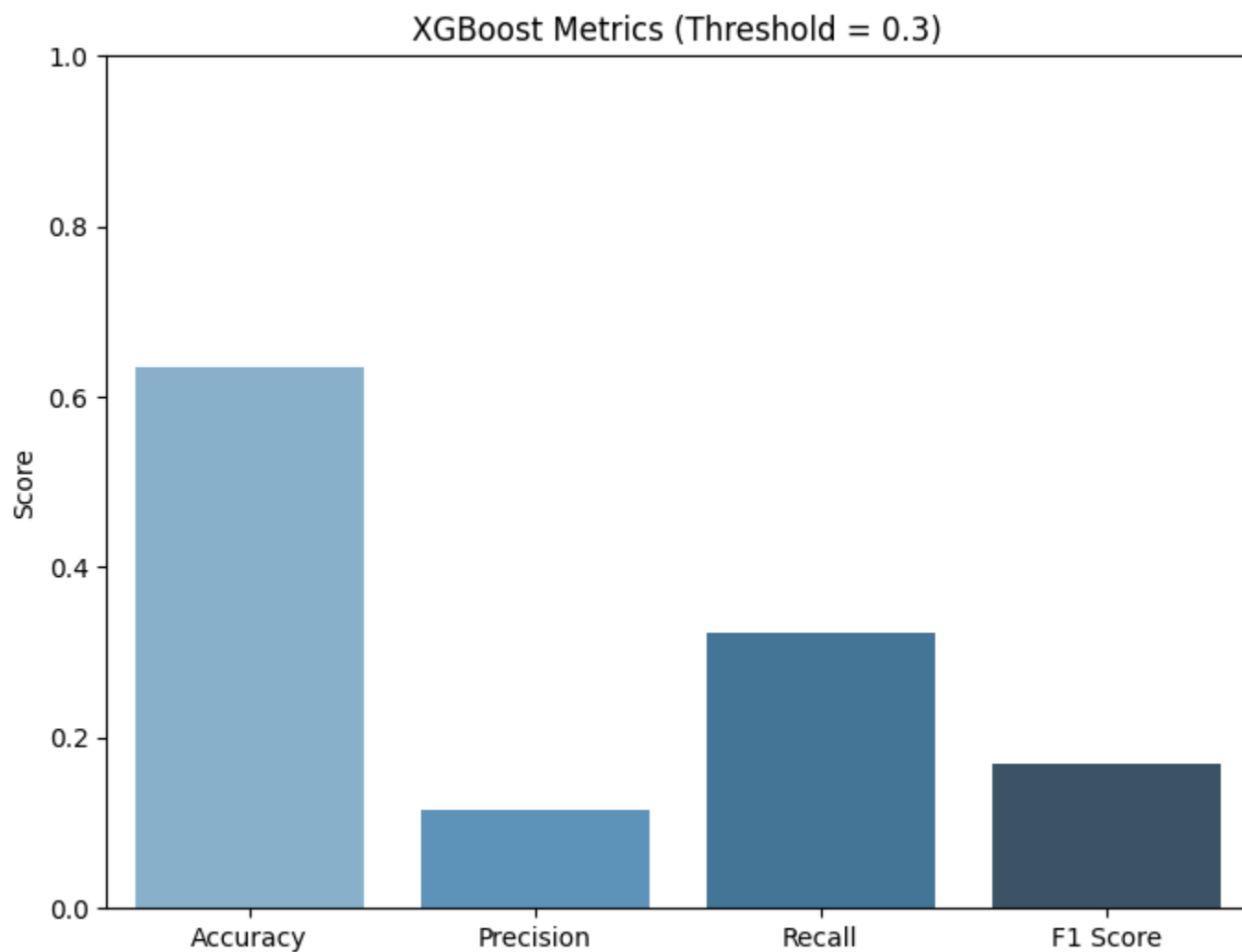
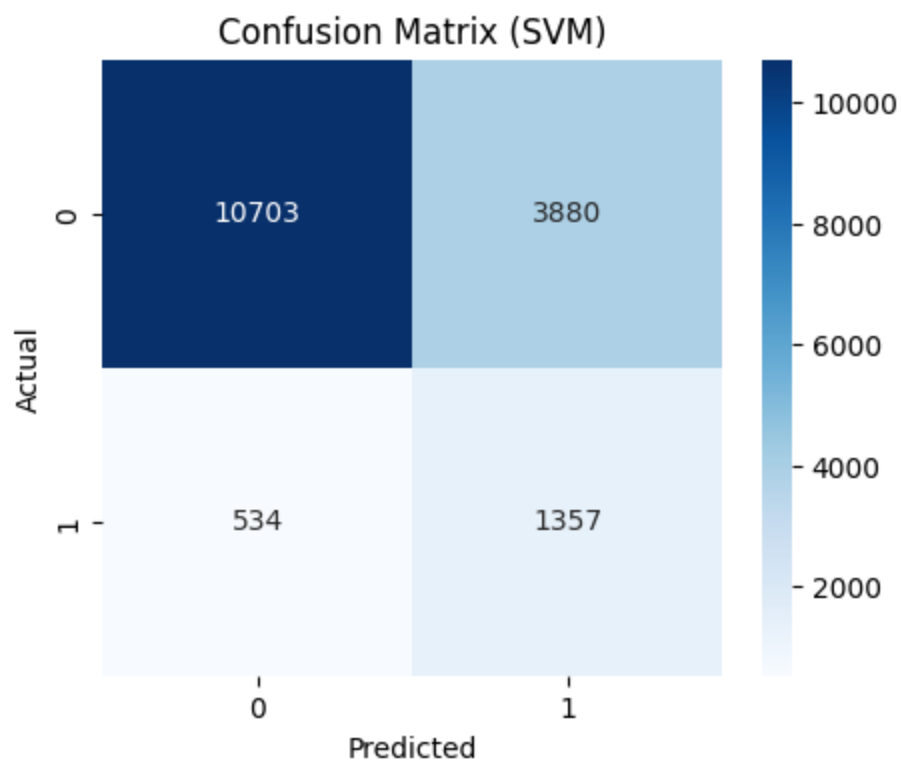


Top 10 Important Features - Decision Tree

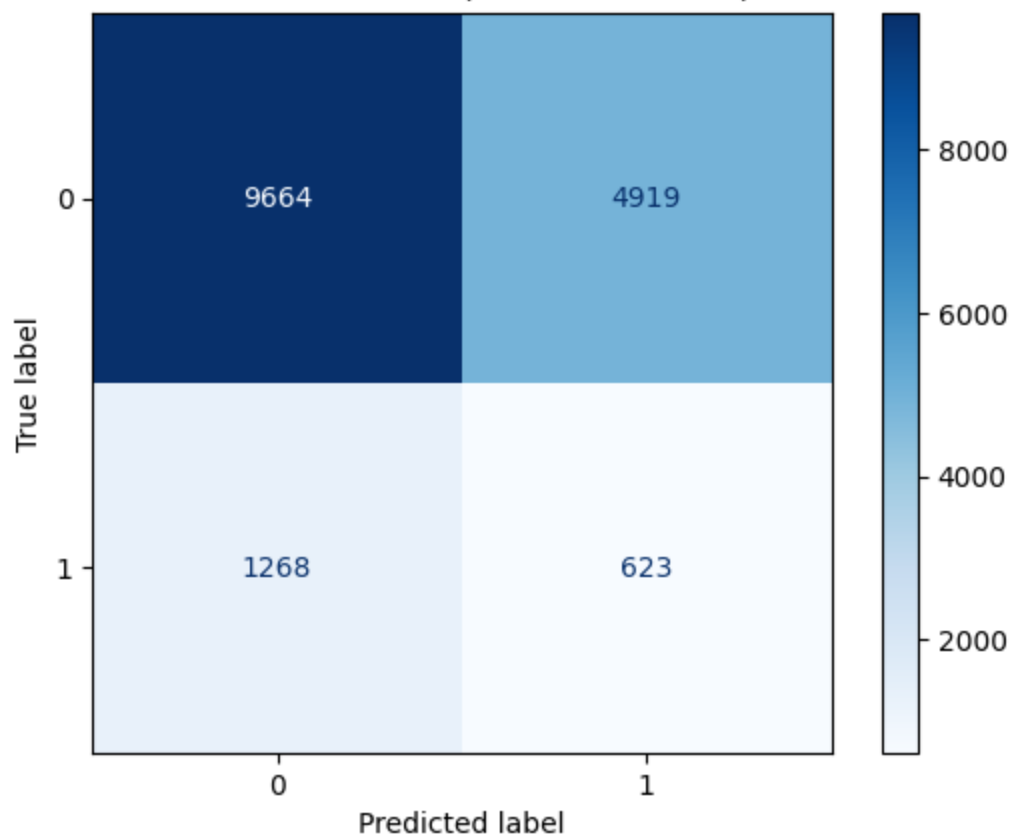


Decision Tree Visualization

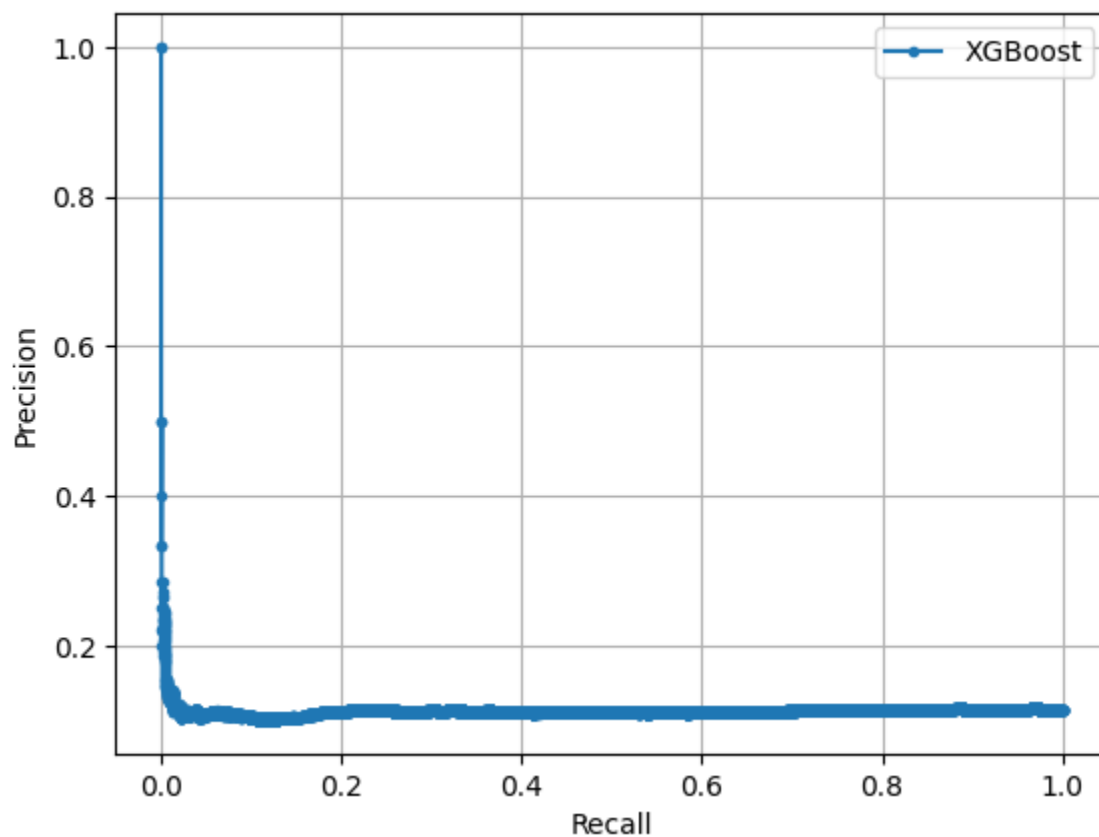




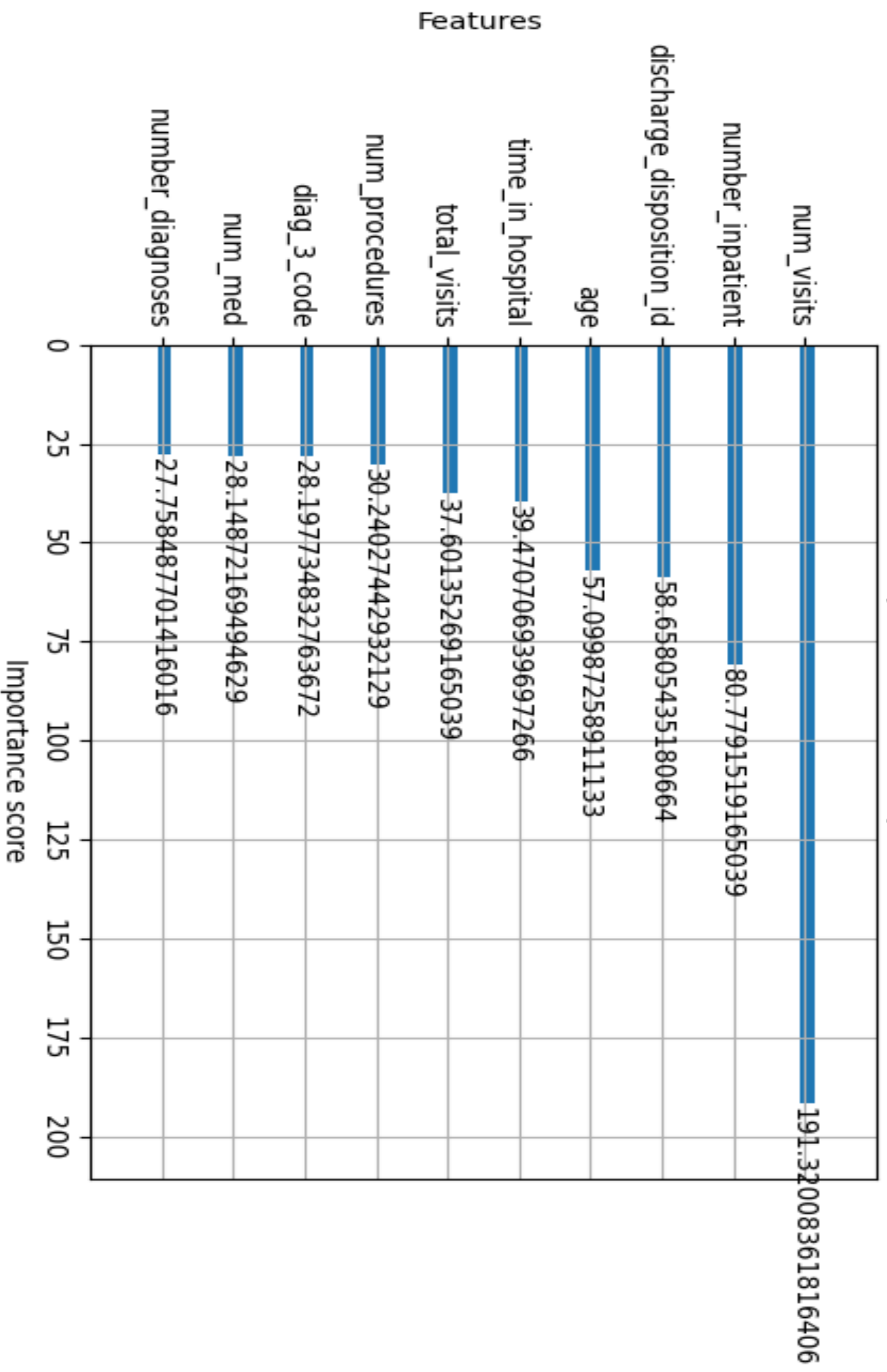
Confusion Matrix (Threshold = 0.3)



Precision-Recall Curve



Top 10 Feature Importances



Conclusion

The project successfully demonstrated the ability to preprocess complex healthcare data, perform necessary EDA, and build predictive models that can aid in forecasting diabetic patient readmissions. Future work could include hyperparameter tuning, exploring additional models, or incorporating more features to enhance prediction performance.